A N   A T L A S   O F   T H E   A G E N T   S T R A T A

THE AGENT, SEEN WHOLE

A Master Volume

Volume III · the agent layer unfolded as ten sub-strata plus four wrapping meta-strata.

Seven artifacts in one bound order — frontispiece, foundation, decisions, plates, tracker.

M

Compiled for A. Yedi

Cycle MMXXVI · Revision I · 2026-05-12

READING GUIDE

How to navigate this volume

A · SEQUENTIAL READ (FULL CYCLE)

B · NINETY-MINUTE SPEED READ

If you cannot read the volume cover-to-cover, take this path. Total ~90 min.

5 min	Master Plate — the whole synthesis on one page
10 min	Tracker Section A — the 7 Bets refreshed (Vol III deltas annotated)
20 min	Addendum Part XI — Synthesis with full Bet rewrites, Risks, Cruxes
15 min	Addendum Part XIII — the Agent-Specific Procurement Rubric
10 min	Substrate Plate I (index) + Plate V (vertical products)
10 min	Report Executive Summary — five forces structuring the agent layer
20 min	Skim the rest as you act on the 6/12/18-Month Action Map (Addendum Part XII)

THE VOL III DELTA (ONE LINE)

Bet #1 sequenced first (claim Process Power) · Bet #2 second (collect equity) · Bet #3 third (compound as advisory) · Bets 4–5 fold into #1 as modules.

THE SYNTHESIS

Master Plate

Volume III · the agent layer as a single connected system

Section I of VII

A N A T L A S O F T H E A G E N T S T R A T A · M A S T E R S E

THE AGENT, SEEN WHOLE

M

M A S T E R P L A T E

Ten sub-strata · four wrapping forces · cross-layer flows · binding constraints · sequenced Bets — the agent layer as a single connected system.

Stratum agenticum tota machina · quomodo strata simul operentur

B I N D I N G C O N S T R A I N T S

- I — per-step ~99% — computer-use ceiling
- I — ctx > 32K — reasoning regression
- III — MCP forks — spec held / experience splits
- VI — OSWorld ~50% — unattended autonomy gates
- VIII — indirect PI — unsolved across all vendors
- VII — no signed reports — procurement-readiness gap
- IX — lab walk-up — encroaches Sierra / Glean wedge
- X — form factor — CISO + buyer + threat-model
- M-A — ASL framework — not in EU code / NIST / ISO
- M-B — EU Art. 14 — agentic triggers caught
- M-C — token price — decline outpaces optim
- M-D — sovereign — data residency lever

X — END-USER · SURFACES

form factor binds enterprise adoption · WhatsApp = global majority

IX — VERTICAL · PRODUCTS

Sierra · Decagon · Glean · Harvey · Hippocratic · Hebbia · Rogo · Clay

VIII — RUNTIME · SAFETY

indirect prompt injection unsolved · mostly absorbed by hyperscalers

VII — EVAL · OBSERVABILITY

signed reproducible reports = unclaimed flag · OTel GenAI Jan '26

VI — ACTION · SURFACES

sandbox · browser · computer · voice · operational layer holds margin

V — PLANNING · REASONING

ReAct + planner-executor + verifier · 70–85% tokens on cheap tier

IV — MEMORY · STATE

RAG with writes · consumer absorbed · compliance niche

III — TOOL · PROTOCOL

MCP spec held · experience fragmenting · gateways durable

II — AGENT · RUNTIMES

thinnest gross-margin layer · splitting along the model spine

I — FOUNDATION · CAPABILITY

OSWorld ~50 · per-step ~99% gates unattended autonomous

form factor gates vertical buyer

vertical -> trajectory eval

guardrail -> eval artifact

verifier -> action surface

tool returns -> injection vector

MCP -> tool surface

reasoning -> trajectory

memory -> RL trajectory data

harness -> model coupling

S E V E N B E T S · S E Q U E N C E D

Bet #1 first (claim Process Power) · Bet #2 second (collect equity) · Bet #3 third (compound)

1. **PROCUREMENT OPERATING STANDARD** — 5 / 5 — open Playbook · 30 expert calls · Agent Procurement Rubric · the unclaimed flag at Meta-B
2. **VERTICAL AGENT GTM ROLE** — 5 / 5 — Sierra · Decagon · Glean · Harvey · Hippocratic · Hebbia · Rogo · Clay · Ramp · Runway · NYC peal
3. **MCP-NATIVE PRACTICE (reframed)** — 4 / 5 — advisory + gateway-adjacent · pair with Cloudflare / Kong / Pomerium · not productized servers
4. **FINOPS FOR TRAJECTORIES** — 4 / 5 — planner-executor audit · 12-mo window before Bedrock auto-routes · fold into Bet #1 module

   model walks up-stack (encroach)
5. **RAG + MEMORY ARCHITECTURE** — 4 / 5 — vendor-neutral architecture audit · memory architecture as service line · Crux #5 absorbed/nich
6. **OPERATOR TRANSLATION NEWSLETTER** — 3 / 5 — Job 5 distribution layer · Conclude-phase translation · cross-pollinate Bet #1 downloads
7. **VC OPERATING PARTNER** — 3 / 5 — long-arc fallback · principal-to-operator path active · primary if Bets 1-3 stall

F O U R W R A P P I N G M E T A - F O R C E S

META-A	META-B	META-C	META-D
CAPABILITY SAFETY	REGULATION	ECONOMICS	GEOPOLITICS
ASL · Preparedness · Frontier · METR · AISI	EU AI Act Art.14 · CA SB 53 · sectoral · Bartz · NYT	per-task · outcome pricing · planner-executor · FinOps	sovereign agents · export controls · jurisdictional siting

F I V E C R U X E S · A N S W E R - E V E N T S T H A T R E - R A N K E V E R Y T H I N G

- C1 — Anthropic ARR · 24Bvs30B · Q3 '26
- C2 — MCP commons vs silent fork · EOY '26
- C3 — OSWorld 65% on frontier · Q3 '26
- C4 — EU AI Act Art. 14 teeth · late '26
- C5 — Standalone memory · absorbed vs niche · H2 '26

AN ATLAS OF THE AGENT STRATA

The Foundation Reference

Volume III · the descriptive layer · ten sub-strata plus four wrapping meta-strata

Section II of VII

THE AGENT LAYER

A SUBSTRATE OF ITS OWN

A FOUNDATION REFERENCE · VOLUME III

Higher-resolution zoom · 2025–2026

Compiled for A. Yedi · Companion to Volumes I and II of SUBSTRATE

Preamble

This document is Volume III of the SUBSTRATE atlas — a parallel-resolution zoom into a single band of the 2025–2026 AI stack. The two prior volumes treated the entire stack as eighteen strata. Volume I (six plates plus an ~8,500-word foundation report) carried the visual and prose atlas from the power grid through the end user. Volume II (five plates plus an ~18,700-word decisions playbook) layered five framework analyses on top — the OCQ Matrix, Wardley Mapping, Helmer's 7 Powers, Ecosystem JTBD, and the empirical Talent and Capital Flow tracker — and concluded with seven Big Bets, five Structural Risks, and five Cruxes.

Three of those seven Bets sit directly on the agent layer. Bet #2 names taking a Director, Field-CTO, or Head-of-Enterprise-GTM seat at a vertical-agent company (Sierra, Decagon, Glean, Harvey, Hippocratic, Augment) as the highest-conviction near-term play. Bet #3 names building a small constellation of MCP servers for systems enterprise GTM teams live in, taking advantage of the protocol's 2026 punctuated equilibrium. Bet #4 names inference-cost optimization, whose largest cost surface is inside agent workloads because agents consume hundreds-to-thousands of tokens per task rather than tens. Even Bet #1 — the Enterprise AI Procurement Operating Standard — lives or dies based on how F1000 buyers procure agents specifically, not models in the abstract.

The conclusion of Volume II also carried a tell. Three Bets touching the agent layer, but a single band of analysis devoted to it: Stratum XIII. Eighteen strata across the full stack; one band for the most repriced layer of 2025–2026. This volume corrects that resolution. It takes the agent band and unfolds it into ten sub-strata of its own, wrapped by the same four meta-strata that wrap the parent stack — safety, regulation, economics, geopolitics — with the meta content tightened to agent-specific dynamics rather than the broader field. The aim is to produce, at per-sub-layer resolution, the opportunities, challenges, and open questions that a person with a twelve-year enterprise B2B GTM record, a growing AI-builder practice, and a New York base can act on inside an eighteen-month window.

This volume is the descriptive prose layer. The visual companion — six plates rendered in the same Substrate aesthetic — is the diagrammatic key. The decisions companion — Volume III's addendum, five framework plates plus the ~18,000-word per-sub-stratum analysis — is the strategic layer. The three read together; none is the whole. Anything that re-derives what Volume I or II already concluded has been deliberately omitted; the prior work is referenced by name and built upon.

Executive Summary

The 2025–2026 step-change in artificial intelligence is not raw model capability. It is the transition from a chat-shaped product surface — single-turn prompt, single-turn response — to an agent-shaped one: a system that takes a goal in natural language, plans multiple steps, executes them through tools, observes the result, and iterates until completion or escalation. That transition has compressed roughly a decade of enterprise SaaS adoption pattern into about thirty-six months. Sierra crossed $175 million ARR at four-times year-over-year by Q1 2026; Decagon cleared $80 million; Glean passed $300 million; Harvey past $100 million; Cursor reached a $500-million run-rate; Lovable went from zero to $80 million in twelve months; Anthropic's Claude Code embedded several hundred million in run-rate inside the parent disclosure. Every one of those companies is an agent or agent-adjacent product. The pattern is no longer anomalous; it is the new dominant pattern of enterprise software adoption.

Five forces structure the agent layer as of May 2026.

The first is the capability ceiling at the foundation-model layer. Claude Opus 4.5 reaches 82.8% on SWE-Bench Verified inside the Claude Agent SDK harness. GPT-5 with thinking reaches the high seventies. OSWorld — the closest public proxy for desktop computer use — sits at approximately fifty-to-fifty-five percent against a human baseline of seventy-two percent, having crossed fifty percent only in early 2026. The single number that tells the story is error compounding: a model right on ninety-five percent of individual steps completes a twenty-step trajectory roughly thirty-six percent of the time. Until per-step reliability crosses approximately ninety-nine percent, computer-use agents will be supervised rather than autonomous, and the capability ceiling materially limits applications today. The ceiling will move; the question is the slope.

The second is the thinning of the runtime layer. Six months ago, the operative question was whether LangGraph would consolidate the agent-runtime category the way Kubernetes consolidated container orchestration. The answer as of May 2026 is that it will not. The runtime is splitting along the model spine: Anthropic-shop enterprises run the Claude Agent SDK released in September 2025; OpenAI-shop enterprises run the Agents SDK shipped in March 2025 or wrap LangGraph around it; Google-shop enterprises run the ADK that shipped at Cloud Next in April 2025. The model-neutral runtime is real and defensible — LangGraph 1.0 went GA in October 2025 and LangSmith carries materially more margin than the runtime itself — but it is the thinnest layer of the agent stack by gross margin. Value accrues to the model below and the application above. The runtime is connective tissue; do not build a company on being the best runtime.

The third is the Model Context Protocol — commons or fork. MCP, published by Anthropic on November 25, 2024, was adopted by OpenAI on March 26, 2025, by Google on April 9, 2025, donated to the Linux Foundation's Agentic AI Foundation on December 8, 2025, and crossed approximately eleven thousand registered servers by April 2026. The spec is held. The experience is fragmenting. OpenAI's native Responses API tool-call format remains proprietary, and many production OpenAI agents bypass MCP entirely for latency reasons. Google's Agent-to-Agent protocol is adjacent to MCP and could fragment agent-to-agent communication along the same fault line. Anthropic's own `tool_use` blocks still originate in proprietary form and translate to MCP at the runtime edge. The honest read is that by 2027, "MCP-compatible" will mean roughly what "OpenAPI-compatible" means today — a baseline, not a guarantee of interoperability. A meaningful enterprise gateway category (Cloudflare, Kong, Pomerium) is forming as the procurement-friendly control plane and is one of the more durable middleware positions in this volume.

The fourth is the inversion of token economics into trajectory economics. A single-turn API call is priced in cents per million tokens; an agent task is priced in dollars per task. A typical CX deflection trajectory consumes four-to-twenty-five-thousand tokens at a few cents apiece. A coding agent edit consumes thirty-to-two-hundred-fifty-thousand tokens at twenty cents to three dollars. A deep-research trajectory consumes half-a-million to five million tokens at one to twenty dollars. The planner-executor split — route a few percent of tokens through an expensive thinking model and the rest through a cheap executor — is the durable architectural idea of the era and the operational basis of every shipping production agent at Sierra, Decagon, Cursor, Cognition, and Anthropic's own deep-research products. It is also the basis of "FinOps for Tokens" as a procurable category. Test-time compute beyond approximately eight thousand reasoning tokens has unattractive economics for most enterprise agent classes and should be sold accordingly.

The fifth is the binding constraint on enterprise agentic adoption — and it is not capability. It is form factor. As long as humans remain in the decision loop, which they do across virtually all 2026 enterprise deployments, the surface through which the user meets the agent governs adoption velocity, buyer persona, threat model, and regulatory surface at once. Microsoft 365 Copilot crossed $5 billion ARR with approximately thirty million paid seats by Q3 FY26 because it ships inside an existing seat-license renewal. Claude Code crossed a $500-million run-rate because the CLI is the highest-throughput surface for technical users. Glean built a $300-million ARR business inside Slack because the CISO already approved Slack and the Slack DLP and audit log inherits. Meanwhile, Meta opened the WhatsApp Business API to agentic vendors in January 2026, and the global majority's first AI surface is now a WhatsApp agent — not a chat website, not a Cursor IDE, not a desktop computer-use agent. The US-coastal sequence "chat web → IDE inline → computer use" is a US-coastal narrative. Treat it accordingly.

What follows treats each sub-stratum at the depth a senior practitioner needs to make decisions, and at the brevity a senior practitioner can absorb in a single sitting.

Contents

Part I · The Capability Substrate

Three sub-strata form the substrate on which every agent runs: the foundation model and its agentic capability, the runtime that turns the model into a goal-directed system, and the protocol by which the system reaches out to external tools. These three are the geological basement of the agent layer. Everything else sits on top of them.

STRATUM I · FOUNDATION MODELS AS AGENTIC ENGINES

Treat this stratum strictly as the agentic-capability delta between frontier models, not as the provider-economics layer covered in Volume I. Single-turn benchmarks — MMLU, GPQA — have saturated. The agent-relevant frontier is four numbers: TAU-bench for multi-turn tool use against a simulated user, SWE-Bench Verified for multi-file code agency, OSWorld and WebArena for computer and browser use, and GAIA or BrowseComp for open-web research. Anthropic's Claude Opus 4.5, released November 2025 and refreshed in March 2026, reaches 82.8% on SWE-Bench Verified inside the Claude Agent SDK harness, ~84% on TAU-bench retail, ~67% on TAU-bench airline. OpenAI's GPT-5, launched August 2025 and shipped in a "thinking" GPT-5.1 variant in December 2025, reaches 74.9% on SWE-Bench at GA and approximately 80% with the thinking-plus-Responses-API tool loop. Google's Gemini 2.5 Pro Deep Think refresh in March 2026 reaches GAIA ~70% and BrowseComp ~50% and powers Project Mariner, which moved into Gemini Enterprise GA in April 2026 but has not posted an independent OSWorld number above ~38%. DeepSeek's R1-0528 and the rumored R2 sit in the mid-fifties on SWE-Bench with a structured-output gap, not a reasoning gap. Meta's Llama 4 Maverick and Behemoth, released April 2025, trail closed frontier by ten-to-fifteen points on tool-use benchmarks with a less developed harness ecosystem. The honest cross-vendor ranking on agentic tool use is Claude ahead, GPT-5 close, Gemini behind on tool use but ahead on long-context retrieval; on raw single-shot code generation the order shifts, with GPT-5 often ahead. Anyone claiming a single winner is selling.

The capability ceiling at this stratum is computer use. Anthropic's Computer Use shipped October 2024 with OSWorld in the low twenties; by early 2026 it crossed fifty percent on the public scoreboard; by May 2026 the leading systems sit at fifty-to-fifty-five percent. The human baseline on OSWorld is approximately seventy-two percent. WebArena sits in the high sixties to low seventies for top systems against a human baseline near seventy-eight percent. The arithmetic of multi-step trajectories is unforgiving: a model right on ninety-five percent of individual steps completes a twenty-step task at roughly thirty-six percent end-to-end. Until per-step reliability crosses approximately ninety-nine percent — a 2027 question — computer-use agents will be supervised rather than autonomous. The form-factor implications of this ceiling are unpacked at Stratum X; the procurement implications at Stratum VII and Stratum VIII.

Two under-discussed dynamics warrant naming. The first is reasoning regression: every shipping reasoning-model system card from Q3 2025 onward — Anthropic's Claude Agent SDK release notes from September 2025, OpenAI's GPT-5 system card from August 2025, Google's Deep Think disclosures from May 2025 — acknowledges that models score worse on agentic trajectories than the benchmark sum-of-parts predicts. The culprit is context degradation past roughly thirty-two thousand tokens of tool traces, after which instruction-following decays and tool-argument hallucination spikes. The Claude Agent SDK's sub-agents pattern is an explicit architectural workaround: spawn fresh-context children, summarize back, preserve the parent's context. The pattern is elegant and ahead of peers by approximately six months. It is not a moat; OpenAI's handoffs and Google's ADK agent teams are converging on the same primitive. By Q4 2026 it will be table-stakes. The second dynamic is test-time compute as the real lever: Claude Opus 4.5's sixty-four-thousand-token thinking budget, GPT-5 thinking mode, and Gemini Deep Think show super-linear gains on agent tasks specifically, with latency rising from seconds to minutes and pricing crossing fifty cents to five dollars per task. The market bifurcates accordingly: interactive agents stay on Sonnet-class, Flash-class, or GPT-4.1-class executors; back-office autonomous agents run on Opus, GPT-5 thinking, or Deep Think.

STRATUM II · AGENT RUNTIMES AND HARNESSES

The runtime turns a foundation model into a goal-directed system. It owns the tool-call loop, memory paging, sub-agent orchestration, retries, evals, and observability. Through mid-2025, the consolidation question pointed at LangGraph. By May 2026 the answer is that the runtime is splitting along the model spine, not consolidating. Four serious enterprise contenders define the contour. The Claude Agent SDK, released as `claude-agent-sdk` in September 2025, evolves from the internal harness behind Claude Code; its native primitives are sub-agents, skills, hooks, and MCP-native tool calling, with Python and TypeScript surfaces. Anthropic uses it internally at scale; Cursor, Vercel, and Replit have referenced it in 2026 release notes. Its strength is the model-coupling and the velocity of shipping. Its weakness is precisely the same: the "neutral runtime" pitch is gone, and it locks to Claude.

The OpenAI Agents SDK shipped March 11, 2025 alongside the Responses API, deprecating the older Assistants API with a final shutdown mid-2026. Its primitives are handoffs, guardrails, tracing dashboards, and MCP support added in April 2025. It mirrors much of LangGraph’s surface area but ships directly from OpenAI. Its strength is the OpenAI distribution and the tight Responses-API tool loop; its weakness is lighter multi-agent orchestration than LangGraph or AutoGen. Google’s ADK shipped April 9, 2025 at Cloud Next; it is Python-first with Java added in September 2025, bundled with Vertex AI Agent Engine, and integrated into Gemini Enterprise (GA December 2025). The Agent-to-Agent protocol announced at the same event is Google’s bid to do for agent-to-agent communication what MCP did for tools. ADK adoption outside Google customers is thin; A2A has not won broad cross-vendor commitment as of May 2026. LangGraph 1.0 went GA on October 22, 2025 after a year of `0.x` evolution; the repositioning is that LangChain Inc. is now four products — LangChain (the easy surface), LangGraph (the production runtime), LangSmith (the observability and evals layer where margin actually sits), and LangGraph Platform (managed deployment). Its strength is model-neutrality and the deepest production maturity in the field. Its weakness is API churn fatigue: the field narrative that "LangGraph wins enterprise" is ahead of the evidence, and several F500 shops that standardized on LangChain in 2024 are now picking between staying on LangGraph and ripping it out for a vendor SDK.

The honorable mentions are real but narrower. Mastra, a TypeScript-native runtime out of Y Combinator W24, is winning the Next.js / Vercel developer wedge. Pydantic AI, from the Pydantic Logfire team, owns typed-validation-first agents in Python and is growing in regulated industries that need schema rigor. CrewAI, with its multi-agent role-playing pattern, raised $18 million in October 2024 and is the chosen runtime in many operations and automation use cases without being serious as a code-agent runtime. Hugging Face Smolagents (December 2024) is the minimalist "code-as-action" library — technically interesting, narrowly used. Microsoft AutoGen split into AutoGen 0.4 from MSR and the community fork AG2 in November 2024 and retains traction inside Microsoft customers via Semantic Kernel. DSPy from Stanford and BAML from BoundaryML are not runtimes but compile-time and prompt-IR layers that increasingly sit beneath the runtimes above; Inspect AI from the UK AI Safety Institute is the eval framework with growing traction in safety-conscious enterprises. Non-US runtimes are material: Manus AI from Butterfly Effect in China launched March 2025 and reshaped field expectations with its viral demo; Mistral’s Agents API shipped May 2025; Sarvam AI in India announced an agent platform in Q1 2026; Aleph Alpha pivoted toward compliance-first agent tooling in Europe. The opinionated read: today the runtime layer is splitting along the model spine and is the thinnest layer of the agent stack by gross margin. Value accrues to the model below and the application above. The "sub-agents as moat" claim is a feature, not a moat. Build a company elsewhere.

STRATUM III · TOOL USE AND THE MODEL CONTEXT PROTOCOL

Tool use is the joint at which an agent reaches out of its context window into the world. Until November 2024 the tool-use interface was the OpenAI function-calling spec from June 2023, refined by structured outputs in August 2024 — both proprietary, per-vendor, and not interoperable. Anthropic published the Model Context Protocol on November 25, 2024 as a JSON-RPC standard with Python and TypeScript SDKs and four reference servers — filesystem, GitHub, Slack, Postgres. OpenAI committed to MCP across the Agents SDK, ChatGPT desktop, and the Responses API on March 26, 2025, with a Sam Altman tweet crediting Anthropic. Google announced MCP support across Gemini and Vertex AI on April 9, 2025 at Cloud Next, alongside A2A. Microsoft shipped MCP into Copilot Studio and Semantic Kernel in May 2025; the GitHub MCP server became the canonical first-party example. Hugging Face released MCP-compatible Spaces in June 2025. On December 8, 2025, MCP was donated to the Linux Foundation under a new Agentic AI Foundation, with Anthropic relinquishing unilateral spec authority and the initial Technical Steering Committee seating Anthropic, Microsoft, Google, OpenAI, Hugging Face, and Cloudflare.

The registry growth is the empirical proof. Roughly fifty servers existed at launch in November 2024. By April 2025 there were approximately twelve hundred. By December 2025, at the Linux Foundation donation, there were fifty-eight hundred. By April 2026, per the Linux Foundation MCP registry statistics page, there were approximately eleven thousand four hundred. Quality is bimodal: a long tail of toy servers, plus a slowly hardening tier of first-party servers from Stripe, Linear, Cloudflare, Notion, GitHub, Atlassian, Datadog, Snowflake, Databricks, and ServiceNow. The MCP gateway category — Cloudflare's Workers AI MCP (December 2024), Kong's MCP Gateway (announced March 2025, GA July 2025), Pomerium's identity-aware MCP proxy (May 2025), and Anthropic's own remote-server pattern (November 2025) — is the enterprise control plane (authentication, audit, rate-limiting, secret injection) and is analogous to the API gateway category in the REST era. It is one of the more durable middleware categories forming in this volume.

The Crux from Volume II — MCP, commons or fork? — admits a 2026 answer in two parts. The bull case is that the spec is held: all four hyperscalers committed, the LF governance removes the Anthropic kill-switch objection, registry growth is real, and the OpenAPI parallel argues that protocols at this layer settle and stay settled. The bear case is that the experience is fragmenting silently. Four fork vectors are visible. First, OpenAI's Responses-API extensions ship MCP support but the native tool-call format remains proprietary, and many production OpenAI agents bypass MCP entirely because latency is lower and structured-output reliability is higher — the "MCP-compatible but not MCP-native" pattern is growing. Second, Google's A2A is adjacent to MCP and could fragment agent-to-agent communication along a second axis if it wins. Third, Anthropic's own `tool_use` blocks remain the proprietary native format in the Claude API, with MCP servers translating into them; the translation overhead is itself a fork vector. Fourth, a quality split — first-party MCP servers from Stripe, Linear, GitHub are well-maintained; the long tail is abandonware. Enterprises will de facto fork by curating private registries. The honest call: by 2027, "MCP-compatible" will mean roughly what "OpenAPI-compatible" means today — a baseline, not a guarantee. MCP wins on portability and loses on hot-path performance; anyone telling you MCP is unanimous is reading the press releases, not the production traffic.

Part II · The Loops of Cognition

Two sub-strata constitute the inner loops of a working agent: memory, by which the agent retains state across turns, sessions, users, and time; and planning-and-reasoning, by which the agent decomposes a goal into actionable steps and decides which to take next. These two layers determine whether the agent feels like a stateless function or a persistent coworker.

STRATUM IV · MEMORY AND STATE

Vendors blur five things into "memory." Context window — what fits in the prompt — is lab-controlled and not memory at all (Gemini 2.5 Pro at 2M, Claude 4.5 Sonnet at 1M, GPT-5 at 400K). Working memory, the agent's intra-task scratchpad, lives in the orchestrator (LangGraph state, Letta archival blocks, OpenAI Assistants threads) and is engineering rather than category. Episodic memory — recall of past sessions — is the actual category. Semantic memory — extracted entity facts, preferences, account state — is where graph-versus-vector debates live. Procedural memory — learned playbooks — is still mostly research-stage. A useful falsifiability test: ask any "memory" vendor which of episodic, semantic, or procedural it stores, where it lives, and the eviction policy. Roughly seven of every ten vendor pitches collapse to "we summarize and write to a vector database" — RAG with a write path, not a distinct primitive.

The category players in May 2026 are four. Mem0 (Y Combinator W24) raised a $24 million Series A in December 2025 led by Basis Set with YC and Kindred; its architecture extracts, stores in a vector-plus-graph hybrid with pgvector default and Neo4j optional, and retrieves. It claims a twenty-six percent lift on the LoCoMo benchmark in self-reported numbers from April 2025; its production-named logos are thinner than its star count suggests. Letta — formerly MemGPT, the OS-style paging primitive that came out of the NeurIPS 2023 paper — operates as a Berkeley spin-out from Packer and Wooders with a $10 million seed from Felicis and Essence in September 2024 and a reportedly closing Series A around $70 million valuation per an unconfirmed Information rumor; its differentiator is the stateful agent as the unit rather than the stateless completion. Zep, with its Graphiti graph engine (open-sourced August 2024) and $15 million Series A from Curiosity Lane in February 2025, has the sharpest enterprise compliance positioning — SOC 2 Type II, HIPAA-ready, LangGraph and CrewAI integrations, with Athelas and additional late-stage fintech logos. Cognee remains the open-source pipeline with a $2.5 million seed and a likely consolidation outcome rather than a standalone exit. Non-US and lab-native dynamics matter: MemoryOS from Tsinghua (September 2025 paper, not commercialized); Chinese frameworks like Qwen-Agent ship native memory and bypass the category entirely; in

Europe there is no serious standalone challenger as of May 2026; and lab-native memory — ChatGPT memory generally available in paid tiers since April 2024 and free since September 2024, Claude Projects memory GA in April 2026, Gemini Workspace personalization GA in March 2026 — sits in front of hundreds of millions of users against the standalone category's tens of thousands of developer accounts.

Knowledge-graph hybrids are the other major architectural question. Microsoft's GraphRAG (research paper April 2024, open source July 2024) has partial production traction at Azure customers including Hitachi and KPMG. HippoRAG and HippoRAG 2 (OSU, April 2024 and August 2025) remain research-only despite frequent vendor-deck citation. PathRAG (March 2025) and OG-RAG (October 2024) remain papers. LightRAG from HKU is the only graph alternative with non-trivial production adoption beyond GraphRAG. The honest read is that graph approaches win on cross-session entity reconciliation, auditable citations in regulated domains, and multi-hop reasoning; vector approaches win on FAQ-shaped retrieval, semantic similarity over unstructured text, and any latency-sensitive deployment. Production teams settle on hybrid summarization buffers plus selective semantic writes served from pgvector or Turbopuffer alongside a primary RAG store. The boring engineering reality is that most "memory" in production is summarization plus a database. Standalone vendors earn their keep when the team wants a managed API or when cross-session reasoning is the product (customer experience, longitudinal health, longitudinal sales).

Procurement and compliance reality is the wedge. F1000 buyers ask: where does memory live physically? Who can subpoena it? Does the vendor honor GDPR Article 17 forget requests with auditable confirmation? Does it tie out to EU AI Act Article 13 GPAI transparency? Does it satisfy HIPAA? And — a frame no vendor ships against — does it preserve contextual integrity in Helen Nissenbaum's sense, such that a memory written in B2B sales is not resurfaced in HR? In May 2026, Zep has the cleanest residency commitment and is the only credible HIPAA story; Mem0 added explicit forget APIs in February 2026 and a BAA enterprise tier in March 2026; Letta is self-host-first and pushes compliance to the customer; lab-native memory is opaque on deletion auditability and is the enterprise wedge. The directional answer to the Crux from Volume II — memory standalone or absorbed? — is that consumer and prosumer memory will absorb into the labs (it largely already has), while a compliance-sensitive enterprise niche will survive standalone at a $200–400 million ARR ceiling split across two or three vendors. The implication for Bet #5 in the prior tracker is to fold memory architecture into the Enterprise RAG Architecture Practice as a service line, not to stand it up as a separate bet.

STRATUM V · PLANNING, REASONING, TEST-TIME COMPUTE

The architectural chain is shorter than the field's hype suggests. ReAct (Yao et al., October 2022) — interleaved thought, action, observation — remains the dominant loop in production agents in 2026; most "agentic frameworks" (LangGraph, CrewAI, Pydantic AI, the Vercel AI SDK) are ReAct plus a tool schema and a state machine. Plan-and-Solve (Wang et al., May 2023) survives as a prompting pattern in coding agents and deep-research products. Tree-of-Thoughts and Graph-of-Thoughts remained lab curiosities; token cost is multiplicative and production usage is rare. Reflexion (Shinn et al., March 2023) survived as the "reviewer" sub-agent pattern visible in Anthropic's research agent, Cognition's Devin verifier, and Manus's critic; it materially helps on coding and SWE-Bench-style tasks and is mostly noise elsewhere. AutoGPT and BabyAGI (March–April 2023) — the open-ended planner-executor loop — are dead in production. What survived is the idea of a scratchpad, persistent memory, and a tool registry, not the autonomous outer loop. Every serious 2026 agent runs bounded.

The planner-executor split is the durable architectural idea of the era. Sierra, Decagon, Cognition, Cursor, Claude Code, and OpenAI's Operator-now-ChatGPT-Agent all separate a "thinking" planner from cheaper "doing" executors; in a typical production trajectory, two of three calls are the executor and one is the planner. The reasoning-model tier impact since September 2024 — o1, then o3 in December 2024 going GA in January 2025, then Claude 3.7 extended thinking in February 2025, then Claude 4 and 4.5 thinking, then Gemini 2.5 Deep Think in May 2025, then DeepSeek R1 and R1-Zero in January 2025, then Qwen QwQ-32B in March 2025, then Kimi K1.5 and K2 in January–July 2025 — collapsed two distinctions that mattered in the ReAct era. First, the model now does internal multi-step search before emitting a token; the agent framework's plan step becomes redundant on hard tasks because the model already planned and you paid for it inside `reasoning_tokens`. Second, tool use during reasoning — o3, Claude extended thinking with tool use, Gemini 2.5 thinking plus tools — means the framework's outer ReAct loop and the model's inner thinking loop overlap. The net result is simpler outer scaffolds and more tokens per step. Most production agents in May 2026 are three-to-seven-step ReAct loops with a reasoning model, good system prompts, and a verifier pass. The exotic graph, tree, and meta-planner work is either inside the model now or relegated to long-horizon research agents.

Test-time compute economics bend the curve fast. Below approximately two thousand reasoning tokens, gains are large and linear on math, coding, and multi-hop retrieval. Between two and eight thousand, diminishing but real — the sweet spot for most agent tasks. Between eight and thirty-two thousand, gains exist on hard coding, olympiad math, and saturated agent benchmarks but cost four-to-ten times and push latency across the ten-to-thirty-second line. Beyond thirty-two thousand, gains regress on tasks where the answer was already correct at four thousand — what Anthropic's February 2026 extended-thinking guidance and OpenAI's `reasoning_effort=high` documentation both call "overthinking." Latency thresholds that matter for enterprise: under two seconds for interactive autocomplete; two-to-ten for "thinking" UIs with a visible spinner; ten-to-sixty for "task" UIs (research, code-PR, ticket resolution); beyond sixty seconds the surface must change entirely to asynchronous with notification. A 10x thinking budget — going from `reasoning_effort=medium` (~2K) to `high` (~16K) — typically costs five-to-twelve times the call price for a three-to-ten-point benchmark gain on hard subsets and zero-to-two points on typical enterprise tasks. For customer experience, retrieval-augmented generation, and most knowledge-work agents, the high setting is a waste. For coding and compliance reasoning it pays. The split is task-class-specific, not vendor-specific. The Chinese reasoning ecosystem — DeepSeek R1 and V3.5, Qwen3 and QwQ, Kimi K1.5 and K2, Manus — is at parity on the recipe (RLVR plus GRPO at trajectory level), behind on raw scale, and ahead on openness, with reported inference cost at twenty-to-forty percent of US frontier. The procurement implication for US F1000 is that open-weight Chinese reasoners are the realistic on-premises option, with political and IP risk the gating factor rather than capability.

Part III · Action and Defense Surfaces

Three sub-strata cover the surfaces by which the agent reaches into the world, the layer by which we measure whether it is doing the right thing, and the layer that defends against the consequences when it does not. Action surface, evaluation, and runtime safety are not three separate problems; they are three views of the same system seen from outside, from above, and from the threat-actor's seat.

STRATUM VI · ACTION SURFACES

Four families of action surface — sandboxed code execution, browser automation, full computer use, and agent-outbound voice — define where agentic systems stop being chatbots and start being labor. The honest read in May 2026: sandboxed code execution is production-ready and commoditizing; browser automation is production-ready in narrow lanes; computer use is still demo-grade despite the marketing; voice agents are production-ready for short scripted calls and collapse on the long tail.

Sandboxed code execution is the most under-discussed procurement requirement in this volume. The category exists because frontier-model code output is non-deterministic and frequently unsafe to run on first-party infrastructure. The vendor set in May 2026 — E2B (open-core, dominant developer mindshare, an estimated $15–25 million ARR doubling year-over-year); Modal (the Modal Labs pivot from serverless GPU to agent workloads, reported north of $50 million in ARR combined); Daytona (newer entrant); Vercel Sandbox (Firecracker microVMs, GA January 2026, tight integration with AI Gateway and Workflow DevKit); Replit Agent sandbox (bundled, prosumer-leaning); and re-purposed Hugging Face Spaces and GitHub Codespaces — converges on the pattern of Firecracker plus Linux plus Python. The non-obvious question is when an agent actually needs a sandbox rather than a direct Python tool call; the honest answer is anytime the code is model-authored and the execution touches anything beyond pure-function math. Sandboxing is a SOC 2 and ISO 27001 control surface, not a feature; Vanta-style frameworks will start mapping a control like "AI-001: AI-generated code must execute in isolated tenancy with default-deny egress" by end of 2026. Vendors shipping "agents that run code" without a named sandbox layer will fail enterprise security review. Commoditization timeline is eighteen-to-twenty-four months before this looks like S3.

Browser automation is distinct from computer use and operates at the DOM level: the agent reads the accessibility tree, dispatches clicks via CDP, parses structured responses, and pixel-perfect rendering does not matter. The vendor map in May 2026 — Playwright with AI patterns as the boring baseline; Browserbase as the fastest-growing managed Chromium at scale at an estimated $50+ million ARR by Q1 2026 with session recording, stealth, and the Stagehand natural-language wrapper; Browserless as the price-led incumbent; Skyvern with a vision-plus-DOM hybrid; Anchor Browser and Hyperbrowser as 2025 MCP-compatible entrants — sits on top of a baseline that frequently outperforms the paid options. A well-prompted Playwright loop with a strong model still wins on cost and latency for sites with stable DOM and no aggressive anti-bot. Reliability inflected in mid-2025; browser automation is production-ready today for sites the agent has seen at training time or has a stable DOM, and still collapses on novel single-page applications with shadow DOM tricks, captcha walls, and SMS-or-2FA auth flows without human in the loop.

Full computer use is loud, hyped, and still demo-grade in May 2026. Anthropic Computer Use shipped October 2024 with OSWorld in the low twenties; Claude for Chrome was a public preview by December 2025 and a wider rollout in April 2026. OpenAI Operator shipped January 2025 and rebranded to ChatGPT Agent after consumer confusion. Google Project Mariner moved to Gemini Enterprise GA in April 2026. Microsoft Copilot Vision and Copilot Studio agents are the Windows-native enterprise default. Amazon acqui-hired Adept in mid-2024 and the ACT-2 lineage now sits inside AWS Q. Multi-On and Twin AI shrank after Operator. Manus from China combines computer use, browser, and voice and gained genuine non-US traction. OSWorld sits at fifty-to-fifty-five percent against a human baseline of seventy-two percent; WebArena sits in the high sixties to low seventies; ScreenSpot-Pro for professional applications sits at forty-five-to-fifty-five percent; AndroidWorld sits near fifty percent. "Is computer use production-ready" admits a clean No for unattended autonomous work. WebArena at seventy percent means roughly one-in-three multi-step tasks fail; OSWorld at fifty percent is a coin flip. For human-in-loop "drive this for me while I watch," it is useful and improving fast. For "run overnight, file the expense reports unsupervised," it is still a 2027 question. Where it does work in May 2026: scripted, narrow, repeated workflows where failure cost is low and a retry is cheap.

Voice agents — the agent makes or receives phone calls, replaces an IVR, performs sales, support, scheduling outbound — are production-ready for short scripted calls and collapse on the long tail. Vapi (Y Combinator W23 origin, $20+ million ARR by Q4 2025), Retell AI, and Bland AI form the trio. LiveKit Agents provides the WebRTC infrastructure layer; ElevenLabs Conversational AI moves up-stack from TTS; PolyAI, Parloa ($66 million Series B 2024), and Regal serve enterprise CX. Foundation pieces include OpenAI Realtime, Cartesia Sonic-2 at approximately forty-millisecond first-byte TTS, Deepgram Nova-3 on accent robustness; telephony substrate splits across Twilio (incumbent), Telnyx (price challenger), and LiveKit Cloud (WebRTC-native, AI-native default). End-to-end voice-to-voice in May 2026 sits at approximately five-hundred-to-eight-hundred milliseconds for the best stacks. Human conversational turn-taking is two-to-three hundred milliseconds; below six hundred the experience feels human, above one second it feels robotic. The field has crossed the natural-cadence line for short turns. Voice collapses on numeric input over phone (credit card digits, account numbers; ASR error rates jump), long proper nouns, heavy accents and code-switching, mid-utterance interruption (most systems still over-talk or freeze for six hundred milliseconds), and calls beyond five minutes where context drift, repetition, and looping emerge. The procurement implication is that voice is where AI-vendor risk surfaces fastest in regulated industries: TCPA, GDPR, state consent laws. The Vapi-Retell-Bland trio is still maturing consent capture, audit recording, and named voice-cloning policy.

A cross-cutting argument frames the layer. The bull case for commoditization: sandboxes converge on Firecracker plus Linux plus Python; browsers converge on managed Chromium plus CDP; voice converges on LiveKit plus OpenAI Realtime plus best-of-breed TTS. In twenty-four months these look like Twilio circa 2018 — boring, multi-vendor, sub-ten-percent gross margin attractive. The counter-argument: the moat is operational tooling around the surface, not the surface itself. Browserbase wins not because Chromium is better but because session recording, replay, anti-bot evasion, and Stagehand DX layer ship together. Vapi wins not on telephony but on a dashboard that lets a non-engineer ship a phone agent in thirty minutes. Both are true. Pure surfaces commoditize inside twenty-four months; the operational layer on top — recording, replay, evaluation, compliance, observability — is where durable value lands.

STRATUM VII · EVALUATION AND OBSERVABILITY

Through 2024, evaluation and observability for language models meant logging the prompt, logging the completion, and scoring for hallucination or toxicity. Agents broke that on four axes simultaneously. One: trajectories, not turns. A single request fans out into twenty-to-eighty tool calls across sub-agents, and final-output evaluation catches roughly thirty percent of real failures — wrong tool selected, right tool called with wrong arguments, infinite loops, a sub-agent that summarized away the load-bearing fact. Trajectory evaluation catches the rest. Two: replay is the debug primitive. "Re-run from step seven with a different model, prompt, or tool stub" became the workflow that Logfire, LangSmith, Braintrust, Langfuse, and Arize Phoenix all converged on by 2025. Three: online evaluation, not just offline. Agents drift inside a session (context degradation past thirty-two thousand tokens of tool traces); the pattern is LLM-as-judge over live traces plus regression sets auto-curated from production failures. Four: cost and latency are first-class. A forty-seven-tool-call trace at Opus pricing is a P&L event; token, dollar, and wall-clock per node are non-negotiable to a modern enterprise buyer.

The vendor map as of May 2026. Tier one with real enterprise traction: LangSmith and LangChain (bundled with LangGraph 1.0 GA in October 2025; Series C of $25 million in February 2024 at approximately $1 billion valuation; ARR an estimated $40–60 million by Q1 2026), tightest LangGraph integration but ecosystem-locked. Braintrust (Series A $36 million from a16z in June 2024; a Series B reportedly closed Q1 2026 around $60 million per an unconfirmed leak), the "evals-as-CI" workflow with publicly named logos including Notion, Airtable, and Brex, the most likely pure-play winner. Langfuse (open-source core under MIT, Y Combinator W23, $4 million seed plus extension from Lightspeed in July 2024), the reference choice for EU and regulated deployments. Arize Phoenix and Arize AX (Phoenix is the Apache 2.0 OSS sibling; Series C of $70 million from Salesforce Ventures in December 2024) for MLOps-mature accounts migrating from SageMaker or Vertex. Galileo (Series B $45 million from Scale Venture in June 2024; Series C reported April 2026 around $60 million), hallucination metrics and agent-trace evaluation with Hubspot and ServiceNow named in Q3 2025. Tier two and credible: Comet Opik (OSS, September 2024); Confident AI / DeepEval (the pytest-style eval-as-code option); HumanLoop (UK, strong in financial services); Patronus AI (Series A $17 million from Notable in March 2024) with Lynx and Glider as open eval models; AgentOps (OSS, CrewAI / AutoGen ecosystem); Helicone (the observability-as-proxy model, losing as runtime-native tracing matures); Promptfoo (acquired by OpenAI in September 2025; OSS continues but strategic neutrality is gone); Inspect AI (UK AISI, OSS, cited explicitly in EU AI Act conformity-assessment drafts from February 2026); and METR as a research organization not a vendor (cite, do not procure).

The benchmark landscape is uneven. TAU-bench and τ²-bench (Sierra, August 2024 and Q1 2025) are the best CX-agent proxy — Sierra-co-authored, read accordingly. SWE-Bench Verified (Princeton plus OpenAI, August 2024) is inflated by harness tricks; SWE-Bench Live is cleaner. GAIA (Meta, November 2023) is saturating at frontier ~70% against a human ~92%. BrowseComp (OpenAI, April 2025) sits at fifty-to-fifty-five percent for Deep Research products. OSWorld (HKU plus CMU, April 2024) and WebArena (CMU, July 2023) read as covered above. SWE-Lancer (OpenAI, February 2025) frames "would I pay for this" honestly — the frontier earns approximately $200K of a $1M envelope, under twenty-five percent. The honest read: no public benchmark predicts enterprise agent ROI; they predict the ceiling of the possible. Every serious enterprise builds internal evaluation sets from production traffic — typically two hundred to one thousand curated trajectories — and the vendors winning are the ones who make that curation cheap. OpenTelemetry GenAI semantic conventions stabilized in January 2026, so trace ingest is no longer a lock-in vector; lock-in is in evaluation logic and judge models (Braintrust YAML, Galileo metric definitions take a quarter to port). The OSS triad of Langfuse plus DeepEval plus Inspect AI survives a vendor switch. Consolidation by 2027 is partial: LangSmith stays because LangGraph stays; Braintrust wins the OpenAI-shop eval-as-CI lane; Langfuse wins EU and self-host; Arize keeps MLOps-mature; two-to-three of the tier-two list (Galileo, HumanLoop, Patronus, Comet, Helicone, AgentOps, Confident AI) get acquired by hyperscalers or runtime vendors by end of 2026; native model-provider tooling eats the bottom.

STRATUM VIII · RUNTIME SAFETY AND GUARDRAILS

Runtime safety — the tool-boundary defenses, distinct from capability-level safety regimes covered in the Meta-strata — addresses three threats in May 2026. Direct prompt injection and jailbreak (user types "ignore your instructions") is largely solved at the frontier model layer with residual low risk. Indirect prompt injection via tool returns — the agent fetches a webpage, email, Jira ticket, Slack message, or PDF containing adversarial instructions — is unsolved. Anthropic's constitutional-classifier research, Simon Willison's continuing reporting, and the EchoLeak and "AgentSmith"-class proofs of concept through 2025–2026 all confirm: any agent that reads attacker-controlled data and has write or action tools is exploitable. Defense is layered (sandbox, action confirmation, output validation) and never complete. Multi-turn and multi-agent injection — compromised sub-agent influencing parent, tool returns shaping later-turn decisions — is emerging and mostly research-stage.

The vendor map splits into pure-play, hyperscaler bundle, and open source. Pure-play: Lakera (Swiss, Series A $20 million from Atomico in October 2023; Series B reported Q4 2025 around $50 million), Guard for real-time PI defense plus Red for red-team-as-a-service, the strongest brand with Atlassian, Dropbox, and Citi publicly named in Q3 2025. Protect AI (Series B $60 million from Evolution Equity in August 2024; acquired by Palo Alto Networks in announcement April 2025, closed Q3 2025 for approximately $700 million) — the first big consolidation signal in the category. Robust Intelligence (acquired by Cisco August 2024, now Cisco AI Defense). HiddenLayer (Series B $50 million from M12 in March 2024), model-supply-chain security earlier in lifecycle than Lakera. Cranium (KPMG spin-out 2023), more GRC than runtime. CalypsoAI (Series B $23 million from Paladin in September 2023), enterprise GenAI gateway with a federal and defense angle. Apex Security (seed from Sequoia in March 2024), an NYC competitor to Lakera, early but credible. Hyperscaler bundles — the real competitor: AWS Bedrock Guardrails with PII redaction, denied topics, and Automated Reasoning for hallucination (GA December 2024); Microsoft Purview AI Hub and Defender for AI (Ignite November 2024 with GA 2025), prompt DLP, AI inventory, posture; Google Vertex AI Safety Filters and Model Armor; Anthropic's constitutional classifiers (baked into Claude, not a sold product). Open source: NVIDIA NeMo Guardrails (Apache 2.0, GTC 2023; 1.0 GA April 2026), Meta Llama Guard 3 and Llama Prompt Guard 2 (open weights), Promptfoo red-team (now under OpenAI), Garak (NVIDIA OSS vulnerability scanner).

The honest assessment on prompt-injection defense is that nobody is solving indirect injection. Vendors quoting "99.X% PI detection" report known-pattern rates against public attack corpora; adaptive-adversary numbers fall to sixty-to-eighty percent (Lakera and Anthropic both publicly admit this). The defense-in-depth stack that does work in production: treat tool returns as untrusted (tag bytes from outside user instructions, degrade or refuse high-privilege actions when planning context is tainted); action confirmation gates for any tool that writes, pays, sends, or escalates, or run them in a reversible sandbox; output validation and schema enforcement (constrain decoding, refuse free-form action when a tool call is expected); privilege separation across sub-agents (a web-reading sub-agent does not carry write tokens — the Claude Agent SDK's sub-agents pattern is the cleanest expression); continuous red-team via Promptfoo, Garak, or Lakera Red on every major prompt or model change. A vendor selling item one only is selling vapor for production touching money; the procurement-grade answer is the whole stack. Runtime safety as a standalone category will be mostly absorbed into hyperscaler bundles within eighteen months — Protect AI to Palo Alto was the bell; AWS and Azure and GCP guardrails ship "good enough" for eighty percent of use cases at zero procurement friction. Lakera survives as the cross-cloud, model-neutral, audit-grade specialist; Apex, Calypso, Cranium likely roll up. The Llama Guard plus NeMo Guardrails plus Promptfoo OSS floor rises and eats the bottom of the paid market. One durable niche remains for a Lakera-shaped vendor that closes the procurement-readiness gap — signed reports, EU AI Act tie-out — and that niche is where Bet #1's Playbook will partner or compete.

Part IV · The Productized Layer

Two sub-strata productize everything below. Vertical agent products are the domain-specific applications where enterprise dollars actually land. End-user surfaces are the form factors through which the buyer, the user, and the regulator meet the agent. The productized layer is where Volume II's Bet #2 (vertical-agent GTM role) and Bet #1 (procurement playbook) most directly apply.

STRATUM IX · VERTICAL AGENT PRODUCTS

The vertical layer is organized by buying motion, not technology. Nine domains define the contour in May 2026.

Coding agents — Claude Code, Cursor (Anysphere, $500-million-plus run-rate, $9.9-billion valuation as of July 2025), Cognition Devin ($80–100M ARR in Q1 2026, having absorbed Codeium's Windsurf residual in a $2.4-billion three-party deal in December 2025), Augment ($40 million ARR, $977-million valuation), Replit Agent ($150-million-plus ARR run-rate), Lovable (Stockholm, fastest EU SaaS curve at $80 million in twelve months, $1.8-billion valuation), Factory, Bolt and v0, Magic.dev, Reflection.AI, GitHub Copilot Workspace, the Windsurf residual — split multi-winner at the prosumer tier and winner-take-most at enterprise, with that winner being Claude Code inside Anthropic's $200-billion parent. The honest read is that Cursor's $500-million run-rate is real but the $9.9-billion valuation prices in a dominance Anthropic is contesting. Outcome-based pricing fails here because outcomes (commits, PRs, merged changes) resist clean attribution. Foundation-lab encroachment is the highest of any vertical: Claude Code, Codex, Gemini Code are the channel.

Customer experience and customer support — Sierra ($175-million-plus ARR at four-times year-over-year by Q1 2026 per Bret Taylor, $10-billion valuation rumored in March 2026 against a confirmed $4.5 billion from August 2025), Decagon ($80-million-plus, $4.5-billion valuation

September 2025), Cresta, Ada, Intercom Fin, Parloa (Berlin, €30+ million ARR), PolyAI, Regal (NYC, $40-million ARR estimate), Forethought, Lorikeet — is the two-winner segment confirmed. Sierra's NYC-plus-SF dual headquarters and Stripe-Ramp-Salesforce alumni pipeline make it the most aligned Bet #2 target. Cresta and Ada are stalling; full-automation has out-competed agent-assist. Pricing inflection is per-resolution standard at one-to-four dollars per resolved ticket — Bret Taylor's outcome thesis is most validated here and likely only here. Encroachment is medium; the moat is multi-system orchestration and outcome SLAs.

Knowledge worker and enterprise search-agents — Glean ($300-million-plus ARR Q1 2026, $7.2-billion valuation September 2025), Hebbia (NYC HQ, $50-million-plus ARR Q1 2026, $700-million valuation July 2024 and overdue for a raise), Sana, Mem (struggling), Notion AI, Coda AI (merged into Grammarly December 2024), Microsoft Copilot 365 agents — is a single dominant winner with niche players. Glean is the category leader and the second-most-aligned Bet #2 target for Alex: NYC growing, sourcing Stripe, Snowflake, Databricks. Hebbia is the best NYC-native pure-play vertical agent for Alex's weekly target list — financial-services-leaning but selling across legal, consulting, and private equity. Pricing is per-seat fifty-to-one-hundred-and-fifty dollars per user per month plus workflow credits; outcome pricing has not landed because knowledge-work outcomes are not measurable. Encroachment is the highest in this volume: ChatGPT Business plus connectors attacks Glean's wedge directly, with governance and cross-system stitching as a twenty-four-month moat.

Legal — Harvey ($100-million-plus ARR Q1 2026, $5-billion valuation February 2026, SF plus NYC office), Hebbia cross-segment, Spellbook, EvenUp ($80 million in PI law), Eve, Robin AI, Ironclad agent products — is winner-take-most at the AmLaw 100 tier with Harvey owning it and selling to Cravath, A&O Shearman, PwC Legal. NYC office real and growing — second-most-aligned Bet #2 target after Sierra. Below AmLaw the market fragments by use case. Pricing is per-seat $250–$500 per lawyer per month; outcome pricing rejected because no firm denominates in "matters won." Encroachment is low because privilege, bar liability, and contract-data RLHF fence the segment. This is the most defensible vertical in the volume and the most overlooked in Alex's network.

Healthcare — Hippocratic AI (NYC HQ plus Palo Alto, $50-million-plus ARR Q1 2026, $2-billion valuation February 2026), Abridge (Pittsburgh plus SF, $120-million-plus ARR Q1 2026, the fastest healthcare-AI ramp on record at $2.75-billion valuation), Ambience, Suki (stalled), Nabla (Paris), DeepScribe (slowed), Iodine Software (mature CDI), Talkdesk Healthcare — is multi-winner with a functional split. Abridge owns hospital ambient scribing; Hippocratic owns voice-agent patient-facing workflows (post-discharge, pre-op intake) with NYC anchor strengthening and Munjal Shah's nine-dollars-per-hour "RN equivalent" pricing line setting the per-call standard. Pricing is per-encounter for Abridge, per-call for Hippocratic, per-seat for Suki. Encroachment is very low because HIPAA, FDA SaMD classification, and EHR tax keep generalists out. The segment is under-discussed in NYC, over-discussed in the Bay — an arbitrage for Alex.

RevOps and sales — Clay (NYC HQ, $80-million-plus ARR Q1 2026, $1.5-billion valuation January 2026), 11x ($20-million flat with churn issues, the cautionary tale), Regie.ai (NYC), Artisan, AiSDR, Nooks, Common Room (Seattle), Default, Lavender, Apollo agent products — is the most over-funded under-performing segment in this volume. 11x's flat ARR is the cautionary tale of synthetic SDRs hitting a quality ceiling. Clay is the exception with NYC HQ, hiring, and a moat in data orchestration and workflow rather than the agent itself. Pricing is per-seat dominant; per-meeting-booked has been tried and rejected on attribution grounds. Taylor's thesis is least validated here because RevOps buyers do not trust outcome attribution from vendors selling the outcome. Encroachment is medium-high. Clay is the only NYC RevOps Bet #2 target.

Finance and financial services — Rogo (NYC HQ, $30-million-plus ARR Q1 2026, $400-million valuation January 2026), Hebbia cross-application, Brex AI, Ramp AI agents (NYC HQ, $13-billion valuation March 2025 with a $20-billion-plus raise rumored Q2 2026), FactSet AI, Bloomberg GPT-derived products — is the most NYC-native segment in the volume and Alex's strongest non-CX bet. Rogo (banker copilot for IB and equity research with ex-Goldman and ex-JPM founders) hires enterprise GTM with banking fluency or not. Hebbia cross-applies. Ramp and Brex are not pure agent companies but their AI-product organizations hire aggressively for senior enterprise GTM and the equity is meaningful. Pricing is per-seat heavy at one-to-five thousand dollars per banker per month at Rogo; per-transaction emerging at Ramp and Brex. Encroachment is low at the regulated tier because FINRA, OCC, and broker-dealer rules fence generalists. Sleeper: Bloomberg may externalize internal agent products as a Terminal add-on in H2 2026.

Creative and media — ElevenLabs ($200-million-plus ARR Q1 2026, $3.3-billion valuation January 2025 with a $5-billion-plus raise rumored), Suno ($60-million-plus ARR with RIAA litigation as a structural risk), Sora-2-derived products embedded in OpenAI, Runway (NYC HQ Chelsea, $100-million-plus ARR Q1 2026), Pika, Higgsfield (fast prosumer), Krea — splits multi-winner with sharp lane separations. ElevenLabs owns voice as the only true platform in segment; Runway owns pro video and is the NYC-creative Bet #2 target. Pricing is per-seat plus usage credits, ElevenLabs API per-character. Outcome pricing absent. Encroachment is medium-high; ElevenLabs is most defensible via voice-cloning data and IVR integrations.

Sovereign and non-US verticals — Mistral Le Chat Enterprise (€600 million Series C at €11 billion in March 2026, the only credible non-US enterprise-agent platform with BNP, Orange, Schneider, and increasingly US F500 EU-subs on data residency), Falcon-derived sovereign in UAE via Humain ties, Sarvam AI and Krutrim in India, Manus AI and Coze on Feishu and the WeChat Mini-Programs ecosystem in China — are non-trivial and accelerating. EU AI Act enforcement in late

2026 reshapes the segment; teeth would make Mistral-equivalents in protected jurisdictions acquisition targets. Bet #1 (procurement playbook) intersects directly.

Cross-cutting calls: multiple-winner at Coding, CX, Healthcare, Creative, Finance; winner-take-most at Knowledge (Glean), Legal AmLaw 100 (Harvey), RevOps (Clay only). Over-funded relative to ARR: 11x, Magic.dev, Mem, Suno, Reflection.AI. Under-funded relative to ARR velocity: Augment, Hebbia, Rogo, Sana. Hiring exactly Alex’s profile right now in NYC: Sierra, Glean, Hippocratic, Hebbia, Harvey, Rogo, Clay, Ramp AI org, Runway, Decagon. The Bret Taylor "outcome-based pricing is how AI eats SaaS" thesis is segment-specific (CX confirmed, healthcare partial, failed elsewhere). Foundation-lab encroachment ranking: Knowledge > Coding > RevOps > Creative > CX > Healthcare > Legal > Finance.

STRATUM X · END-USER SURFACES AND FORM FACTOR

Capability is increasingly form-factor-indifferent; humans are not. Form factor governs four things at once: adoption velocity, buyer persona, threat model, and regulatory surface. As long as humans remain in the decision loop — true across virtually all 2026 enterprise deployments — the surface is the bottleneck. Form factor and model co-evolve: ChatGPT mobile made multimodal vision a product requirement; Claude Code made long-context tool-loop reliability a model-team OKR. Surfaces are inputs to model roadmaps, not outputs. Twelve form factors define the contour.

Chat on web and desktop — ChatGPT at approximately 800 million weekly active users per OpenAI Dev Day October 2025; Claude.ai at an estimated 30 million MAU Q1 2026; Gemini web; Perplexity at approximately 22 million MAU March 2026 — is product-stage and plateauing on novel use cases. Yet a Pew study in January 2026 found forty-one percent of US adults who tried ChatGPT in 2024 churned within six months, citing "did not know what to ask." Chat is product; good prompting is still custom. "Chat is solved" is an SF-bubble artifact.

Chat on mobile and on-device — ChatGPT iOS and Android at approximately 250 million MAU Q4 2025; Claude mobile; Gemini in Android (default Assistant replacement completed March 2026); Apple Intelligence with ChatGPT and Gemini hand-off (iOS 18.4 March 2026); Samsung Galaxy AI — is the user-acquisition wedge of the era. Apple Intelligence ships with OpenAI in slot and added Gemini in March 2026; Anthropic talks were confirmed by Bloomberg in April 2026 but no ship. Claude mobile is feature-thin: no advanced voice, no live video, no Apple Intelligence slot. Real form-factor disadvantage independent of model quality.

CLI and terminal — Claude Code (GA February 2025, approximately $500-million ARR run-rate per Anthropic March 2026); OpenAI Codex CLI (April 2025); Aider (open source, large mindshare); Plandex; Warp Agent Mode; Cursor CLI (April 2026) — is the highest-throughput, lowest-friction surface for technical users. A senior engineer in Claude Code does in ninety minutes what took a day in May 2024. The bottom-up developer buyer plus a $100–$200-per-month tier moved enterprise IT from "skeptical" to "PO written" in one quarter. Product, racing to commodity for easy edits; long-tail multi-repo work still custom.

In-IDE inline — Cursor (Anysphere, approximately $500-million ARR March 2026, $9.9-billion valuation); Windsurf (Cognition acquired Codeium at approximately $3 billion in March 2026 after the OpenAI deal collapsed July 2025); GitHub Copilot (more than $1.5 billion ARR per Microsoft FY25 earnings, approximately 22 million paid seats March 2026); Cline (OSS); Continue; Augment Code — is the dominant form factor for software work. It does not make the user smarter; it removes the IDE-to-chat context-switch tax. Rare buyer mix of developer-led bottom-up and CIO top-down. Threat model largely solved via enterprise tenancy (Copilot Enterprise, Cursor for Teams).

Agentic triggers via webhook, cron, and event — n8n agent nodes (€55 million raise March 2025, more than 300,000 self-hosted instances); Zapier AI Actions (more than 2.2 million paid users, Agents GA April 2025); Make.com AI modules; Vercel Cron plus AI Gateway; Temporal-orchestrated flows; Pipedream — is the "agent runs while you sleep" pattern, no synchronous human approval unless engineered. EU AI Act Article 14 (human oversight) bites here: a cron-triggered Claude auto-replying to support tickets is, under one Article 14 reading, a "high-risk AI system" without compliant oversight. Test cases by Q4 2026. Highest velocity for SMB, highest risk for regulated enterprise.

In-browser via extension, sidebar, and overlay — Claude for Chrome (preview December 2025, wider rollout April 2026, paying-tier only); ChatGPT browser extension; Perplexity Comet (GA October 2025, 5 million downloads by March 2026); Arc Browser AI (now under The Browser Company plus Atlassian as of April 2026); Sigma AI; Brave Leo — partially validates Aravind Srinivas's "browser is the new OS" thesis: the browser is the only universal cross-SaaS surface, and a sidebar agent sees workflow context without forty OAuth integrations. The threat model is severe: extensions can read passwords, session cookies, every form field. Chrome Web Store added an "AI agent" disclosure label in August 2025. Regulated verticals — financial services, healthcare, legal — block by default. Comet's growth is consumer, not enterprise.

Embedded in incumbent SaaS — Microsoft 365 Copilot (more than $5 billion ARR per Microsoft Q3 FY26 commentary, approximately 30 million paid seats); Google Workspace Gemini (bundled into Workspace Business+ from January 2025); Notion AI; Slack AI ($10 per seat add-on, more than 1 million paid seats March 2026); Linear AI; Salesforce Agentforce (more than $300 million ARR per Benioff Q4 FY25, approximately 10,000 customers March 2026); HubSpot Breeze; Zendesk AI Agents — is the CIO buyer, existing-seat renewal. Microsoft will win more agentic enterprise revenue than any pure-play in 2026. Copilot 365 still runs GPT-4/5-class under the hood; the surface owns the workflow, not the model. Claude is not embedded as default in any top-10 enterprise SaaS. No model-quality story closes that distribution gap.

Remote messaging on Slack, Teams, Discord, WhatsApp, SMS, email — Glean (Slack-first, more than $300 million ARR Q1 2026, $7.2 billion valuation); Slack AI; Microsoft Copilot in Teams; Notion AI in Slack; ClickUp Brain; Sierra (channel-agnostic); HoneyHive; WhatsApp-native agents (Meta opened the WhatsApp Business API to agentic vendors January 2026); Ramp's SMS card-control agent (`ramp.com/ai`, February 2026) — is headless, asynchronous, and conversational. A Slack-deployed agent inherits the Slack DLP and audit log, which is exactly why Glean's Slack-first strategy is a procurement wedge: it lands inside a system the CISO already approved. WhatsApp is the single most under-weighted surface in US discourse.

Voice inbound (user-to-agent) — ChatGPT Advanced Voice (September 2024 to standard May 2025); AirPods Pro 2 with Apple Intelligence voice mode (iOS 18.4 March 2026); Siri rebuilt on LLM (delayed, GA promised September 2026 per WWDC 2025); Gemini Live; Vapi for consumer-voice agents; Wispr Flow ($30 million Series A February 2026) — is real for low-tech-fluency users, commute, hands-free, accessibility. Claude has no first-party voice product in May 2026. Wrappers exist; a native Anthropic surface does not. Real gap.

Wearable, AR, and ambient — Meta Ray-Ban (Wayfarer plus Display, Display variant September 2025, more than 2 million units sold Q1 2026 per Meta); Apple Vision Pro 2 (rumored Q4 2026); Humane AI Pin (shipped April 2024, HP acquired assets February 2025, line dead); Rabbit r1 (effectively dead by Q3 2025); Limitless pendant (~50K units estimate); Friend.com (DOA); Plaud Note (more than 1 million units sold per company January 2026); Brilliant Labs Frame — is bifurcating. Smart glasses real, AI-only pendants mostly dead. Plaud is the dark horse: recorder plus summarizer is "ambient" without "always-on listening" creep. Genesis for AR-with-AI; custom for ambient recording.

Computer use as form factor (UX, not action) — Anthropic Computer Use, ChatGPT Agent, Google Mariner — Stratum VI covers reliability; Stratum X is what it feels like to watch. As of May 2026, every shipped surface still requires the user to watch for high-stakes approvals; average task runs three-to-eight minutes; users context-switch and miss approval prompts; "agent in your machine" creep is real and unresolved. No regulated enterprise (financial services, healthcare, government) has greenlit unattended computer use in production. Form-factor problem outranks capability here. Winning UX pattern: agent runs in a named ephemeral VM the user can re-enter and audit, not on the primary machine. ChatGPT Agent's hosted-browser approach is closer to right than Claude for Chrome's local-extension approach for high-trust work.

API and SDK (no UI) — Anthropic API, OpenAI API, Gemini API, Vercel AI Gateway, AWS Bedrock, Cohere, AI21 — has no form factor; another product consumes the model. Two-thirds of OpenAI revenue per Reuters October 2025 and the majority of Anthropic's per The Information March 2026 come through the API. The implication: every first-party surface above is also a competitor to the API tier — every successful first-party UI is one fewer integration partner.

Anthropic's "API-led, Claude.ai as reference UX" posture is more honest about this than OpenAI's UI-first flywheel; OpenAI's Apps SDK (December 2025) is a forced concession to the platform shift.

Non-US weight is the under-priced point. WhatsApp at more than 2.9 billion monthly actives (Meta Q1 2026), India alone more than 500 million, Brazil more than 120 million, plus the WhatsApp Business API opening to agentic vendors in January 2026 means a Mumbai SMB owner's first agent is a WhatsApp agent — not ChatGPT, not Copilot, not Claude. Regional incumbents: Yellow.ai, Haptik (Jio), Gupshup. Chinese surface evolution is parallel-track: Manus, Coze on Feishu (ByteDance), WeChat Mini-Programs ramping agentic capability Q2 2026. The US-coastal sequence "chat web → IDE inline → computer use" is a US-coastal narrative. The actual global path includes WhatsApp-native, Feishu and WeChat-native, and voice-first surfaces in Hindi, Portuguese, and Bahasa where keyboard prompting is the barrier.

Part V · The Meta-Strata

Four meta-strata wrap the agent layer the way they wrap the parent stack: capability-level safety regimes, regulation, economics, and geopolitics. The content is agent-specific rather than the field-wide treatment from Volume I.

META-A · CAPABILITY-LEVEL SAFETY REGIMES

Distinct from Stratum VIII's runtime defenses, capability-level safety is the lab-side regime that determines whether an agent capability tier is released, restricted, or held back. Three frameworks define the field. Anthropic's Responsible Scaling Policy and the AI Safety Level (ASL) framework — ASL-3 invoked for Claude Opus 4 in May 2025 with the November 2024 RSP update binding through deployment — ties capability evaluations to deployment authorizations and creates an institutional commitment that survives ARR pressure. OpenAI's Preparedness Framework, with its tracked categories of cybersecurity, CBRN, persuasion, and model autonomy, contains an adjustment clause that permits OpenAI to relax safeguards "if competitors deploy" — the clause that Volume II flagged as Risk #3 and that remains an open question for 2026. Google's Frontier Safety Framework arrived in May 2024 and continues to mature; DeepMind's capability evaluations focus on autonomous replication, cybersecurity, and persuasion-and-deception. Pre-deployment evaluations are now run, with varying public disclosure, by the UK AI Safety Institute (founded November 2023), the US AI Safety Institute (now AISI, restructured in early 2026 under Trump-era reorganization), and METR (research org, with the time-horizon doubling result as the most-cited agent capability finding). The European AI Office is the EU's structural counterpart and gains regulatory teeth under the AI Act. Capability-level safety is the bottleneck on the highest-autonomy agent products — long-horizon coding agents, computer-use systems, and any system with broad action authority. The two questions the field is betting on: when does ASL-4 capability emerge (Anthropic's description points at

2026–2027), and what happens when one lab's framework triggers a deployment freeze while another's does not.

META-B · REGULATION

The EU AI Act's GPAI obligations under Article 55 took effect August 2, 2025; first conformity-assessment guidance drafts circulated February 2026; first enforcement guidance on Article 14 human oversight published April 2026. The Article 14 lever is the most actionable near-term: unattended agentic systems (cron, webhook, event-driven) in regulated sectors face direct regulatory exposure, and the agentic-trigger form factor (Stratum X) is the lightning rod. California SB 53 (the Transparency in Frontier Artificial Intelligence Act) went live January 1, 2026 with frontier-model disclosure requirements. New York S5641 and adjacent state-level proposals advance through 2026. Federal preemption volatility — the Trump December 2025 executive order, court rulings ongoing — re-prices compliance practice every quarter; the Volume II Risk #4 frame holds. Sectoral overlays bite hardest: FINRA and broker-dealer rules for finance agents; HIPAA and FDA Software-as-a-Medical-Device classification for health agents; TCPA and state consent for voice; Bartz v. Anthropic settlement ($1.5 billion training-data provenance) and NYT v. OpenAI dynamic re-prices vendor terms across the board. The Volume II Crux #4 — EU AI Act teeth or paper tiger? — is still open; the answer-event is the first GPAI fine, expected late 2026. Either way, agent-specific procurement becomes the wedge regardless of which way the crux resolves: with teeth, it becomes a $10-billion-plus advisory category; without teeth, it remains niche but defensible.

META-C · ECONOMICS

Agent economics invert token economics. Per-token pricing fell roughly four-to-ten times for frontier APIs from Q1 2024 through Q1 2026; per-task agent economics did not fall the same way because trajectory length grew. A typical CX deflection trajectory in 2024 consumed two-to-five thousand tokens; the same trajectory in 2026 consumes four-to-twenty-five thousand because the model thinks longer and verifies more. A coding-agent edit consumes thirty-to-two-hundred-fifty thousand. A deep-research trajectory consumes half-a-million to five million. The planner-executor split routes seventy-to-eighty-five percent of those tokens through cheap executors (Sonnet 4.5, GPT-5-mini, Gemini Flash) and reserves the expensive planner (Opus, GPT-5 thinking, Deep Think) for the calls that change the outcome. Vercel AI Gateway, AWS Bedrock, Martian, and OpenRouter expose this routing as a first-class product. Outcome-based pricing — per-resolved-ticket in CX, per-encounter in healthcare scribing, per-call in voice agents — is the new commercial form-factor; per-seat survives where outcomes resist attribution (coding, knowledge work, finance). The Bret Taylor "outcome-based pricing is how AI eats SaaS" framing is confirmed in CX, partial in healthcare, and not yet validated elsewhere. The Volume II Bet #4 — Inference Cost Optimization / FinOps for Tokens — finds its largest cost surface inside agent workloads and remains a twelve-to-eighteen-month window before hyperscaler bundling (Bedrock auto-optimization, Azure routing) absorbs the layer.

META-D · GEOPOLITICS

The agent-specific geopolitical layer differs from the broader-stack treatment in Volume I. Sovereign agents are the new operational form. Mistral Le Chat Enterprise (€600 million Series C at €11 billion in March 2026) is the only credible non-US enterprise-agent platform, selling against US incumbents on EU data residency and AI Act-aligned posture. UAE's G42 and the Humain (KSA) arrangement extend a sovereign AI thesis into the agent layer with Falcon-derived deployments and AMD MI300/350 hardware commits. India's Sarvam ($41 million at $300 million December 2024, raise rumored Q2 2026) and Krutrim (Ola, $50 million at $1 billion January 2024) advance sub-scale agent platforms with political-funding tailwinds. China's Manus AI (Butterfly Effect, viral March 2025, $500 million valuation rumored Q1 2026) and Coze (ByteDance) are non-investable from a US-LP perspective but real. The export-controls dimension extends from chips (the existing regime) to capability: US Commerce authorizations for advanced-chip exports to G42 and Humain in November 2025 set a precedent; restrictions on agent capabilities (full computer use, autonomous decision-making) are a 2026–2027 regulatory front. The jurisdictional siting of agent workloads — where the trajectories run, where the memory persists, where the action logs accrue — becomes a procurement lever for EU and Asian buyers comparable to where data resides for SaaS today. Sovereign-agent counter-positioning is one of the durable strategic positions in the volume.

Closing · How to Read the Agent Stack

Three integrative observations close the descriptive layer and hand off to the Decisions Playbook (Volume III addendum). First, the agent layer is not one stack but two coupled stacks: an inner cognition loop (model, runtime, MCP, memory, planning) that determines what the agent CAN do, and an outer productization loop (action surface, eval, runtime safety, vertical product, end-user surface) that determines whether the agent gets deployed, used, and trusted. The inner stack is dominated by model labs and their adjacent infrastructure; the outer stack is dominated by vertical applications and the operational tooling around them. The inner stack carries higher capability optionality; the outer stack carries higher near-term enterprise revenue. Most of the high-conviction bets in Volume II — and the refreshed bets that Volume III's addendum will articulate — sit on the outer stack for someone with a GTM operator profile.

Second, the binding constraint shifts as you move up the stack. At the inner stack, capability is the binding constraint: error compounding, reasoning regression, context degradation past 32K, sub-99% per-step reliability on computer use. At the outer stack, capability is largely sufficient and the binding constraint is form factor, procurement, and trust. The implication for someone with twelve years of buyer-side procurement scar tissue and growing AI-builder fluency is uncommon-leverage at the seam: capability is enough; deployment is not. The Volume II Bet #1

(Enterprise AI Procurement Operating Standard) sits exactly on this seam; this volume sharpens its agent-specific lens.

Third, the timing window for high-conviction agent-layer flag-planting is tighter than the parent stack's. Anthropic's Claude Agent SDK is approximately fourteen months old at this writing; OpenAI Agents SDK shipped fifteen months ago; Google ADK fourteen months ago; MCP went from Anthropic-only to Linux Foundation governance in fourteen months. The number of named, well-funded computer-use agent enterprise companies in 2024 was effectively zero; in May 2026 it is approximately nine. The cycle time inside the agent layer is materially shorter than the cycle time at the full-stack level, and the windows for planting flags are correspondingly shorter — twelve to eighteen months for most positions, six to twelve for the highest-velocity ones (Bets #2 and #4). The Decisions Playbook layered on top of this report ranks those positions, names their falsifiability, and assigns 6/12/18-month sequencing.

Sources

Primary research, May 2026. Specific citations are inline; sources cluster as follows. Vendor disclosures (Anthropic, OpenAI, Google, Microsoft, Meta, Mistral, Sierra, Decagon, Glean, Harvey, Hippocratic, Cursor, Lovable, Cognition, Augment, ElevenLabs, Runway, Clay, Rogo, Hebbia, Ramp, Mem0, Letta, Zep, Lakera, Protect AI, Braintrust, Langfuse, Arize, Galileo, Browserbase, E2B, Modal, Vapi, LiveKit). Benchmark releases and updates (TAU-bench, SWE-Bench Verified and Live, OSWorld, WebArena, ScreenSpot-Pro, AndroidWorld, GAIA, BrowseComp, SWE-Lancer, METR Task Time-Horizon). Trade press and operator newsletters (The Information, Reuters, Bloomberg, TechCrunch, SemiAnalysis, Latent Space, Stratechery, No Priors, Air Street State of AI). Regulatory and standards artifacts (EU AI Act with Article 14 and Article 55 implementing guidance, CA SB 53, Anthropic Responsible Scaling Policy and ASL framework, OpenAI Preparedness Framework, Google Frontier Safety Framework, UK AISI evaluations, OpenTelemetry GenAI semantic conventions, Linux Foundation Agentic AI Foundation MCP registry statistics). Research papers (ReAct, Plan-and-Solve, Tree-of-Thoughts, Reflexion, MemGPT, GraphRAG, HippoRAG, Process Reward Models / Lightman et al., DeepSeek R1, Promptfoo, Garak). Cross-reference Volume I and Volume II of SUBSTRATE for the upstream stratigraphy and the framework analyses; the OCQ Living Tracker (`OCQ_TRACKER.md`) maintains rolling talent and capital signals.

VISUAL ATLAS

Substrate · Volume I

Plates I–VI · the agent column · index + sub-strata + meta

Section III of VII

THE AGENT COLUMN

A higher-resolution zoom into Stratum XIII of the parent atlas — the agent layer unfolded as ten sub-strata plus four wrapping meta-strata.

Stratum agenticum — denso compositum, ordo descendens, gradibus decem cum quattuor strata circumdantia

X SUB-STRATA · IV META · DESCENDING

X	END-USER · SURFACE form factors	chat · cli · ide · trigger · slack · whatsapp · voice · ar	⑩
IX	VERTICAL · PRODUCTS productized agents	sierra · decagon · glean · harvey · hippocratic · clay · rogo	⑨
VIII	RUNTIME · SAFETY tool-boundary defense	lakera · nemo · llama-guard · bedrock · purview	⑧
VII	EVAL · OBSERVABILITY trajectory analytics	braintrust · langsmith · langfuse · arize · galileo · inspect	⑦
VI	ACTION · SURFACES agent reaches into world	e2b · modal · browserbase · computer-use · vapi	⑥
V	PLANNING · REASONING cognition loop	react · planner-executor · o-series · extended-thinking	⑤
IV	MEMORY · STATE persistence	mem0 · letta · zep · graphrag · lab-native	④
III	TOOL · PROTOCOL interface to world	mcp · function calling · gateways · a2a	③
II	AGENT · RUNTIMES harness	claude agent sdk · openai agents · adk · langgraph · mastra	②
I	FOUNDATION · CAPABILITY the agentic delta	claude · gpt-5 · gemini · deepseek · tau · swe · osworld	①

META · WRAPPING FORCES

A	CAPABILITY · SAFETY	asl · preparedness · frontier · aisi · metr
B	REGULATION	eu ai act · ca sb 53 · sectoral · bartz · nyt
C	ECONOMICS	per-task · outcome pricing · finops
D	GEOPOLITICS	sovereign agents · export controls · jurisdiction

KEY

sub-stratum · component · ¶ meta-stratum · — dependency

CAPABILITY · SUBSTRATE

Strata I–III · the model, the runtime, the protocol — the geological basement on which every agent runs.

Stratum primum, secundum, tertium · capacitas, harnesia, protocollum tooluse

III

TOOL · PROTOCOL

MODEL CONTEXT PROTOCOL · ADOPTION · FORK VECTORS

- **MCP — Anthropic Nov '24** — JSON-RPC; Py + TS SDKs
- **OpenAI adoption — Mar '25** — Agents SDK · Responses API
- **Google adoption — Apr '25** — Gemini · Vertex AI · A2A
- **Microsoft adoption — May '25** — Copilot Studio · SK
- **Hugging Face — Jun '25** — MCP-compatible Spaces
- **Linux Foundation — Dec '25** — Agentic AI Foundation
- **Registry · ~11.4K servers** — Apr '26 LF stats
- **Cloudflare Workers AI MCP** — Dec '24 · gateway
- **Kong MCP Gateway** — GA Jul '25 · F500 logos
- **Pomerium identity proxy** — May '25
- **Function calling (legacy)** — OpenAI Jun '23
- **Structured outputs** — OpenAI Aug '24
- **Stripe MCP** — first-party · canonical
- **Linear · Cloudflare MCP** — first-party tier
- **GitHub MCP** — MSFT canonical
- **A2A protocol — Apr '25** — agent-to-agent (Google)
- **Fork vector — proprietary tool calls** — OAI latency wins
- **Fork vector — quality split** — long tail abandonware

REGISTRY · MAY '26	HYPERSCALERS COMMITTED	GATEWAY CATEGORY
~11.4K	4 / 4	GA

stratum tertium · the joint to the world

MCP · JSON-RPC · GATEWAYS

II

RUNTIMES

AGENT RUNTIMES · HARNESSES · ORCHESTRATORS

- **Claude Agent SDK** — Sep '25 · sub-agents · skills
- **OpenAI Agents SDK** — Mar '25 · handoffs
- **Google ADK** — Apr '25 · Vertex Agent Engine
- **LangGraph 1.0** — GA Oct '25 · production
- **LangSmith** — obs layer · margin lives
- **Mastra** — TypeScript · YC W24 · Vercel
- **Pydantic AI** — typed validation · Logfire
- **CrewAI** — multi-agent · $18M Oct '24
- **Smolagents (HF)** — minimalist · Dec '24
- **AutoGen / AG2** — MSR + community fork
- **DSPy · BAML** — compile-time / IR layer
- **Inspect AI (UK AISI)** — eval framework · OSS
- **Mistral Agents API** — May '25 · EU
- **Manus AI (CN)** — Butterfly Effect · viral Q1 '25
- **Sarvam AI (IN)** — Q1 '26
- **Aleph Alpha** — compliance-first · DE

SHIPPED · LAST 14 MO	LANGGRAPH ARR	RUNTIME GROSS MARGIN
4 SDKs	$40–60M (inf)	THINNEST

stratum secundum · the harness

SDK · LOOP · SUB-AGENTS

I

FOUNDATION · CAPABILITY

AGENTIC CAPABILITY DELTAS · NOT PROVIDER ECONOMICS

- **Claude Opus 4.5** — Nov '25 · 82.8% SWE-V
- **Claude Sonnet 4.5** — interactive executor
- **GPT-5 / 5.1 thinking** — ~80% SWE-V w/ tools
- **ChatGPT Agent** — Operator+DR · OSWorld claim
- **Gemini 2.5 Pro Deep Think** — GAIA ~70 · BrowseComp ~50
- **Project Mariner** — GA Apr '26 · OSWorld ~38%
- **DeepSeek R1 / R2** — open weights · mid-50s SWE
- **Qwen QwQ / Qwen3** — RLVR · open · CN
- **Kimi K1.5 / K2** — long-context CN · Moonshot
- **Meta Llama 4** — Apr '25 · -10-15 pts vs frontier
- **Anthropic sub-agents** — fresh ctx x children · 6mo behind
- **Reasoning regression** — context degr. past 32K tokens
- **Error compounding ceiling** — 0.95^20 ≈ 36%
- **Computer-use ceiling** — OSWorld ~50% vs human 72%
- **Extended thinking 64K** — interactive vs back-office
- **Tool use in thinking** — Claude 4.5 · GPT-5 · Deep Think

SWE-BENCH VERIFIED	OSWORLD (MAY '26)	REASONING-MODEL TIERS
82.8%	~50%	9

stratum primum · the agentic delta

TAU · SWE · OSWORLD · BROWSECOMP

COGNITION · LOOPS

Strata IV–V · the inner cognition of the working agent — what it remembers and how it plans the next step.

Stratum quartum et quintum · memoria et ratiocinatio · circuitus interni

V

PLANNING · REASONING

WHAT SURVIVED · TEST-TIME COMPUTE · ECONOMICS

- ReAct (Yao Oct '22) — still dominant 2026 loop
- Plan-and-Solve (May '23) — prompting · alive
- Tree-of-Thoughts (May '23) — lab curiosity · rare prod
- Reflexion (Mar '23) — verifier sub-agent · alive
- AutoGPT / BabyAGI — dead in prod · idea surv
- Planner-executor split — durable architectural ide
- Meta-planning / hierarchical — emerging 2025-26
- o1 / o3 (Sep–Dec '24) — reasoning model tier opens
- Claude extended thinking — Feb '25 · 64K budgets
- Gemini Deep Think (May '25) — super-linear on hard
- DeepSeek R1 / R1-Zero — Jan '25 · pure RLVR
- Qwen QwQ-32B · Kimi K1.5 — open reasoning
- Tool use IN thinking — outer ReAct collapses
- `reasoning_effort` knob — OpenAI · buyer-controlled
- PRM (Lightman May '23) — in-model · not exposed
- GRPO at trajectory — DeepSeek · default RL

CX TRAJECTORY · TOKENS	CODE AGENT · TOKENS	DEEP RESEARCH · TOKENS
4–25K	30–250K	0.5–5M

stratum quintum · the inner loop

REACT · PLANNER-EXECUTOR · TTC

IV

MEMORY · STATE

STANDALONE OR ABSORBED · COMPLIANCE WEDGE

- Mem0 (YC W24) — $24M A Dec '25 · 40K sta
- Letta (MemGPT) — $10M seed · A rumored '26
- Zep — Graphiti — $15M A Feb '25 · HIPAA
- Cognee — OSS pipeline · seed Dec '24
- Microsoft GraphRAG — Apr '24 paper · Jul '24
- LightRAG (HKU) — OSS '24 · cheaper graph
- HippoRAG · PathRAG · OG-RAG — research-only
- MemoryOS (Tsinghua) — Sep '25 paper · CN
- ChatGPT memory — Apr '24 paid · Sep '24 f
- Claude Projects memory — GA Apr '26
- Gemini personalization — GA Mar '26
- Working memory (LangGraph) — engineering not category
- Procurement — Article 17 — Mem0 forget API Feb '26
- HIPAA story (Zep) — Athelas · F500 health
- Contextual integrity — Nissenbaum · no vendor
- Falsifiability test — 70% pitches = RAG + writes

STANDALONE CEILING (INF)	LAB MEMORY MAU	CRUX #5 (INHERITED)
$200–400M	~800M	OPEN

stratum quartum · persistence

EPISODIC · SEMANTIC · PROCEDURAL

ACTION · DEFENSE

Strata VI–VIII · where the agent reaches into the world, how we observe that it did the right thing, and what defends when it does not.

Stratum sextum, septimum, octavum · operatio, observatio, custodia

VIII

RUNTIME · SAFETY

GUARDRAILS · INJECTION DEFENSE · BUNDLING

- **Lakera (Guard · Red)** — Series A '23 · B Q4 '25
- **Protect AI · to Palo Alto** — ~$700M Q3 '25 · the bell
- **Robust Intelligence · to Cisco** — Aug '24 · AI Defense
- **HiddenLayer** — $50M B Mar '24
- **Apex Security** — Sequoia seed Mar '24
- **CalypsoAI** — $23M B · federal angle
- **Cranium** — KPMG spin · GRC-leaning
- **AWS Bedrock Guardrails** — Apr '24 · Auto Reasoning
- **Purview AI Hub · Defender** — MSFT Ignite Nov '24
- **Vertex AI Model Armor** — Cloud Next '24
- **NeMo Guardrails 1.0** — GA Apr '26 · OSS
- **Llama Guard 3 · Prompt Guard 2** — Meta open weights
- **Promptfoo (now OAI)** — red-team OSS · Sep '25
- **Garak (NVIDIA OSS)** — vulnerability scanner
- **EchoLeak CVE class** — Mar '26 · indirect PI
- **Constitutional classifiers** — Anthropic · baked in

stratum octavum · the tool boundary

PURE-PLAYS ABSORBED	INDIRECT PI	DEFENSE STACK · LAYERS
18 MO	UNSOLVED	5

PI · ACTION GATES · SANDBOXING

VII

EVAL · OBSERVABILITY

AGENT EVAL · OBS · BENCHMARKS

- **Braintrust** — $36M A · B rumored '26
- **LangSmith / LangChain** — $25M C · LangGraph 1.0
- **Langfuse** — OSS · YC W23 · EU choice
- **Arize Phoenix / AX** — $70M C Dec '24
- **Galileo** — $45M B · C reported Apr '26
- **HumanLoop** — UK · FSI
- **Patronus AI · Lynx · Glider** — $17M A · defensible judge
- **Comet Opik · DeepEval** — OSS · pytest-style
- **AgentOps** — CrewAI / AutoGen ecosystem
- **Helicone · proxy** — losing as native matures
- **Promptfoo (acq OAI)** — Sep '25 · neutrality gone
- **Inspect AI (UK AISI)** — EU AI Act conformity
- **METR · time-horizon** — Mar '25 · cite not procure
- **TAU-bench / tau-2** — Sierra-co · CX proxy
- **SWE-Bench Verified · Live** — code agency · ~75–80%
- **OSWorld · WebArena · GAIA** — computer/web/research

stratum septimum · trajectory analytics

OTEL GENAI CONVENTIONS	CONSOLIDATION (2027)	PROCUREMENT GAP
JAN '26 GA	PARTIAL	OPEN

TRAJECTORY · REPLAY · ONLINE

VI

ACTION · SURFACES

SANDBOXED CODE · BROWSER · COMPUTER · VOICE OUTBOUND

- **E2B** — OSS-core · ~$15–25M ARR
- **Modal** — agent-workload pivot · >$50M
- **Daytona** — …er · price-aggressive
- **Vercel Sandbox** — GA Jan '26 · Firecracker
- **Replit Agent sandbox** — bundled · prosumer
- **HF Spaces · Codespaces** — ad-hoc · weaker fit
- **Browserbase + Stagehand** — >$50M ARR Q1 '26
- **Browserless · Skyvern** — incumbent · OSS
- **Playwright + AI** — the boring baseline
- **Anchor · Hyperbrowser** — MCP-compat 2025
- **Anthropic Computer Use** — Oct '24 · Claude for Chrome
- **ChatGPT Agent** — Operator rebrand mid '25
- **Google Project Mariner** — GA Apr '26 · Gemini Ent
- **MSFT Copilot Vision** — Windows-native
- **Vapi · Retell · Bland** — voice trio · $20M+ ARR e…
- **LiveKit Agents** — WebRTC substrate
- **ElevenLabs Conv. AI** — voice up-stack
- **Cartesia Sonic-2** — ~40ms first-byte
- **Deepgram Nova-3** — ASR · accent
- **Twilio · Telnyx · LiveKit Cloud** — telephony

stratum sextum · agent reaches into world

VOICE-TO-VOICE FLOOR	SANDBOX TIMELINE	CU PRODUCTION
500–800ms	18–24 MO	NO

SANDBOX · BROWSER · COMPUTER · VOICE

PRODUCTIZED · LAYER

Strata IX–X · where enterprise revenue lands and where the human meets the agent — the productized application and its surface.

Stratum nonum et decimum · productum agenticum et facies hominis

X

END-USER · SURFACES

TWELVE FORM FACTORS · THE BINDING CONSTRAINT

- Chat — web/desktop
  ChatGPT 800M WAU · plateau
- Chat — mobile/on-device
  Apple Intelligence slot
- CLI / terminal
  Claude Code ~$500M RR
- In-IDE inline
  Cursor · Copilot · Windsurf
- Agentic trigger
  n8n · Zapier · cron · Arc
- In-browser sidebar
  Claude for Chrome · Comet 5M
- Embedded in SaaS
  Copilot 365 ~30M seats
- Slack / Teams · headless
  Glean wedge
- WhatsApp · SMS · email
  Meta API Jan '26 · 2.9B
- Voice — inbound
  Advanced Voice · AirPods
- Wearable · AR · ambient
  Ray-Ban >2M · Plaud >1M
- Computer use (UX)
  watched only · no reg ent
- API · SDK · no-UI
  2/3 OAI rev · majority Anthropic
- WhatsApp Business API
  Jan '26 · GLOBAL MAJORITY
- Coze on Feishu · WeChat
  CN parallel-track
- Manus · Yellow.ai · Haptik
  non-US agent surfaces
- Pew chat-churn 41%
  Jan '26 · 'chat solved'
- Anthropic mobile gap
  no SAI slot · no voice

stratum decimum · form factor

MICROSOFT COPILOT ARR	WHATSAPP MAU	BINDING CONSTRAINT
$5B+	2.9B	FORM

BUYER · THREAT · ADOPTION

IX

VERTICAL · PRODUCTS

ENTERPRISE REVENUE LAYER · BUYING MOTION BY DOMAIN

- Sierra — $175M+ ARR
  $10B (Mar '26) · NYC dual
- Decagon — $80M+
  $4.5B Sep '25 · NYC/SF
- Glean — $300M+
  $7.2B Sep '25 · NYC growing
- Harvey — $100M+
  $5B Feb '26 · NYC office
- Hippocratic — $50M+
  $2B Feb '26 · NYC HQ
- Augment Code — $40M
  $977M Nov '25 · enterprise
- Cursor — $500M RR
  $9.9B Jul '25 · IDE leader
- Cognition (Devin)
  $3B Codeium absorbed Mar '26
- Replit Agent — $150M
  $3B Sep '25 · prosumer
- Lovable — $80M Q1'26
  fastest EU SaaS · Stockholm
- Abridge — $120M+
  $2.75B Jun '25 · scribe
- Hebbia — $50M+
  NYC HQ · ex-2Sigma · raise
- Clay — $80M+
  NYC HQ · $1.5B Jan '26
- Rogo — $30M+
  NYC HQ · $400M Jan '26
- Ramp AI org
  13B·NYC· 20B raise rumor
- Runway — $100M+
  NYC Chelsea · creative
- ElevenLabs — $200M+
  voice platform · $3.3B
- Cresta · Ada · Intercom Fin
  CX maturing/stalling
- Parloa · PolyAI · Regal
  EU/UK + NYC sales voice
- EvenUp · Eve · Spellbook
  legal long tail
- 11x · Artisan · AiSDR
  synthetic SDR · 11x flat
- Mistral Le Chat Ent.
  €600M Mar '26 · EU sovereign

stratum nonum · productized agents

WINNER-TAKE-MOST SEGMENTS	SET #2 NYC TARGETS	OUTCOME PRICING (TAYLOR)
3 / 9	10+	CX ONLY

CX · CODE · LEGAL · HEALTH · FINANCE

META · WRAPPING FORCES

Meta A–D · the four forces that wrap the agent layer: capability-safety regimes, regulation, economics, geopolitics.

Quattuor stratorum meta · circumscribunt agenticam stratam

D
GEOPOLITICS

WHERE AGENTS RUN · EXPORT CONTROLS · SIT-CONTROLS

- **Mistral Le Chat Ent.** — €600M @ €11B Mar '26
- **G42 · Falcon · UAE** — Humain ties · sovereign
- **Humain (KSA)** — MI300/350 commits · $10B
- **Sarvam AI (IN)** — $41M · Q1 '26 raise rumor
- **Krutrim (Ola IN)** — $50M · political tailwind
- **Manus (CN)** — Butterfly · $500M rumor
- **Coze on Feishu** — ByteDance · parallel-track
- **WeChat Mini-Programs** — agentic Q2 '26
- **US Commerce auth — Nov '25** — G42 + Humain chips
- **Agent-capability export rule** — 2026-27 regulatory front
- **EU residency lever** — AI Act + GDPR + Schrems II
- **Trajectory siting** — Where memory + actions reside

meta · sovereign agents

SOVEREIGN · EXPORT · JURISDICTION

C
ECONOMICS

FROM PER-TOKEN TO PER-TASK · OUTCOME PRICING

- **Token price · –4× / –10×** — Q1 '24 to Q1 '26
- **CX trajectory · 4–25K tok** — ~0.02–0.30 / task
- **Code agent · 30–250K tok** — ~0.20–3 / task
- **Deep research · 0.5–5M tok** — ~1–20 / task
- **Planner-executor split** — 70–85% on cheap tier
- **Vercel AI Gateway routing** — first-class · GA '26
- **AWS Bedrock · OpenRouter** — tier routing standard
- **Outcome pricing — CX** — per-resolution · $1–4 (Taylor)
- **Outcome — healthcare** — per-encounter · per-call
- **Per-seat survivors** — $9/hr knowledge · finance
- **FinOps for Tokens (Bet #4)** — 12–18 mo window
- **Hyperscaler bundling** — Bedrock auto-optim threat

meta · the inversion

PER-TASK · OUTCOME · FINOPS

B
REGULATION

EU ARTICLE 14 · CA SB 53 · BARTZ · SECTORAL

- **EU AI Act · Art. 55 GPAI** — enforceable Aug 2 '25
- **EU AI Act · Art. 14 oversight** — draft Apr '26 · agentic
- **EU AI Act · conformity drafts** — Feb '26 · Inspect AI cited
- **CA SB 53 frontier disclosure** — effective Jan 1 '26
- **NY S5641 · state-level** — advancing 2026
- **Trump Dec '25 preemption EO** — courts through '26–'27
- **Bartz v. Anthropic** — $1.5B training-data settle
- **NYT v. OpenAI** — vendor terms repricing
- **FINRA · OCC · BD rules** — finance agent fence
- **HIPAA · FDA SaMD** — health agent fence
- **TCPA · state consent** — voice agent fence
- **EU AI Office** — structural regulator

meta · the gauntlet

EU AI ACT · CA SB 53 · SECTORAL

A
CAPABILITY · SAFETY

LAB-SIDE SAFETY REGIMES · EVAL INSTITUTES

- **Anthropic ASL framework** — ASL-3 Opus 4 May '25
- **Anthropic RSP — Nov '24** — binding policy
- **OpenAI Preparedness Framework** — tracked categories
- **Preparedness adjustment clause** — Risk #3 · open question
- **Google Frontier Safety** — May '24 · maturing
- **DeepMind capability evals** — auto-replication · cyber
- **UK AISI** — Nov '23 · Inspect framework
- **US AISI (post-restruct)** — early '26 · narrower scope
- **METR (research org)** — time-horizon · 7-mo doubling
- **EU AI Office** — regulatory counterpart
- **ASL-4 capability emergence** — 2026-27 question
- **Model cards for agents** — in-flight standardization

meta · the deploy gate

ASL · PREPAREDNESS · FRONTIER

FRAMEWORK ANALYSES + DECISIONS

The Decisions Playbook

Volume III · five frameworks · seven Bets refreshed · five Risks · five Cruxes

Section IV of VII

THE AGENT LAYER

A DECISIONS PLAYBOOK

ADDENDUM TO THE FOUNDATION REFERENCE · VOLUME III

Five frameworks · ten sub-strata · four meta · 2025–2026

Compiled for A. Yedi · Five plates accompany this volume

Preamble

This document is the decisions companion to the Volume III foundation report. Where the report descriptively unfolds the agent layer into ten sub-strata plus four wrapping meta-strata, this addendum layers five framework analyses on top — the OCQ Matrix, Wardley Mapping, Helmer's 7 Powers, Ecosystem Jobs-to-be-Done, and the empirical Talent and Capital Flow tracker — and resolves them into a single playbook: seven refreshed Bets, five Structural Risks, five Cruxes, a 6/12/18-month Action Map, and an Agent-Specific Procurement Rubric that becomes the bridge to Volume II's Bet #1 (Enterprise AI Procurement Operating Standard).

The principle of composition is to refresh the prior seven Bets at agent-layer resolution rather than to add new ones for the sake of count. Where this volume's analysis materially changes conviction, sequencing, or falsifiability of a Bet, the change is named explicitly with a delta annotation. Where it confirms the prior conclusion, conviction is updated but the Bet is left intact. The aim is a working surface a sharp operator can act on inside eighteen months without re-reading the whole atlas.

Volume III's strongest synthetic finding — the single observation that emerges from the five frameworks but did not from the prior single-band treatment of agents — is that the highest-conviction near-term opportunity for a senior enterprise GTM operator in New York is Bet #1, not Bet #2 as the prior addendum implied by ranking order. The 7 Powers screen names procurement Process Power at Meta-B as the only durable position Alex can claim rather than rent; the Wardley map places agent-procurement controls at Genesis as the unclaimed flag; the Ecosystem JTBD shows Job 6 (passing the agent-specific procurement gauntlet) as the most-underserved enterprise job; and the OCQ Matrix surfaces multiple high-claimability opportunities (signed reproducible eval reports, Article 14 oversight, trajectory cost audits, form-factor procurement diagnostics) that all compound into Bet #1 rather than into separate productized bets. The implication is a sequencing change: Bet #1 first to claim a durable position; Bet #2 second to collect equity at a power-holder; Bet #3 third as advisory practice compounding from both. The rest of this volume defends this sequencing layer by layer.

Contents

Part VI · Methodology

Five frameworks were applied to the agent layer as defined in the foundation report — ten sub-strata (foundation models with agentic capability; agent runtimes; tool-use protocol; memory and state; planning and reasoning; action surfaces; evaluation and observability; runtime safety and guardrails; vertical agent products; end-user surfaces) plus four meta-strata (capability-level safety regimes; regulation; economics; geopolitics). The lens definitions inherit from Volume II's Part VI Methodology and are tightened here to agent-specific application.

The lenses, precisely

OPPORTUNITY (agent layer): where in the agent stack is value being created faster than the field's prevailing narrative reflects, AND where can someone with Alex's profile (twelve years enterprise B2B GTM, growing AI-builder fluency, NYC base) claim it inside a 12–18 month window before the layer commoditizes or consolidates? Score on three dimensions, 1–5: Confidence (how robust the evidence), Time-to-Monetize (5 = inside six months, 1 = beyond twenty-four), Claimability (5 = uncommon-leverage fit, 1 = wrong profile).

CHALLENGE (agent layer): what is the binding constraint or latent feedback loop that, if it tightens or fires, materially reprices the layer above? Score on three dimensions, 1–5: Severity (5 = total reprice, 1 = nuisance), Probability (5 = near-certain inside eighteen months, 1 = unlikely), Alex's exposure (5 = directly hits bets he is positioned for, 1 = orthogonal).

OPEN QUESTION (agent layer): what is the agent-specific crux the field is betting on without admitting? Score on three dimensions, 1–5: Decidability (5 = answer-event likely inside six months, 1 = open for years), Asymmetry (5 = answer reprices more than two Bets, 1 = marginal), Bet-size (5 = greater than one-billion-dollar category outcome, 1 = niche).

Discipline applied: if more than two of any group's twelve opportunity scores hit 15/15, the lens has stopped working and is re-tightened. The OCQ outputs in Part VII were vetted against this rule.

Inheritance and delta

Where Volume II covered a layer at the full-stack level — model providers, retrieval, regulation broadly, hyperscaler economics, geopolitics broadly — that work is referenced by name and not redone. The agent-specific Part VII matrix is additive at higher resolution. The seven Bets, five Risks, and five Cruxes from Volume II Part X are inherited as the baseline and explicitly delta'd in Part XI. The Talent and Capital Flow tracker (`OCQ_TRACKER.md`) is refreshed for agent-specific signal and feeds Part XI and Part XII; the agent-specific subset will be maintained as `AI_AGENTS_TRACKER.md` going forward.

Part VII · The OCQ × Agent Sub-Stratum Matrix

Each of the ten sub-strata and four meta-strata is treated below with a compressed framing paragraph, the top three Opportunities (the field-narrative-lag thesis), the top two Challenges (the binding constraint), and the top two Open Questions (the agent-specific crux). Every entry ties to a 2025–2026 datapoint and carries a falsifiability mark for opportunities, a watch-signal for challenges, and an answer-event for open questions. Scoring follows the lens definitions in Part VI.

Stratum I · Foundation models with agentic capability

The agent-relevant frontier is four numbers — TAU-bench, SWE-Bench Verified, OSWorld, GAIA/BrowseComp — and one ceiling: error compounding under context degradation past 32K tokens of tool traces. Capability advances continue but the bottleneck has shifted to per-step reliability for unattended use. The Anthropic sub-agents pattern is ahead by approximately six months and will be table-stakes by Q4 2026. Test-time compute economics bifurcate the market into interactive (Sonnet / Flash / 4o class) and autonomous back-office (Opus / GPT-5 thinking / Deep Think). Reasoning regression past 32K is a real, under-discussed limit.

Opportunities

- **Skills-catalog authorship for the Claude Agent SDK** — the SDK's skills primitive (Sep 2025) is ahead-of-peers by approximately six months and is publicly authorable. A 12-month window exists to publish a small, sharp set of enterprise-GTM skills that get cited in field accounts. Score 4/4/5 (Conf/TTM/Claim). Falsifiability: if OpenAI Agents SDK and Google ADK ship handoff/team primitives at functional parity within 6 months and Anthropic does not extend the catalog model, the authorship window closes.
- **Thinking-budget cost-arbitrage advisory** — OpenAI's `reasoning_effort` knob exposed Feb–Apr 2026 and Anthropic's extended-thinking budgets make per-task economics buyer-controllable. Most enterprise deployments default to high; the typical task does not earn it. Audit-and-route advisory stacks onto Bet #4. Score 4/5/4. Falsifiability: if hyperscaler auto-routing ships by Q4 2026 with default-on behavior, this collapses into platform-bundled feature.

- **Computer-use procurement audit** — the gap between OSWorld 50–55% and the human 72% / per-step ~99% required for unattended autonomous use will not close in 2026. Buyer-side audit framework for "what computer-use tasks pass our reliability bar" is sellable today to F500 risk officers exploring AI BPO. Score 4/4/5. Falsifiability: a frontier model crossing OSWorld 65% by Q3 2026 collapses the watch-only / supervised distinction.

Challenges

- **Foundation labs walking up-stack into vertical applications** — ChatGPT Business connectors, Claude for Work, Gemini Workspace agents directly compress the value of vertical-agent companies (Sierra, Glean, Harvey targets). Severity 5 / Probability 4 / Alex exposure 5 (hits Bet #2 directly). Watch signal: connector marketplace launches and named-customer deflection from vertical incumbents.
- **Reasoning regression past 32K context** — acknowledged by Anthropic Sept 2025 SDK release notes and OpenAI Aug 2025 GPT-5 system card. Until sub-agent or long-context architectures fully solve this, every long-horizon agent is bottlenecked at trajectory length. Severity 4 / Probability 4 / Alex exposure 3. Watch: Claude 5 / GPT-6 / Gemini 3 long-context benchmarks at trajectory level.

Open questions

- **OSWorld 65% by Q3 2026 — yes or no** — the threshold that moves computer use from supervised to deployable for narrow back-office. Decidability 4 / Asymmetry 5 / Bet-size 5. Answer event: a frontier-system score on the public scoreboard.
- **When does ASL-4 capability emerge** — Anthropic's capability tier triggers deployment freezes; if 2026 vs. 2027 differs by twelve months, capability-tier-based competitor advantage windows shift. Decidability 3 / Asymmetry 4 / Bet-size 4. Answer: Anthropic capability disclosure or evaluation publication.

Stratum II · Agent runtimes and harnesses

The runtime layer is the thinnest by gross margin in the agent stack and is splitting along the model spine rather than consolidating. Four serious enterprise contenders (Claude Agent SDK, OpenAI Agents SDK, Google ADK, LangGraph 1.0) plus a credible OSS / TS-native middle (Mastra, Pydantic AI, CrewAI, Smolagents, AutoGen / AG2). The "neutral runtime" promise is real but losing ground in greenfield 2026 builds because vendor SDKs ship faster than neutral wrappers. Value accrues to the model below and the application above.

Opportunities

- **Migration audit between LangChain 2024 and a vendor SDK** — several F500 shops standardized on LangChain in 2024 and now face a stay-or-rip decision. Sellable advisory, six-month window before LangGraph 1.0 stabilizes the migration tooling. Score 4/4/4.

- **Runtime selection rubric for AI procurement** — folds into the Bet #1 Playbook as the "which SDK does your vendor ship on" check; the SDK choice gates capability surface, observability hooks, and lock-in. Score 4/4/5.
- **Sub-agents pattern training** — Anthropic's sub-agents primitive is the cleanest expression of the privilege-separation pattern (web-reading sub-agent without write tokens). Six-month window to be the canonical voice on enterprise application. Score 3/4/5.

Challenges

- **Runtime commoditization** — LangGraph 1.0 GA Oct 2025 made the field competitive; LangSmith carries the margin. Pure runtime opportunities collapse to twelve months. Severity 4 / Probability 5 / Exposure 3.
- **Vendor-SDK lock-in within the model-shop** — Anthropic shops increasingly cannot run on OpenAI; F500 multi-model strategy gets harder. Severity 3 / Probability 4 / Exposure 3.

Open questions

- **Does LangChain Inc. sustain $40–60M+ LangSmith ARR** — Harrison Chase's "LangSmith is the margin layer" public posture (Sep 2025+) is an honest concession against interest; the question is whether observability sustains where the runtime did not. Decidability 4 / Asymmetry 3 / Bet-size 3. Answer: 2026 ARR disclosure or leak.
- **Does Google A2A win cross-vendor commitment** — without major non-Google adoption it dies or stays Genesis. Decidability 3 / Asymmetry 3 / Bet-size 2.

Stratum III · Tool-use protocol (MCP)

The MCP spec is held; the experience is fragmenting silently. Eleven thousand four hundred registered servers as of April 2026 with bimodal quality; four hyperscalers committed; Linux Foundation governance (December 2025) removed the Anthropic kill-switch objection. Fork vectors are real: OpenAI Responses-API proprietary tool calls bypass MCP for latency; A2A overlaps MCP for agent-to-agent; Anthropic `tool_use` blocks remain proprietary native; long-tail server quality is bimodal. The durable MCP power locates not in the protocol itself but in the gateway control plane (Cloudflare Workers AI MCP, Kong MCP Gateway GA July 2025, Pomerium identity-aware proxy).

Opportunities

- **MCP-gateway buyer advisory** — the gateway category is the durable middleware position. Pair-with-Cloudflare/Kong/Pomerium as the AI-procurement specialist on gateway deployment, auth flows, audit, server quality grading. Stacks onto Bet #3 with the reframe that the gateway sub-category is the punctuation, not the spec. Score 4/4/5.
- **Private-registry curation for F500** — enterprises will de facto fork by maintaining private MCP registries (first-party tier from Stripe, Linear, GitHub, Cloudflare, Notion, Atlassian,

Datadog, Snowflake, Databricks, ServiceNow plus internal). Curation rubric is sellable. Score 4/4/4.

- **MCP server constellations for enterprise GTM SaaS** — Salesforce, Outreach, Gong, Highspot, Zoominfo, Linear — many incumbent SaaS systems ship zero or one MCP server. Window for productization is Q2–Q4 2026; defensibility is in matching server design to the Job 4 / Job 1 underserved-outcome list. Score 3/3/4.

Challenges

- **Silent MCP fork via Responses-API proprietary path** — production OpenAI agents bypass MCP for latency; "MCP-compatible but not MCP-native" is the growing pattern. If this becomes dominant by 2027, Bet #3 thesis weakens materially. Severity 4 / Probability 4 / Exposure 4.
- **SaaS incumbents shipping first-party MCP servers** — Snowflake, Databricks, ServiceNow first-party shipping (Q1–Q2 2026) commoditizes the productized-server thesis the moment it lands. Severity 4 / Probability 5 / Exposure 3. Reframe Bet #3 to advisory.

Open questions

- **MCP commons or fork by EOY 2026** — inherited Crux #3. The protocol is held; the experience is the open question. Decidability 4 / Asymmetry 5 / Bet-size 5. Answer: major-vendor proprietary tool-use schema announcements; Linux Foundation governance stability.
- **Does the MCP gateway category cross ten F500 logos by Q3 2026** — if Cloudflare + Kong + Pomerium cumulatively cross ten named F500, the durable middleware thesis is confirmed. Decidability 4 / Asymmetry 4 / Bet-size 4.

Stratum IV · Memory and state

Memory is RAG-with-writes, not a distinct primitive. Approximately seven of ten vendor pitches collapse to "we summarize and write to a vector database." The standalone category ceiling is $200–400 million ARR split across two or three vendors. Lab-native memory (ChatGPT, Claude Projects April 2026, Gemini personalization March 2026) absorbs the consumer and prosumer tier; Zep retains a defensible compliance-sensitive enterprise niche; Mem0 and Letta straddle. Knowledge-graph hybrids (Microsoft GraphRAG, LightRAG) hold partial production traction; HippoRAG, PathRAG, OG-RAG remain research-only.

Opportunities

- **Memory architecture as a service line inside Bet #5** — fold memory selection (lab-native vs. Mem0 vs. Letta vs. Zep vs. pgvector-and-summarize) into the Enterprise RAG Architecture Practice rather than as a separate Bet. Score 4/4/5.

- **Procurement memory checklist (Article 17, Schrems-II, contextual integrity)** — no vendor ships a contextual-integrity primitive. Buyer-side audit checklist is sellable as a Bet #1 module. Score 4/4/5.
- **Healthcare and finance memory deployment with Zep** — Zep is the only credible HIPAA-ready story; vertical advisory pairs cleanly with healthcare-agent vendor selection. Score 3/3/4.

Challenges

- **Lab-native memory absorbs consumer and prosumer** — Claude Projects memory GA April 2026, ChatGPT 800M MAU, Gemini personalization GA March 2026. Standalone category compresses to compliance-niche by 2027. Severity 4 / Probability 5 / Exposure 2.
- **Lock-in risk per vendor** — Claude memory cannot be queried by a GPT agent. Multi-model enterprise stacks need portable memory; nobody ships one cleanly. Severity 3 / Probability 4 / Exposure 2.

Open questions

- **Standalone memory absorbed by 2027** — inherited Crux #5. Directional answer from this volume: absorbed for consumer/prosumer, standalone-niche for compliance enterprise. Decidability 4 / Asymmetry 3 / Bet-size 3. Answer: Mem0 or Letta acquisition vs. independent Series B at materially higher valuation.
- **Does contextual integrity become a procurement standard** — if Helen Nissenbaum's "memory written in B2B sales must not surface in HR" frame enters EU AI Act enforcement guidance, a new category opens. Decidability 2 / Asymmetry 5 / Bet-size 4.

Stratum V · Planning, reasoning, test-time compute

Most production agents in May 2026 are three-to-seven-step ReAct loops with a reasoning model, a verifier, and a planner-executor split routing seventy-to-eighty-five percent of tokens through cheap executors. Tree-of-Thoughts and meta-planning are lab curiosities. The reasoning-model tier (o1/o3/Claude extended thinking/Gemini Deep Think/DeepSeek R1/Qwen QwQ/Kimi K1.5) collapsed two distinctions: the model now does internal multi-step search before emitting; tool use during reasoning collapses the outer ReAct loop. Test-time compute beyond approximately eight thousand reasoning tokens has unattractive economics for most enterprise classes.

Opportunities

- **Trajectory cost-and-latency audit (FinOps for Trajectories)** — the named highest-score opportunity in B2: planner-executor-verifier routing as a sellable F500 audit. 12-month window before Bedrock auto-routing defaults on. Stacks onto Bet #4 with explicit decay. Score 5/5/5.

- **Reasoning-budget procurement rubric** — buyer-side "should this task be on `reasoning_effort=high` or medium" decision tree. Bet #1 module. Score 4/4/5.
- **Open-weight reasoning deployment advisory (DeepSeek / Qwen / Kimi)** — for F500 with political risk acceptance, twenty-to-forty-percent of US frontier inference cost makes on-premises reasoning realistic. Advisory window is 12 months. Score 3/3/3.

Challenges

- **Hyperscaler auto-routing of planner-executor** — Vercel AI Gateway, AWS Bedrock, Martian, OpenRouter all expose tier-routing as config; default-on auto-routing plausible by H2 2027. Severity 4 / Probability 4 / Exposure 5 (hits Bet #4 window directly).
- **Reasoning-model opacity to enterprise buyers** — buyers cannot reason about internal CoT faithfulness or RLVR-pipeline generalization. Treat reasoning marketing as opaque. Severity 3 / Probability 5 / Exposure 2.

Open questions

- **Does test-time compute hit diminishing returns at thirty-two thousand reasoning tokens** — inherited Crux. Anthropic Feb 2026 and OpenAI `reasoning_effort=high` documentation already warn about "overthinking." Decidability 4 / Asymmetry 4 / Bet-size 4.
- **Does open-weight reasoning narrow the gap to single digits by H2 2026** — DeepSeek R2 leak suggestion. If real, on-prem reasoning advisory inflates. Decidability 3 / Asymmetry 3 / Bet-size 3.

Stratum VI · Action surfaces

Sandboxes are production-ready and commoditizing on a twenty-four-month timer. Browser automation is production-ready in narrow lanes; the boring baseline (Playwright plus a good model) still wins for sites with stable DOM and no aggressive anti-bot. Full computer use is still demo-grade in May 2026 — OSWorld 50–55%, WebArena 70%, AndroidWorld 50%, with error compounding making unattended autonomous use a 2027 question. Voice agents are production-ready for short scripted calls and collapse on the long tail. The moat is operational tooling around the surface (recording, replay, evaluation, compliance) not the surface itself.

Opportunities

- **Sandboxing as a procurement control** — SOC 2 / ISO 27001 control like "AI-001: AI-generated code MUST execute in isolated tenancy with default-deny egress." Vanta-shaped mapping plus buyer audit. Bet #1 module. Score 4/4/5.
- **Vertical voice agent packaging (debt collection, intake, dental, field-service)** — substrate is commoditizing (LiveKit + Cartesia + Realtime API); vertical packaging plus compliance plus integrations is the moat. Advisory to vendors or as deployment partner. Score 3/3/3.

- **Computer-use reliability rubric for buyers** — "what tasks meet our internal reliability threshold given OSWorld 50–55%" is a structured-rubric advisory. Bet #1 module. Score 4/4/5.

Challenges

- **Surface commoditization on twenty-four-month timer** — Browserbase $50M ARR is real but the underlying Chromium is open. Margin moves to the operational layer. Severity 4 / Probability 5 / Exposure 3.
- **Voice regulatory enforcement (TCPA, GDPR, state consent)** — Q1 2026 enforcement uptick already forced Vapi / Retell / Bland to ship consent capture. Cost-of-compliance is rising; small voice-agent vendors squeeze. Severity 3 / Probability 5 / Exposure 2.

Open questions

- **OSWorld 65% on a frontier system by Q3 2026** — shared with Stratum I; the binding metric for computer-use deployability. Decidability 4 / Asymmetry 5 / Bet-size 5.
- **Voice-to-voice floor crosses three-hundred-millisecond turn-taking** — the natural-cadence line. Current floor approximately 500–800 ms. If sub-300 by H2 2026, voice consolidates faster. Decidability 4 / Asymmetry 3 / Bet-size 3.

Stratum VII · Evaluation and observability

Agents broke LLM evaluation on four axes: trajectories not turns, replay as debug primitive, online evaluation not just offline, cost and latency as first-class signals. Tier-1 vendors with enterprise traction are LangSmith, Braintrust, Langfuse, Arize, Galileo. OpenTelemetry GenAI semantic conventions stabilized in January 2026; trace ingest is no longer a lock-in vector. Lock-in is in evaluation logic and judge models (regression sets as Braintrust YAML or Galileo metric definitions take a quarter to port). Partial consolidation by 2027 with two-to-three pure-plays acquired by hyperscalers or runtime vendors by end of 2026.

Opportunities

- **Signed reproducible eval reports — the load-bearing Bet #1 wedge** — no vendor ships turnkey signed reports tied to model pin, dataset hash, harness version that survive a regulator subpoena. Inspect AI closest but not turnkey. The procurement-readiness gap. Score 5/4/5.
- **Internal-eval-set curation as a service** — every serious enterprise builds 200–1,000-trajectory regression sets from production traffic. Curation methodology as advisory plus tooling pair with Braintrust / Langfuse. Score 4/3/4.
- **Multi-party agent audit (vendor + customer + auditor on one trace with redaction)** — Genesis stage; no vendor ships. Long-cycle build; advisory-first beachhead. Score 3/2/4.

Challenges

- **Native model-provider eval eats the bottom** — Anthropic Claude Agent SDK eval templates (March 2026), OpenAI Traces, Anthropic eval dashboards. Pure-play eval pressed from above and below. Severity 3 / Probability 5 / Exposure 2.
- **Hyperscaler bundling of observability** — AWS Bedrock, Azure, GCP guardrails ship "good enough" at zero procurement friction. Severity 4 / Probability 5 / Exposure 3.

Open questions

- **Does Braintrust win as eval consolidator by EOY 2026** — rumored Series B $60M Q1 2026 leak. If real, M&A path consolidates the middle. Decidability 4 / Asymmetry 3 / Bet-size 3.
- **Does Inspect AI become the EU AI Act conformity standard** — cited in February 2026 conformity drafts. If formally adopted, audit-grade vendors pivot to Inspect-compatible. Decidability 4 / Asymmetry 4 / Bet-size 4.

Stratum VIII · Runtime safety and guardrails

Indirect prompt injection via tool returns remains unsolved. Vendors quoting 99.X% detection report against known-pattern corpora; adaptive-adversary numbers fall to 60–80%. Defense-in-depth is the only honest answer: untrusted tool returns, action confirmation, output validation, sub-agent privilege separation, continuous red-team. Protect AI to Palo Alto for $700M in Q3 2025 was the bell. Standalone runtime safety mostly absorbed into hyperscaler bundles by 2027 except a Lakera-shaped cross-cloud, model-neutral, audit-grade niche.

Opportunities

- **Adaptive-adversary red-team audit as Bet #1 module** — buyer-side framework for testing agent vendors against attacker-aware injection. Promptfoo, Garak, Lakera Red as substrate; advisory wraps. Score 4/3/4.
- **Tool-boundary policy template** — "which tools the agent can invoke unsupervised, which require approval, which are blocked when context is tainted." No vendor ships a turnkey template. Bet #1 module. Score 4/3/5.
- **OSS guardrail orchestration consulting** — NeMo Guardrails 1.0 GA April 2026 plus Llama Guard 3 plus Prompt Guard 2. Open-source floor risen; F500 deployment advisory. Score 3/3/3.

Challenges

- **Hyperscaler bundles absorb the middle** — AWS Bedrock Guardrails, Purview AI Hub, Vertex Model Armor cover 80% at zero friction. Pure-plays roll up. Severity 4 / Probability 5 / Exposure 2.

- **Indirect prompt injection cannot be solved by any vendor today** — the EchoLeak / AgentSmith class will recur. Defensive stack always layered, never complete. Severity 4 / Probability 5 / Exposure 4 (hits any production-touching-money deployment).

Open questions

- **Does Lakera survive as last independent at scale** — or get acquired (Cisco, Palo Alto, hyperscaler) by EOY 2026. Decidability 4 / Asymmetry 2 / Bet-size 2.
- **Does adaptive-adversary red-team enter EU AI Act conformity assessment** — if so, the procurement-readiness gap is closed by regulation, not the market. Decidability 3 / Asymmetry 4 / Bet-size 4.

Stratum IX · Vertical agent products

Nine domains. Multiple-winner at Coding, CX, Healthcare, Creative, Finance. Winner-take-most at Knowledge (Glean), Legal AmLaw100 (Harvey), RevOps (Clay only). The Bret Taylor outcome-pricing thesis is segment-specific (CX confirmed; healthcare partial; failed elsewhere). Foundation-lab encroachment ranking is Knowledge > Coding > RevOps > Creative > CX > Healthcare > Legal > Finance. The NYC Bet #2 target list refreshes to ten companies: Sierra, Decagon, Glean, Hippocratic, Hebbia, Harvey, Rogo, Clay, Ramp AI org, Runway — plus Mistral Le Chat Enterprise for the EU sovereign play.

Opportunities

- **Sierra Field-CTO / Head-of-Enterprise-East seat** — NYC dual-HQ, $175M+ ARR at 4× YoY, Bret Taylor monthly NYC cadence, Stripe/Ramp/Salesforce alumni pipeline at peak migration. Highest-aligned Bet #2 target. Score 5/5/5.
- **Hebbia or Rogo NYC-native vertical-agent role** — under-funded relative to ARR velocity (both overdue for next round); NYC HQ; financial-services-adjacent; consulting alumni pipeline less competed than Stripe/Ramp. Hebbia's VP Revenue NYC hire March 2026 is the signal. Score 4/4/5.
- **Pricing-model advisory for vertical agent vendors** — per-seat vs. per-resolution vs. per-encounter is segment-specific. Two-year operational scar tissue compresses to a usable advisory framework. Score 3/3/4.

Challenges

- **Foundation-lab encroachment, Knowledge segment first** — ChatGPT Business connectors hit Glean directly; Knowledge is highest-encroachment. Severity 4 / Probability 4 / Exposure 4.
- **Synthetic SDR quality ceiling (11x cautionary tale)** — RevOps over-funded and under-performing; only Clay survives. Severity 3 / Probability 5 / Exposure 2.

Open questions

- **Anthropic ARR — $24B or $30B** — inherited Crux #1. Single load-bearing variable for Bet #2 valuation and timing. Decidability 5 (Q3 2026 disclosure or leak) / Asymmetry 5 / Bet-size 5.
- **Does outcome pricing extend beyond CX into knowledge or finance** — Bret Taylor thesis is segment-specific so far. Decidability 3 / Asymmetry 4 / Bet-size 3.

Stratum X · End-user surfaces and form factor

Form factor is the binding constraint on enterprise agentic adoption in May 2026, not capability. Twelve form factors: chat web, chat mobile, CLI, IDE inline, agentic triggers, browser sidebar, embedded SaaS, remote messaging, voice inbound, wearable/AR, computer-use UX, API. Microsoft 365 Copilot ($5B+ ARR, ~30M paid seats) is the largest single agentic-revenue position in 2026. WhatsApp Business API opening to agentic vendors in January 2026 means the global majority's first AI surface is a WhatsApp agent. The US-coastal sequence "chat web → IDE inline → computer use" is a US-coastal narrative.

Opportunities

- **Form-factor procurement audit** — buyer-side framework for "which form factor passes our CISO" (Slack inherits DLP; extensions block by default; computer use no-go in regulated). Single highest-scoring opportunity in B4. Score 5/4/5.
- **Agentic-trigger compliance advisory under Article 14** — EU AI Act Article 14 enforcement guidance draft April 2026 directly implicates cron/webhook unattended agents in regulated sectors. Bet #1 module with named regulatory anchor. Score 5/4/4.
- **WhatsApp-native enterprise agent advisory** — US F500 with India/LatAm/Africa operations need consent-compliant, audit-capable WhatsApp deployment. Yellow.ai / Haptik / Gupshup ecosystem advisory. Score 3/3/3.

Challenges

- **Microsoft Copilot 365 distribution moat** — $5B+ ARR, existing-seat renewal; pure-plays trail by an order of magnitude. Severity 4 / Probability 5 / Exposure 2.
- **Anthropic first-party surface deficit** — no Apple Intelligence slot, no native voice product, Claude mobile feature-thin, no top-10 enterprise SaaS embedding. Structural shape, not temporary. Severity 3 / Probability 5 / Exposure 3 (if Alex bets Bet #2 inside Anthropic-channel verticals).

Open questions

- **Does Apple Intelligence ship Claude integration by EOY 2026** — Bloomberg April 2026 confirmed talks; no ship. Decidability 4 / Asymmetry 4 / Bet-size 4.

- **Does unattended-agent reliability cross 99% per step by H2 2027** — if so, form factor stops being binding and capability returns as the constraint. Decidability 2 / Asymmetry 5 / Bet-size 5.

Meta-A · Capability-level safety regimes

Anthropic ASL framework (ASL-3 invoked for Claude Opus 4 May 2025), OpenAI Preparedness Framework with adjustment clause, Google Frontier Safety Framework, UK AISI Inspect framework, US AISI restructured 2026, METR time-horizon doubling. The ASL framework is Anthropic's most defensible single-vendor brand position but is not baked into EU AI Code of Practice, NIST AI RMF GenAI Profile, or ISO 42001. UK AISI Inspect cited explicitly in EU AI Act conformity-assessment drafts February 2026 — the quiet vendor-neutral standard.

Opportunities

- **Capability-safety translation for enterprise procurement** — "what does ASL-3 mean for our deployment review" / "what triggers an OpenAI Preparedness adjustment clause" translated for F500 risk officers. Bet #1 module. Score 4/3/4.
- **Inspect AI deployment advisory** — UK AISI framework as the vendor-neutral choice cited in EU conformity drafts. Position pre-formal-adoption. Score 3/3/3.
- **METR time-horizon framework for capability briefing** — most-cited single chart in the field. Bet #6 newsletter target as a translation lever. Score 3/3/3.

Challenges

- **OpenAI Preparedness adjustment-clause activation** — inherited Risk #3 from Volume II. If invoked, voluntary safety regimes destabilize. Severity 4 / Probability 3 / Exposure 3.
- **Lab-internal safety as competitive disadvantage** — Anthropic's deployment discipline could lose share to less-restrictive competitors. Severity 3 / Probability 3 / Exposure 3.

Open questions

- **When does ASL-4 capability emerge** — 2026 vs 2027 differs by twelve months of competitor window. Decidability 3 / Asymmetry 4 / Bet-size 4.
- **Does any major lab invoke the Preparedness adjustment clause publicly** — if yes, the voluntary regime story changes. Decidability 3 / Asymmetry 4 / Bet-size 4.

Meta-B · Regulation

EU AI Act Article 55 GPAI obligations enforceable August 2025; conformity-assessment guidance drafts February 2026; Article 14 human-oversight enforcement guidance draft April 2026. CA SB 53 effective January 2026. Federal preemption volatility from the Trump December 2025 executive order resolves through 2026–2027. Sectoral overlays bite hardest at FINRA / broker-dealer (finance), HIPAA / FDA SaMD (healthcare), TCPA / state consent (voice). Bartz v. Anthropic settlement at $1.5 billion and NYT v. OpenAI dynamic re-price vendor terms.

Opportunities

- **EU AI Act Article 14 human-oversight playbook** — April 2026 draft directly implicates agentic-trigger form factor. Buyer-side playbook the highest-actionability regulatory lever in the volume. Score 5/4/4.
- **Sectoral overlay advisory (finance, health, voice)** — FINRA, HIPAA, TCPA agent-specific applications. Pairs with vertical agent vendor selection. Score 4/4/4.
- **Bet #1 expansion into regulated geography** — EU residency wedge for F500 EU-subsidiaries; Mistral-equivalent pairing. Score 4/4/4.

Challenges

- **EU AI Act paper-tiger outcome** — inherited Crux #4. If first enforcement late 2026 stays small and procedural, Bet #1 TAM stays niche. Severity 4 / Probability 3 / Exposure 5.
- **US federal preemption volatility** — Trump EO court fights resolve through 2026–2027; CA SB 53 lawsuits active. Severity 3 / Probability 4 / Exposure 4.

Open questions

- **Does EU place autonomous agents in Annex III high-risk** — named highest-asymmetry open question in B3. If yes, Bet #1 TAM 3×. Decidability 3 / Asymmetry 5 / Bet-size 5.
- **When does first GPAI fine land** — Crux #4 answer event. Decidability 4 / Asymmetry 5 / Bet-size 5.

Meta-C · Economics

Per-token pricing fell roughly four-to-ten times for frontier APIs from Q1 2024 to Q1 2026; per-task agent economics did not fall the same way because trajectory length grew. CX trajectory four-to-twenty-five-thousand tokens at a few cents; coding agent thirty-to-two-hundred-fifty-thousand at twenty cents to three dollars; deep research half-a-million to five million at one to twenty dollars. Planner-executor split is the durable architectural idea; per-resolution pricing dominates CX; per-encounter dominates healthcare; per-seat survives where outcomes resist attribution.

Opportunities

- **FinOps for Trajectories — distinct from FinOps for Tokens** — per-task planner-executor routing audit. 12-month window before hyperscaler auto-routing defaults. Score 5/5/5 (named max in B2 and B4).
- **Outcome-pricing rubric for vertical-agent buyers** — per-resolution validation, per-encounter rate-setting, per-FTE-equivalent benchmarking. Bet #1 module. Score 4/4/4.

- **Cross-cloud trajectory cost transparency** — Vercel AI Gateway, AWS Bedrock, OpenRouter trajectory-level cost reporting. Procurement input. Score 3/3/3.

Challenges

- **Token-price decline outpaces optimization** — named max in B4 challenges. If frontier inference compresses another four-to-ten times in 2026, the "optimization" thesis goes below the line of caring for many enterprise workloads. Severity 4 / Probability 4 / Exposure 5.
- **Hyperscaler bundling of routing** — Bedrock auto-optim, Vertex routing, Azure routing default-on plausible H2 2027. Severity 4 / Probability 4 / Exposure 5.

Open questions

- **Inference compute — 10x growth or flat** — inherited Crux #2. Determines whether Bet #4 / Meta-C window stays open or closes faster. Decidability 4 (Q4 2026 hyperscaler earnings) / Asymmetry 5 / Bet-size 5.
- **Does outcome-based pricing become procurement standard outside CX** — Bret Taylor thesis. If yes, vertical-agent ACVs reshape. Decidability 3 / Asymmetry 4 / Bet-size 4.

Meta-D · Geopolitics

Sovereign agents are the new operational form. Mistral Le Chat Enterprise (€600M Series C at €11B March 2026) is the only credible non-US enterprise-agent platform. UAE G42 plus Humain (KSA) extend a sovereign AI thesis into the agent layer with Falcon-derived deployments. India Sarvam and Krutrim politically funded sub-scale. China Manus (Butterfly Effect, $500M rumored valuation Q1 2026) and Coze (ByteDance) non-investable from US-LP perspective but real. US Commerce November 2025 advanced-chip-export authorizations to G42 and Humain set precedent; agent-capability export restrictions are a 2026–2027 regulatory front.

Opportunities

- **EU residency wedge for F500 EU-subsidiaries** — Mistral-equivalent partner play for US F500 with German, French, Nordic operations on AI Act-aligned posture. Bet #1 expansion. Score 4/3/4.
- **Sovereign-agent procurement framework** — "where does the trajectory run, where does the memory persist, where do action logs accrue" as the new procurement axis comparable to data residency for SaaS. Score 4/3/4.
- **Non-US capability-safety translation** — UK AISI Inspect, Singapore AISI, EU AI Office as the non-US capability-safety counterparts. Bet #1 international module. Score 3/3/3.

Challenges

- **US-LP non-investability of Chinese verticals** — Manus, Coze, WeChat agent platforms are real and growing but cannot enter Alex's capital path. Severity 2 / Probability 5 / Exposure 1.
- **Export-control regime expansion to agent capability** — full computer use, autonomous decision-making could enter export controls 2026–2027. Severity 3 / Probability 3 / Exposure 2.

Open questions

- **Does Mistral cross $500M ARR by Q3 2027** — validates the sovereign enterprise-agent thesis. Decidability 4 / Asymmetry 3 / Bet-size 3.
- **Does any major non-US sovereign cap-safety institute become a F500 procurement reference** — if UK AISI Inspect is adopted by F500 procurement, the international reference architecture shifts. Decidability 3 / Asymmetry 4 / Bet-size 4.

Part VIII · Wardley Mapping the Agent Stack

A Wardley map plots components by user-visibility (vertical axis) and evolution stage (horizontal axis: Genesis → Custom-built → Product/Rental → Commodity/Utility). The discipline of the lens is to read the slope, not the snapshot: where each component is going next, not just where it sits. The Volume II Wardley analysis covered the full AI stack; this part refreshes placement at agent-stratum resolution and identifies the 2026–2027 punctuated equilibria specific to the agent layer.

What Wardley says about agents now

Most of the agent stack is not where its press releases place it. The narrative reads "MCP won, computer use is here, agents are deploying." The map shows MCP is a stable spec with a fragmenting experience, computer use is Genesis-bordering-Custom for unattended autonomous work, and most "production agents" are three-to-seven-step ReAct loops with a verifier. Evolution is bifurcated: infrastructure beneath the runtime (sandboxes, browsers, gateways, TTS, telephony) is on a twenty-four-month commodity trajectory; the operational layer above (evaluation, observability, guardrails, procurement controls, vertical workflow integration) is Custom-to-early-Product and is where durable margin lives. The field is not consolidating; it is stratifying. Surfaces below the runtime commoditize fast; vertical and compliance layers harden into moats. That is the strategic terrain.

Anchored user needs (top of map)

Four user-need anchors define the dependency chains for the agent map. The first is completing a multi-step research-and-reporting task autonomously — the GAIA / BrowseComp / Deep Research use case (Anthropic Research, OpenAI Deep Research, Gemini Deep Research, Manus, Genspark). The second is operating a browser or computer to complete a back-office workflow — claims processing, expense, vendor onboarding; the CFO-credible "AI BPO" pitch. The third is handling a customer query end-to-end with voice, tools, and escalation — the Sierra / Decagon / Hippocratic / Vapi pattern and the only domain where outcome pricing has landed. The fourth is writing, testing, and merging a production code change — Claude Code / Cursor / Codex / Devin / Augment; the only category where AI labor is priced like FTE displacement at scale.

Layer-by-layer placement

At the end-user surface layer (Stratum X), chat web and desktop sit at Product racing toward Commodity for casual use and back toward Custom for high-stakes; chat mobile is Product trending Commodity-bundled-with-OS; CLI surfaces are Product with slow commoditization; IDE inline is Product stratifying with Cursor and Copilot most durable; browser sidebar is Custom-to-Product for consumer but stalled at Custom for regulated; embedded SaaS surfaces sit at Product trending Commodity-bundled; computer-use UX is Genesis-to-Custom and will not cross to Product without OSWorld at sixty-five percent; voice inbound is Custom-to-Product crossing the natural-cadence line for short calls; wearable and AR is Genesis with smart glasses progressing and pendants dead; API / SDK is Product trending Commodity.

At the vertical-product layer (Stratum IX), customer experience products sit at Custom-to-Product with a confirmed two-winner shape (Sierra, Decagon); knowledge search-agents are Custom-to-Product trending Product with ChatGPT Business encroachment; legal stays Custom-to-Product with the lowest encroachment risk; healthcare sits at Custom with HIPAA / FDA SaMD fencing; RevOps is Custom-struggling with only Clay durable; coding is Product multi-winner at prosumer and winner-take-most at enterprise; finance is Custom with NYC-anchored players (Rogo, Hebbia, Ramp AI) holding; creative is Product (ElevenLabs, Runway, Suno with RIAA risk); sovereign vertical products (Mistral, Sarvam, Manus) sit at Custom with EU AI Act enforcement as the catalyst that could move Mistral-equivalents to Product inside Europe.

At the action-surface layer (Stratum VI), sandboxes are Product converging to Commodity within twenty-four months (Firecracker plus Linux plus Python); browser automation is Product with the operational layer on top retaining durability (Browserbase plus Stagehand plus session-recording); voice substrate is Product trending Commodity (LiveKit + Cartesia + Realtime API + Twilio/Telnyx); voice orchestration is Custom-to-Product consolidating to one or two winners with compliance as moat. At the runtime layer (Stratum II), Claude Agent SDK and OpenAI Agents SDK sit at Custom-to-Product racing to Product; Google ADK is Custom and thin outside GCP; LangGraph 1.0 is Product with margin moving to LangSmith; Mastra, Pydantic AI, CrewAI are Custom-niche-Product. At the protocol layer (Stratum III), the MCP spec is Product (held); MCP gateways are Custom-to-Product crossing in H2 2026; first-party MCP servers are Product with the long tail abandonware; A2A is Genesis.

At the memory layer (Stratum IV), lab-native memory is Custom-to-Product with Claude Projects April 2026 GA, absorbing consumer and prosumer; standalone memory (Mem0, Letta, Zep) is Genesis-to-Custom with the compliance wedge as the only path forward; GraphRAG and LightRAG sit at Custom with selective production traction. At the planning layer (Stratum V), reasoning models are Product with `reasoning_effort` as buyer-controllable; planner-executor split is Product (gateway config flag); RLVR / GRPO trajectory fine-tunes are Genesis-to-Custom open to vertical-agent vendors who own their verifier data. At the eval/obs/safety layer (Strata VII–VIII), OpenTelemetry GenAI is Product (commodity ingest); eval platforms (Braintrust, LangSmith, Galileo) are Custom-to-Product with eval logic as lock-in and partial consolidation by 2027; Inspect AI is Custom and audit-grade; runtime guardrails (Lakera, NeMo) are Custom-to-Product mostly absorbed by 2027. At the procurement-grade controls "layer" — signed eval reports, multi-party audit, EU Act tie-out — components sit at Genesis. No vendor ships turnkey. This is the unclaimed flag.

The five-to-seven punctuated equilibria for 2026–2027

1. MCP gateways — Custom to Product, H2 2026

Kong GA July 2025; Cloudflare expansion; Pomerium identity-aware proxy. The enterprise control plane (authentication, audit, rate-limiting, secret-injection) hardens. Reprices custom MCP builds; opens adjacent-possible for MCP-native iPaaS, policy-firewall products, F500-private registries. This sharpens Bet #3: the gateway sub-category, not the spec itself, is the punctuation.

2. Computer use — Genesis to Custom, mid-to-late 2026

OSWorld 50–55% crosses approximately sixty-five percent on a frontier system; end-to-end reliability sits at roughly thirty percent at fifty steps. The stage jump is real but it is piloted-in-narrow-lanes, not deployable for unattended back-office. Reprices RPA incumbents (UiPath, Automation Anywhere); opens adjacent-possible for AI-augmented BPO at a fifth the cost for reversible, auditable click sequences. Unattended autonomous computer use stays a 2027–2028 question.

3. Voice telephony substrate — Product to Commodity, mid-to-late 2026

LiveKit + Cartesia + OpenAI Realtime + Twilio/Telnyx is a stack default with sub-six-hundred-millisecond voice-to-voice for short calls. Substrate commoditizes; voice orchestration (Vapi, Retell, Bland) consolidates to one or two winners with compliance as moat. Reprices Genesys, NICE, Five9 incumbents; opens vertical voice (healthcare intake, debt collection, dental, field service).

4. Standalone memory — forced binary, resolves H2 2026

Either Mem0, Letta, and Zep harden the compliance wedge — Zep most likely, approximately $50 million ARR by 2027 — or absorbed: lab-native at the consumer end, runtime primitives at the developer end. The directional answer for this volume: absorbed for consumer and prosumer, niche-standalone for compliance enterprise. Memory's access pattern is RAG-with-writes, not a distinct primitive. Refreshes Crux #5; rolls memory architecture into Bet #5 as a service line.

5. Procurement-grade agent controls — Genesis to Custom, Q4 2026

No vendor today ships signed eval reports plus multi-party audit plus EU AI Act conformity plus action-rollback plus adaptive-adversary red-team as a turnkey bundle. EU AI Act GPAI enforcement starts August 2026; the first conformity-assessment draft (February 2026) names Inspect AI explicitly. First Custom solutions emerge from Big-4 consulting plus a Lakera-shaped partner. This is the unclaimed flag for Bet #1. The Wardley-defensible position is not "Drata for AI" (Settle-quadrant SaaS, harder cold-start) but the buyer-side Playbook that names the gap between what vendors ship and what regulators will require. Plant the flag with the open Playbook; SaaS is a downstream option, not the first move.

6. Eval / observability consolidation — Product, Q4 2026 to H1 2027

OpenTelemetry stabilization killed ingest lock-in; eval logic remains. The category ceiling is approximately $300 million ARR. LangSmith stays because LangGraph stays; Braintrust is the most likely consolidator; Langfuse wins EU and self-host; two-to-three of {Galileo, Patronus, HumanLoop, Comet, Helicone, AgentOps} are acquired by hyperscalers or runtime vendors by end of 2026. The pure-play eval thesis reprices from defensible to acquisition-target.

7. Foundation labs walking up-stack — continuous 2026

ChatGPT Business connectors, Claude for Work, Gemini Enterprise reprice Glean, Notion AI, and Copilot 365's monopoly continuously. Vertical-agent companies with deep workflow integration, outcome SLAs, and custom guardrails (Sierra, Decagon, Harvey, Hippocratic) are the least vulnerable — the moat is integration plus change-management, not the model. This confirms Bet #2 and sharpens the target list around vertical workflow integration depth.

Strategic quadrants

Pioneer (Genesis — build / explore, expect failures)

Procurement-grade signed eval bundles (no vendor turnkey; Inspect AI closest). Computer use for unattended back-office (still thirty percent at fifty steps). Multi-party agent audit (vendor + customer + auditor on one trace with redaction). Contextual-integrity primitives for memory (no vendor ships "written in B2B sales, refuse to surface in HR"). Sovereign agent stacks for EU AI Act–exposed F500 EU-subsidiaries (Mistral-equivalent partner play). MCP-native iPaaS replacing Zapier (technically possible; no winner).

Settle (Custom to Product — productize what works)

MCP server constellations for enterprise GTM SaaS (Salesforce, Outreach, Gong, Highspot, Zoominfo, Linear) — Bet #3's buildable target. Vertical voice agents (debt collection, healthcare intake, NPS-detractor recovery) — substrate is commodity, vertical packaging plus compliance is the moat. Vertical AI workers with deep workflow integration — the Sierra / Decagon / Harvey / Hippocratic pattern; Bet #2 = take a GTM role inside one rather than build one. Enterprise-acceptance evaluation and observability (SOC 2, model-risk, signed reports, red-team artifacts) — "Drata for AI agents." AI procurement / deal-desk SaaS — Bet #1's downstream product. Memory architecture as a service line (folded into Bet #5).

Consume (Product to Commodity — rent, do not build)

Foundation models via API or AI Gateway. Sandboxes (E2B, Modal, Vercel Sandbox). Headless browsers (Browserbase plus Playwright). Voice substrate (LiveKit, Cartesia, Deepgram, Twilio / Telnyx). Agent runtimes — use the SDK of your dominant model provider. MCP transport plus first-party servers. AI gateways (Vercel, Cloudflare, OpenRouter). Vector databases, embeddings, rerankers (pgvector or Turbopuffer plus Voyage or Cohere). OpenTelemetry GenAI tracing.

Utility / Build-around (ubiquitous)

Linux, Postgres, Chromium, WebRTC, Firecracker, Python and TypeScript, PyTorch, the public internet, Kubernetes, OpenTelemetry, Markdown, JSON-RPC.

Implications for Alex

Alex's compounding advantage is enterprise GTM judgment — twelve years of F1000-buyer-side procurement scar tissue — plus AI-builder fluency. That fits Settle, with a Pioneer flag on procurement-grade controls and a disciplined Consume stance on everything below the runtime. The agent-layer map sharpens the prior bet stack in three specific ways. First, the unclaimed Pioneer flag is procurement-grade agent controls sold first as a buyer-side audit framework, not a vendor product. This is Bet #1's Wardley-defensible position. Second, Settle holds the highest-conviction zone with Bet #2 (vertical agent GTM role) and Bet #3 (MCP-native enterprise integration practice) intact but timing-sharpened around the gateway sub-category. Third, Consume names the silent leverage: anything below the agent runtime is a rent decision, with the twenty-four-month commoditization timer favorable to buyers and the build-cost unfavorable to builders.

Where the map disagrees with the field, two corrections matter. "MCP won" is half-true — the spec is held and the experience is fragmenting. Bet #3 plans for "MCP-compatible baseline," not "MCP-native interop guarantee." "Computer use is here" is wrong for unattended autonomous work through 2026. Error compounding (ninety-five percent per step across twenty steps is thirty-six percent end-to-end) is the hard ceiling. Bet on watched and narrow products; do not bet on autonomous back-office agents to clear F500 procurement before mid-2027.

Part IX · Helmer’s 7 Powers and Agent-Layer Durability

Helmer's 7 Powers — Scale Economies, Network Economies, Counter-Positioning, Switching Costs, Branding, Cornered Resource, Process Power — require both benefit and barrier. Mindshare is not a power. Hype is not a power. Applied strictly to the agent layer, the screen is brutal: capability gaps close in months; the runtime is the thinnest gross-margin layer; MCP is by design a non-moat. The real power positions are narrow.

Per-stratum power assessment

At the foundation-model layer (Stratum I), the moat that endures is Process Power on the RL-from-trajectory-data pipeline (RLVR plus GRPO plus PRMs plus verifier harness) — not the architecture. The capability lead is rentable; DeepSeek R2 and Qwen3 close the gap at four-to-six times lower cost. Trajectory-data Process Power strengthens as labs accumulate proprietary RL data from their own products (Claude Code, ChatGPT Agent). The Cornered Resource is the approximately two-hundred senior researchers who have shipped large-parameter frontier models with reasoning loops. Branding is narrow: Anthropic owns "safety plus coding"; OpenAI owns "consumer." At the runtime layer (Stratum II), almost no powers exist. Switching costs are shallow (LangGraph-to-Claude-SDK migration is a quarter). LangChain Brand is eroding to fatigue. Vendor SDKs derive their pull from the model, not the runtime. Do not build a company here.

At the MCP / tool-use protocol layer (Stratum III), Network Economies are forming at the registry level (eleven thousand four hundred servers in April 2026; first-party Stripe / Linear / GitHub / Snowflake servers compounding). But open protocols are by design non-moats. The durable power at this layer locates at the MCP gateway control plane (Cloudflare, Kong, Pomerium) — Switching Costs through authentication, audit, and secret-injection enterprise plumbing. Anthropic's Counter-Positioning win is permanent and underweighted: shipping MCP first, getting it adopted, then donating to the Linux Foundation in December 2025 — locking the standard while disclaiming control. No US lab can now re-claim spec authority. At the memory layer (Stratum IV), powers are thin. Zep has nascent Switching Costs in compliance-sensitive accounts. Lab absorption is the dominant force; only Zep holds a durable niche on compliance Process Power.

At the planning and reasoning layer (Stratum V), powers exist inside frontier labs only (Process Power via PRM, RLVR, GRPO pipelines that buyers cannot inspect). External to labs, there is none — planner-executor-verifier is a config flag in gateway products. At the action-surface layer (Stratum VI), powers are mixed. Browserbase has emerging Switching Costs through session recording, replay, and the Stagehand DX layer — but Chromium is not proprietary. LiveKit at the WebRTC infrastructure layer is the closest thing to a Cornered Resource — the only credible AI-native real-time substrate. The Anthropic / OpenAI / Google computer-use products are demoware-grade and will not generate power until OSWorld crosses approximately seventy percent.

At the eval and observability layer (Stratum VII), Switching Costs are real in evaluation logic (regression sets as Braintrust YAML or Galileo metric definitions take a quarter to port). OpenTelemetry GenAI conventions stabilized in January 2026 killed lock-in on trace ingest. LangSmith has tight LangGraph Switching Costs; Braintrust has eval-as-CI workflow stickiness. The durable niche is signed and auditable evaluation reports tied to EU AI Act and NIST AI RMF — and nobody ships this turnkey today. That is Bet #1's opening. At the runtime-safety layer (Stratum VIII), powers are thin. Lakera Brand plus early Switching Costs in regulated accounts; no prompt-injection moat. Mostly absorbed into hyperscaler bundles within eighteen months.

At the vertical-product layer (Stratum IX) — the upper-stack power concentration — Process Power plus Switching Costs are real at Sierra (customer-experience outcome-pricing operational scar tissue plus per-resolution SLA enforcement infrastructure that took two years to build), Harvey (BigLaw RLHF on contract data plus privilege and bar liability fence), Abridge (Epic embedding depth plus clinician workflow integration), Hippocratic (per-call labor-substitute pricing plus state licensing buy-in plus Cerner and Epic integrations), Glean (organization-wide data graph that compounds with usage — the closest thing to Network Economies in the vertical layer). Cornered Resource is rare here; Harvey's Cravath and A&O Shearman design-partner relationships are the closest. F1000 buying is RFP-driven, not brand-driven; Branding matters less than usually claimed. Vertical-specific Process Power strengthens for the two-to-three winners per vertical and erodes for the long tail.

At the end-user-surface layer (Stratum X), surface-by-surface differentiation. Microsoft Copilot 365 has Switching Costs (enterprise seat lock-in, M365 graph integration) plus Scale Economies (existing channel); it is the largest agentic-revenue position of 2026 with $5 billion ARR and approximately thirty million paid seats. Apple Intelligence has Cornered Resource (Neural Engine plus iOS distribution) plus Switching Costs. WhatsApp (Meta) has Cornered Resource — the dominant AI surface for the global majority once the agentic API opened in January 2026. Anthropic has Branding plus Process Power on the model but a first-party-surface deficit (no Apple Intelligence slot; Claude mobile feature-thin; no native voice product; Claude for Chrome paying-tier only). Anthropic's enterprise power is real but routes through API, IDE, and CLI surfaces, not consumer surfaces — a structural shape, not a temporary gap.

Meta-strata

At Meta-A (capability safety), Anthropic holds Branding (uniquely owns "safety" in enterprise mind) plus Process Power (Constitutional AI, interpretability program) plus Cornered Resource (alignment researchers, approximately two hundred globally; Anthropic retained Jan Leike). Strengthening with EU AI Act conformity-assessment drafts (February 2026) citing Inspect AI explicitly. The most defensible single-vendor position in the entire stack. At Meta-B (regulation), Counter-Positioning for Mistral and EU-resident challengers under EU AI Act enforcement (late 2026): a US frontier lab cannot match "data never leaves the EU" without re-architecting its training pipeline. Palantir holds Process Power (accreditation craft) plus Cornered Resource (cleared personnel plus government relationships) at the FedRAMP / IL5-IL6 / HIPAA frontier — decades-deep. Strengthening with EU AI Act enforcement and US federal preemption volatility. This is where Bet #1 sits.

At Meta-C (economics), powers are eroding. Token prices compress four-to-ten times per cycle; OpenRouter and Bedrock commoditize switching. Hyperscalers retain Scale Economies on inference infrastructure but pure-play inference (Groq, Together, Fireworks) has Process Power on kernels without durable Switching Costs. Anthropic's valuation depends on capability-lead-pricing surviving the next two cycles; it might not. At Meta-D (geopolitics), Counter-Positioning for Mistral (EU sovereignty), Sarvam (India), Manus (China-language-data Cornered Resource — Mandarin trajectory data US labs cannot legally train on). Manus's Cornered Resource on Chinese-language agent-trajectory data is real and US-discourse-underweighted.

The five most durable agent-layer power positions, May 2026

NVIDIA at the substrate layer — Cornered Resource (CoWoS, HBM allocation) plus Process Power (CUDA) plus Scale Economies — three powers stacked, agent-stack-relevant via inference economics, durable through 2030 on training and eroding faster at inference. Anthropic's alignment plus trajectory-data Process Power — Constitutional AI plus Claude Code RL-from-production-trajectories plus the approximately two-hundred-person alignment talent pool. The capability lead is rentable; the Process Power on internal RL pipelines compounds with own-product usage. Endures past 2027 because the data flywheel is self-reinforcing.

Sierra in CX — vertical Process Power plus Switching Costs. Outcome-pricing operational scar tissue, per-resolution SLA infrastructure, two-year head start on the operational moat. Helmer-strict: this is Process Power, not Brand. Endures because the operational practices took two years to build, are not codifiable in a runbook, and Decagon (the closest competitor) is one cycle behind on the same curve. Glean's organizational data graph — Network Economies, vertical-bounded. Value increases with users inside the same enterprise; cross-system graph is hard to replicate; ChatGPT Business connectors attack the wedge but cannot replicate the org-graph density. Endures eighteen-to-twenty-four months past 2027 if Microsoft does not ship a credible cross-tenant equivalent.

MCP gateway control plane (Cloudflare, Kong, Pomerium) — Switching Costs via auth, audit, rate-limit, and secret-injection enterprise plumbing. Once an F500 has thirty-plus MCP servers wired through Kong's gateway with SSO and audit, ripping it out is a year of work. Strengthens as the registry grows. This is the durable MCP power — not "build at the MCP layer" but "own the enterprise control plane around it." Honorable mentions: Harvey at AmLaw 100 (Process

Power plus privilege and bar fence; smaller TAM); Microsoft Copilot (Scale plus Switching, inherits OpenAI dependency); Palantir at the federal frontier; Browserbase's operational layer.

The five most over-rated power positions

Cursor at $9.9 billion valuation as a durable position. Real ARR (more than $500 million run-rate), but the moat is mindshare plus developer experience, not power. Anthropic owns the model underneath (Claude Code is the encroacher); IDE inline is the form factor most exposed to foundation-lab walk-up. Switching Costs are low — engineers move IDEs in a week. Branding is not a moat. LangChain / LangGraph as the "neutral runtime moat." API churn fatigue is real; vendor SDKs ship faster; the field narrative is ahead of evidence. The durable LangChain bet is LangSmith observability, not the runtime. Memory companies (Mem0, Letta) as standalone categories. "Pinecone for memory" is the wrong analogy. Lab absorption plus RAG-with-writes economics plus opaque enterprise compliance story. Letta's pivot to "stateful agent runtime" is the implicit concession.

11x and the "synthetic SDR" category. $20 million flat ARR, churn spike, Hayden Sukkar publicly conceding the quality ceiling. RevOps buyers reject vendor outcome attribution. No power. Clay survives by owning data, not by being an agent. Standalone runtime-safety vendors (excluding Lakera's red-team niche). Indirect prompt injection is unsolved; hyperscaler bundles cover eighty percent at zero procurement friction; Protect AI to Palo Alto was the consolidation bell. Apex, Calypso, Cranium most likely roll up. Pure-play guardrails are not a durable power.

Implications for Alex — career-bet calibration

The 7 Powers screen produces a sequencing insight no other framework does. Bet #1 (Procurement Operating Standard) is the highest-power-density career bet in the portfolio — Counter-Positioning on EU AI Act enforcement plus signed-eval-report gap plus tool-boundary policy gap plus action-rollback procurement frame. This is Process Power that compounds with each F500 engagement and that no incumbent (Vanta, Drata, Secureframe) currently fills. Bet #2 (vertical agent GTM seat) is the highest-equity bet but it rents power from Anthropic, Sierra, Harvey, Glean. Bet #3 (MCP-native enterprise integration practice) is advisory power, not platform power — operational Process Power flavor from doing it ten-plus times, not software moat. The sequence the screen produces: Bet #1 first to claim the durable position, Bet #2 second to collect equity at a power-holder while the procurement scar tissue compounds, Bet #3 third as advisory practice that draws from both.

The 7 Powers target list for Bet #2 sharpens. The most durable upper-stack positions are Sierra (Process Power on outcome-pricing operations plus emerging vertical Switching Costs), Harvey (lowest-encroachment-risk vertical due to privilege and bar liability fence), Glean (vertical-bounded Network Economies), Abridge (Epic embedding depth), Hippocratic (state licensing buy-in plus per-call labor-substitute pricing), Hebbia (NYC-native financial-services-adjacent), Rogo (banking workflow Process Power developing). The over-rated list eliminates 11x, Cresta, Lavender, and any RevOps target other than Clay from serious consideration — they do not pass the power screen even where ARR is real.

Part X · Ecosystem-Level Jobs to be Done

Christensen, Ulwick, and Moesta framed Jobs-to-be-Done as the top-down lens for what users hire a system for. Applied to the agent layer specifically, seven ecosystem-level jobs emerge. Each is mapped through Ulwick's eight-phase Job Map (Define, Locate, Prepare, Confirm, Execute, Monitor, Modify, Conclude) with the most-overlooked outcome at each phase named and the top three underserved desired outcomes flagged with importance-satisfaction gaps.

Job 1 · Complete a discrete back-office task without babysitting

An operations or finance leader has a multi-step rules-governed workflow — invoice triage, payroll exception, contract abstraction, refund, KYC — and wants the agent to take it inbox-to-archive over hours-to-days, fail safely, and report. Buyer is the VP of Operations, Controller, or Shared Services lead. Willingness to pay is $25,000–$150,000 per workflow per year mid-market and $500,000–$2 million F500, often priced per-resolution, per-document, or per-FTE-replaced.

The phase map surfaces the gaps. Define: time to translate an SOP into an agent specification — n8n and Zapier serve partially; no "SOP-to-agent" owner exists. Locate: percentage of source data reachable — Glean and Hebbia for search; ETL bespoke. Prepare: confidence tools have correct write-scope and rollback — sandboxes for code; no SaaS-write equivalent. Confirm: pre-run probability the run will not stall at step twelve — PRMs are internal at labs and not exposed to buyers. Execute: trajectories that succeed-as-judged but write wrong — the silent-failure problem, unserved. Monitor: time to detect a looping run before token blow-out. Modify: mid-flight intervention by a non-engineer — replay UIs are engineer-only. Conclude: audit-grade run summary a controller will sign — universally unserved. The three highest-importance underserved outcomes: minimize time to diagnose why a partial-failure run stopped (importance 9 / satisfaction 3); increase percentage of unattended runs producing tamper-evident, signed evidence packs (8/2); minimize trajectories that complete-as-judged but write the wrong value to a system of record (10/4 — the silent-failure problem).

Job 2 · Run a customer-facing conversation to resolution

A CX or growth leader needs every contact — chat, voice, WhatsApp, SMS, email — to either resolve to a stated outcome or hand off warm with full context. Buyer is the VP of CX (incumbent) or CRO when retention-coupled. Willingness to pay anchors at $1–$4 per resolved ticket (Sierra, Decagon), $9 per hour "RN equivalent" (Hippocratic), $0.05–$0.15 per minute outbound voice (Vapi, Retell).

The phase map: Define — time to agree what "resolved" means with the business (Sierra Outcomes leads); Locate — percentage of identity and history found across CRM/billing/help-desk; Prepare — channel compliance (TCPA, GDPR, state consent, HIPAA) pre-call, unserved; Confirm — cold-start hallucination on policies told but not tested; Execute — first-contact resolution without misrepresenting policy; Monitor — time to detect a class of failure; Modify — non-engineer CX lead ships a fix same-day, universally unserved; Conclude — handoff with no re-asking, ticket linked, full context (Sierra leads, the rest lose CSAT). The three highest-importance underserved outcomes: minimize percentage of escalations where the human re-asks something the agent already asked (10/3); increase confidence a WhatsApp or SMS agent is consent-compliant across jurisdictions pre-launch (9/2 — WhatsApp Business API opened January 2026 with almost no compliance tooling); minimize time for a non-engineer CX lead to ship a policy correction (9/3). The non-US note matters: WhatsApp-native (Yellow.ai, Haptik, Gupshup, Meta first-party) is the dominant global shape with the same phase map and a materially worse threat model — no end-to-end encryption guarantees, no audit-log inheritance.

Job 3 · Execute a multi-step coding change including PR review and merge

An engineering lead has a specified change and wants the agent to branch, code, test, open the PR, satisfy CI, respond to review, and merge — without senior babysitting. Buyer is the Engineering Director, VP of Engineering, or CTO. Willingness to pay is $200–$500 per seat per month enterprise (Cursor Teams, Copilot Enterprise, Claude Code); success-priced is not stable.

The phase map: Define — time from ticket to fully-scoped agent task (spec quality is the bottleneck); Locate — percentage of repo, doc, PR, and test context surfaced; Prepare — secrets and keys exposed during execution; Confirm — pre-PR confidence the change will not break unknown downstream; Execute — trajectories that pass tests but regress in production (SWE-Bench Verified is gameable); Monitor — wall-clock to know if CI failure is flaky vs. real; Modify — reviewer redirects mid-PR without restart; Conclude — residue (dead branches, abandoned WIPs post-merge), unserved. The three highest-importance underserved outcomes: minimize percentage of agent PRs that pass review and CI but need senior rewrite within thirty days (9/4); reduce time to disambiguate flaky CI vs. agent regression vs. environment (8/3); increase percentage of code-agent runs producing audit-grade authorship and provenance metadata (model pin, prompt, dataset hash) for compliance and IP defense (7/2 — becomes 10 in regulated industries post-Bartz and NYT v. OpenAI). The over-served caution: "make any code change from a prompt" is over-served (fourteen-plus funded players). The gaps live at the end of the job map.

Job 4 · Operate a SaaS application on the user's behalf

A line-of-business user knows the tool (Salesforce, Workday, SAP, NetSuite, ServiceNow, Notion, Linear) but does not want to drive it. Buyer is the line manager who owns the workflow — RevOps Director, FP&A lead, Service Desk lead — not IT. IT is compliance veto. Willingness to pay anchors as embedded in seat (Agentforce $30–$125, Copilot $30) or standalone $50–$200 per seat for cross-tool orchestrators (Glean, Hebbia).

The phase map: Define — natural-language intent to executable plan (embedded copilots lead); Locate — percentage of user's actual app state correctly perceived (computer-use fails in production; embedded cheats via API); Prepare — OAuth blast radius for a transient task, universally unserved; Confirm — user trust the agent will not touch out-of-scope; Execute — first-try success on multi-screen workflows (Mariner approximately thirty-five-to-forty percent on OSWorld); Monitor — cognitive load of watching a SaaS-UI agent in real time (the form-factor problem); Modify — pause, edit goal, resume mid-task (ChatGPT Agent leads); Conclude — unambiguous record of what changed (embedded ships change-logs; cross-tool does not). The three highest-importance underserved outcomes: reduce OAuth blast radius — increase percentage of tasks running with least-privilege, time-boxed, scope-bounded credentials (9/2 — MCP 0.3 auth closes this eventually; today unserved; the Bet #3 wedge); increase percentage of cross-tool tasks (Salesforce → Outreach → Gong → Slack) that complete without a human reconnecting the chain (8/3); minimize calendar time a non-IT line manager waits to deploy a new agent into a sanctioned tool (8/3 — procurement gauntlet hits hardest here because the buyer is not the seat owner). Persona honesty: vendor marketing targets the CIO; the true buyer is the line manager. IT is procurement, not customer. Vendors who pitch the line manager and bring IT along sell faster (Glean's Slack-first wedge).

Job 5 · Stay current on a domain and act on what changes

An operator (PM, AE, RevOps, compliance, architect, investor) needs a fast-moving domain monitored, summarized, prioritized, and surfaced only for decisions they would make. Buyer is the operator ($20–$200 per month) or the employer for team tier ($500–$5,000 per seat per year — AlphaSense, Hebbia, Perplexity Enterprise, Glean). The job inherits and narrows from Volume II's Job 3: same shape, any domain.

The three highest-importance underserved outcomes: minimize time from event to operator-deciding (not just notified) (9/2 — Bet #6's target); increase confidence the filter is not silently dropping contrarian sources (8/3); minimize duplicate processing across email, Slack, RSS, podcast, newsletter, and X for the same event (7/3).

Job 6 · Pass agent-specific enterprise procurement and risk review

An AI-agent vendor enters an F1000 cycle and must satisfy InfoSec, Legal, Privacy, AI Governance, Procurement, and the business sponsor on agent-specific risks — autonomy scope, tool-boundary policy, indirect prompt injection, action rollback, sub-agent privilege, Article 14 oversight, eval reproducibility — to close within the buyer's planning cycle. The job inherits and narrows from Volume II's Job 4. Buyer is the AI vendor's sales leader; the real customer is the F1000 committee. Willingness to pay anchors at $50,000–$250,000 per year for a tool that closes the gap (Vanta-shape).

The phase map is the Playbook's table of contents. Define — map the buyer's six counterparty checklists to vendor evidence (Vanta and Drata for SOC 2; nothing AI-specific). Locate — artifacts (model cards, evaluations, DPAs, AI policy) in one place (TrustCenter partial). Prepare — agent-trajectory eval evidence the buyer accepts (no turnkey signed report). Confirm — pre-call confidence AI Governance will sign, unserved. Execute — time in queue at each of six counterparties (Vanta-style workflow needed). Monitor — visibility into which counterparty is blocking, unserved. Modify — counterparty-specific evidence packs on demand. Conclude — evidence freshness through annual review. The three highest-importance underserved outcomes: minimize calendar time from security questionnaire to signed AI Governance sign-off (10/2 — Bet #1's target); increase percentage of agent-specific risk questions (indirect injection, tool-boundary, action rollback, sub-agent privilege) the vendor answers in standardized form (9/2); reduce bespoke effort for an EU AI Act / NIST AI RMF / ISO 42001 conformity tie-out (8/2 — rising fast). The honest demand check: buyer-side demand is real and growing (Volume III A6 lists the five questions). Vendor-side demand is partly marketing-induced. Buyer-pull is sufficient — the falsifiability test (500 downloads / 50 inbound conversations in 60 days) is the measurement.

Job 7 · Onboard or ramp a new role using an agent-augmented training stack

A manager hires into an AI-native role (AE, FDE, Applied AI engineer, CS, ops analyst) and wants the hire to reach productive output in weeks rather than quarters by pairing them with an agent that scaffolds workflow, coaches tool-by-tool, and produces an observable trace for ramp assessment. Buyer is the hiring manager (primary), L&D / People Ops (channel). Willingness to pay anchors at the L&D budget of $1,200 per employee per year (ATD 2024); compressing a six-month ramp to three is worth $20,000–$80,000 per hire. Most "AI for onboarding" is repackaged LMS — this job is different. An agent that does the work alongside the new hire and tapers as the hire ramps. Nobody ships it cleanly in May 2026.

Cross-job synthesis

Three patterns underserve universally. First, Conclude. Across all seven jobs, the Conclude phase — signed evidence pack (Job 1), warm handoff (Job 2), provenance metadata (Job 3), change-log (Job 4), decision routing (Job 5), counterparty evidence (Job 6), agent taper (Job 7) — is the worst-served phase. Vendors compete on Execute. Buyers feel pain at Conclude. Single highest-leverage pattern in the volume. Second, Modify by a non-engineer. Replay UIs exist for engineers (LangSmith, Braintrust); cockpit UIs for the line manager, the CX lead, the FP&A director do not. Every job map shows a Modify gap of seven or higher. This is the operator-translation gap inside the runtime — adjacent to Bet #6, productizable. Third, Confirm (pre-flight). Process-reward models exist inside frontier labs but are not exposed to buyers. Pre-flight checks are technically feasible and commercially absent. Closing this turns Job 1 from "agent attempts, sometimes succeeds" into "agent commits, reliably succeeds" — the difference between $25,000 and $250,000 ACV.

Three patterns over-serve. Execute on coding tasks: Cursor, Claude Code, Codex, Augment, Cognition, Lovable, Replit, Bolt, v0, Magic, Reflection, Factory, Plandex, Aider, Cline, Continue — fourteen-plus funded players, marginal demand at Execute exhausted. Locate / enterprise search-agents: Glean has the moat; Hebbia is the NYC specialist; Notion AI, Copilot, ChatGPT Business connectors, Slack AI, Box AI all attack — twelve vendors for a Locate problem consolidating to two-to-three winners. Execute on inbound chat: web and mobile chat-as-form-factor is over-served; the next hundred chat UIs will not move enterprise demand. The win is Conclude (escalation handoff) and Confirm (pre-deployment trust), not more chat.

Recommendations tied to Bets. Bet #1 (Procurement Playbook) maps precisely to Job 6. The Job 6 phase map is the Playbook's table of contents. The five risk-officer questions from Volume III A6 (signed reproducible eval reports; adversarial-test suite; tool-boundary policy; action-rollback story; EU AI Act / NIST AI RMF / ISO 42001 conformity tie-out) are the five chapter prompts. The Opportunity-10 on "calendar-time to AI Governance sign-off" is the falsifiable hypothesis. Position eventual SaaS as "Vanta for the agent-specific gauntlet" with the Conclude-phase evidence pack as the demo. Bet #2 (vertical agent GTM seat) maps to Jobs 2, 4, and 7. Sierra and Decagon win Job 2; Glean and Hebbia win Job 4 in knowledge; Hippocratic wins Job 2 plus Job 7 in clinical; Harvey wins Job 4 in legal; Rogo and Ramp AI win Job 4 in finance. The Field-CTO or Director-of-GTM role is paid to close the Confirm + Modify + Conclude gaps for one job in one vertical. Frame interviews around "your sales team is selling Execute; the buyer is signing on Confirm and Conclude — here is how I close that." Sierra (NYC dual-HQ) is the cleanest fit; Hebbia and Rogo are NYC-native sleepers worth equal weight. Bet #3 (MCP-native integration) maps to Job 4's OAuth blast radius and Job 1's missing write-action sandbox. Job 4's "least-privilege, time-boxed credentials" is what MCP 0.3 auth enables but no incumbent SaaS has shipped. Filter the Bet #3 audit by which Job 4 / Job 1 outcomes each unlocks. Do not ship for completeness; ship for the Confirm / Conclude gap. The cross-Bet through-line: Conclude is the connective tissue. Bet #1 sells the Conclude gap; Bet #2 closes it inside a vertical; Bet #3 ships the substrate; Bet #6 is the distribution layer that makes the rest reach.

Part XI · Synthesis — Seven Bets Refreshed, Five Risks, Five Cruxes

Five frameworks were applied at agent-layer resolution. Where they converged, conviction strengthens. Where they diverged, the open questions are honest. The seven Bets from Volume II carry over rather than multiply; each is refreshed with the agent-specific delta. Five Structural Risks and five Cruxes update accordingly.

How the agent-layer frameworks converged

The single observation all five agent-layer frameworks produced is that the highest-power-density, highest-claimability, highest-asymmetry near-term opportunity for Alex sits in agent-specific procurement (Bet #1), not the vertical-agent GTM role (Bet #2). The OCQ Matrix surfaces multiple high-claimability opportunities — signed reproducible eval reports, EU AI Act Article 14 human-oversight playbook, tool-boundary policy template, agent-trajectory cost audit, form-factor procurement audit, sandboxing-as-control mapping, sectoral overlay advisory — that all compound into Bet #1 as modules rather than into separate productized bets. The Wardley map places procurement-grade agent controls at Genesis stage — the only unclaimed flag at Pioneer. The 7 Powers screen names procurement Process Power at Meta-B as the only durable position Alex can claim rather than rent. The Ecosystem JTBD analysis maps Job 6 phase-by-phase to the Playbook's table of contents. The Talent and Capital Flow tracker confirms NYC peak hiring window for Bet #2 but also confirms that the Stripe / Ramp / Datadog / Snowflake migration is creating its own marketplace at the procurement-readiness layer where the buyers are. The sequencing implication is unambiguous: Bet #1 first, Bet #2 second, Bet #3 third.

The seven Bets — refreshed

Bet 1 · The Enterprise AI Procurement Operating Standard (agent-specific)

Hypothesis (refreshed). Every F1000 stood up an AI council in 2024–2025 and is procuring AI agents specifically — not just LLMs — with six non-overlapping approvers (InfoSec, Legal, Privacy, AI Governance, Procurement, business sponsor) plus an agent-specific overlay that no incumbent owns: tool-boundary policy, indirect prompt-injection adaptive red-team, action-rollback documentation, sub-agent privilege separation, signed reproducible eval reports, EU AI Act Article 14 human-oversight tie-out, sectoral overlay (FINRA, HIPAA, TCPA). A commercial operator with twelve years of buyer-side procurement scar tissue plus AI-builder fluency can plant the canonical flag in 90 days by publishing the open Playbook that names these gaps. Falsifiability. If the open Playbook gets fewer than 500 downloads or fewer than 50 inbound conversations in 60 days, or if Vanta, Drata, or Secureframe ship a credible AI-specific agent-procurement product in that window, the productized SaaS thesis is dead. Conviction. Five stars — refreshed up from the prior addendum on the strength of five-framework convergence. Delta from Volume II. Adds the seven agent-specific overlays (tool-boundary, action-rollback, etc.); adds the EU AI Act Article 14 anchor as the most actionable near-term regulatory lever; reframes from "publish, then advise, then productize" to "publish, observe, advise on procurement audits, decide on productization at month nine." Next action. Outline the Playbook in Week 1 using the Job 6 phase map and the A6 five-questions structure. Thirty expert interviews in Weeks 2–6. Publish in Week 12 with the open downloadable Procurement Rubric attached.

Bet 2 · Vertical Agent GTM Leadership Role

Hypothesis. Sierra ($175M+ ARR at $10B rumored March 2026), Decagon ($80M+ at $4.5B), Glean ($300M+ at $7.2B), Harvey ($100M+ at $5B), Hippocratic ($50M+ at $2B), Hebbia (NYC

HQ, $50M+, raise overdue), Rogo (NYC HQ, $30M+ at $400M), Clay (NYC HQ, $80M+ at $1.5B), Ramp AI org (NYC HQ, $20B+ raise rumored Q2 2026), Runway (NYC Chelsea HQ, $100M+) plus Augment ($40M at $977M) and Decagon are hiring exactly Alex's profile at $300–$400K base plus 0.05–0.40% equity. NYC vertical-agent hiring is at historic peak Q2 2026. Falsifiability. If 6 months of focused NYC search yields no offers in this band, the bet on Alex's profile is wrong or the market is more SF-anchored than the talent flow suggests. Conviction. Five stars — held. Delta from Volume II. Expanded NYC target list adds Hebbia, Rogo, Clay, Runway, Ramp AI org alongside the existing Sierra / Decagon / Glean / Harvey / Hippocratic / Augment / Decagon set. The 7 Powers screen sharpens the highest-power targets (Sierra, Harvey, Glean) and eliminates the over-rated (Cresta, Lavender, any RevOps target other than Clay). The Anthropic ARR Crux (Q3 2026 disclosure or leak) becomes the single load-bearing valuation variable. Next action. Time the offer before the up-round at Hebbia, Rogo, or Augment. Open consulting-alumni pipeline (BCG / McKinsey / Bain AI practice partners) as a less-competed feed-stock into Sierra / Hebbia / Harvey. Sign before Anthropic ARR resolution Q3 2026 if conviction is high; after if wobbly.

Bet 3 · MCP-Native Enterprise Integration Practice (reframed)

Hypothesis (refreshed). MCP crossed Genesis-to-late-Custom in December 2025 (Linux Foundation governance); the spec is held and the experience is fragmenting silently along four fork vectors. The durable power is at the gateway control plane (Cloudflare, Kong, Pomerium) — Switching Costs via auth / audit / rate-limit / secret-injection — and at first-party MCP servers from SaaS incumbents (Stripe, Linear, GitHub, Snowflake, Databricks, ServiceNow). Productized servers commoditize the moment Salesforce or HubSpot ships first-party. Alex's claimable position is advisory / practice, not platform — Process Power flavor (operational scar tissue from doing it ten-plus times) plus Branding flavor (canonical voice via the Playbook). Falsifiability. If MCP forks (Anthropic-vs-OpenAI proprietary tool-use schemas diverge materially, or A2A wins agent-to-agent without MCP), the entire MCP-layer thesis weakens. Watch for any major vendor unilaterally extending MCP without spec coordination. Conviction. Four stars — held with the reframe. Delta from Volume II. Reframed from "build MCP servers for enterprise GTM SaaS" to "advisory and practice layer around MCP gateway deployment plus private-registry curation plus first-party server quality grading." Productize at month twelve or eighteen if scar tissue justifies. Next action. Q2 2026 audit of the ten SaaS systems most lacking MCP servers in enterprise GTM, filtered by Job 4 / Job 1 outcome unlock. Pair-with-Cloudflare / Kong / Pomerium as the AI-procurement specialist on gateway deployment.

Bet 4 · Inference Cost Optimization / FinOps for Trajectories

Hypothesis (refreshed). The median enterprise runs un-tuned vLLM on over-provisioned H100s. EAGLE-3 at three-to-six-point-five times, FP4 quantization, prompt caching, and aggregator routing can deliver three-to-ten times cost reduction at the token level. At the trajectory level, the planner-executor-verifier split — route seventy-to-eighty-five percent of tokens through cheap executors — is the durable architectural idea and the FinOps-for-Trajectories advisory layer. The procurement-side audit is sellable today; the technical work is straightforward for a two-person services team. Falsifiability. AWS Bedrock auto-optimization features ship in 2026, slamming the twelve-month window. Or token prices fall fast enough that "optimization" is below the line of caring for many enterprise workloads. Conviction. Four stars. Delta from Volume II. Splits into two layers: per-token FinOps (the Volume II framing) and per-trajectory FinOps (the agent-specific reframing). The latter has a sharper twelve-month window because hyperscaler auto-routing of planner-executor is plausible by H2 2027. Next action. Free first audit for five mid-market AI-spending companies; build case studies; decide by Q3 2026 whether to scale or fold into Bet #1 as a module.

Bet 5 · Enterprise RAG plus Memory Architecture Practice

Hypothesis (refreshed). F500 enterprises are paralyzed by eight-plus vector DB options, five embedding choices, three reranker options, four retrieval paradigms, plus the memory-architecture question (lab-native vs. Mem0 vs. Letta vs. Zep vs. summarize-and-pgvector). The decision is $50,000–$500,000 per workload and they have no vendor-neutral guidance. Pure architecture sale; commercial fluency is the asset. Falsifiability. Long-context (Gemini 2M tokens, Claude prompt caching) eats RAG at the low end faster than enterprise-scale grows at the high end. Or vector DB consolidation makes the choice trivial. Conviction. Four stars. Delta from Volume II. Folds memory architecture into the service line per the Crux #5 directional answer (consumer/prosumer absorbed; compliance-enterprise niche-standalone). Next action. Bundle as "Enterprise AI Architecture Audit" alongside Bet #1's Playbook and Bet #4's FinOps. Three-product practice, one buyer.

Bet 6 · Operator’s Translation Newsletter / Public Voice

Hypothesis. Weekly "operator's translation" newsletter — translate frontier capabilities into specific operator implications (enterprise AE, RevOps lead, GTM head). Compounds positioning for everything else. JTBD Job 5 distribution layer; the "what changed → I should change Y" gap. Falsifiability. If subscriber growth is below 2,000 in 6 months and no inbound role or advisory comes from it, kill at month 6 or commit harder. Conviction. Three stars. Delta from Volume II. Confirmed as the distribution layer for the rest of the bets; the JTBD analysis sharpens the angle around Conclude-phase translation ("what should you do next?") rather than executive-brief summarization. Next action. Kit v1 published Week 2. Decide on cadence by Week 4.

Bet 7 · VC Operating Partner / Platform Path

Hypothesis. VC platform teams need someone who can read technical updates and translate them to portfolio-company GTM teams. NYC funds (Lerer Hippeau AI-platform rumored Q1 2026, FirstMark MAD AI specialist, Insight NYC AI-advisory, Lux sci-tech / AI) are visibly hiring for this profile. Falsifiability. If Bets 1, 2, 3 land, this is downstream and lower priority. If none land by month 12, this becomes the primary path. Conviction. Three stars; four if it becomes primary. Delta from Volume II. The VC-principal-to-operator pattern (Carey Lai Insight → Sierra April 2026 and similar) confirms platform and operator paths converging at the senior level. Next action. Build relationships at three NYC funds via RAAIS, Betaworks, and FirstMark MAD events. No active applications until Q4 2026 unless Bets 1–3 stall.

The five Structural Risks — agent-layer refreshed

Risk 1 · Foundation labs walking up-stack into vertical applications

Threatens Bet #2 most directly. ChatGPT Business connectors, Claude for Work, Gemini Workspace agents continuously compress vertical-agent vendor moats. Watch the Connectors / Skills / Apps store on each platform. Mitigate by targeting verticals with regulatory or workflow moats (legal, healthcare, financial services) where horizontal foundations cannot easily land. The 7 Powers screen confirms the mitigation: Harvey, Abridge, Hippocratic, Rogo have the deepest vertical Process Power.

Risk 2 · MCP silent fork via Responses-API proprietary path

Threatens Bet #3 thesis materially if "MCP-compatible but not MCP-native" becomes dominant by 2027. Watch for OpenAI Responses-API extensions outpacing MCP-spec evolution; watch A2A adoption outside Google; watch quality split between first-party MCP servers and the long tail. Mitigate by reframing Bet #3 to advisory plus gateway-adjacent rather than productized servers; this is now done.

Risk 3 · Anthropic ARR resolution (single load-bearing variable for Bet #2 timing)

Anthropic ARR ($24B vs. $30B disputed) resolves Q3 2026. Lower bound: vertical-agent valuations compress twenty-to-thirty percent (Sierra $10B to $6–$8B); equity bands tighten; timing slows. Upper bound: equity strengthens; timing accelerates. Both ends are defensible in May 2026. Mitigate by signing Bet #2 before resolution if conviction is high, after if wobbly.

Risk 4 · Hyperscaler bundling of runtime safety, eval, sandbox, and routing

Threatens Bet #4's per-token / per-trajectory window and the standalone eval / observability category. AWS Bedrock auto-optim, Vertex routing, Azure routing default-on plausible H2 2027. Pure-play eval-and-observability pure-plays roll up. Mitigate by folding FinOps into Bet #1 modules and treating eval as a Bet #1 procurement-rubric input rather than a standalone build.

Risk 5 · EU AI Act paper-tiger outcome

Threatens Bet #1 advisory TAM. If first enforcement late 2026 stays small and procedural, Bet #1 stays niche but defensible. Sell EU compliance (stable) plus contractual compliance (always required) rather than CA-specific or US-state-specific programs. The Wardley placement of procurement-grade controls at Genesis holds regardless; the productized SaaS path is the part that gates on EU teeth.

The five Cruxes — agent-layer refreshed

Crux 1 · Anthropic ARR — $24B or $30B

Decidability Q2–Q3 2026 (audited reports or leaks). Lower bound = vertical-agent valuations compress twenty-to-thirty percent. Upper bound = Bet #2 timing accelerates. Watch The Information (Stephanie Palazzolo), Anthropic disclosure cadence, Stripe data leaks.

Crux 2 · MCP — commons or silent fork by EOY 2026

Decidability H2 2026. Fork = Bet #3 dies in productized-server form; commons = advisory practice accelerates. The reframe to gateway-adjacent advisory means Bet #3 is defensible either way; the productization decision gates on this Crux. Watch major-vendor proprietary tool-use schema announcements; watch A2A adoption.

Crux 3 · OSWorld 65% on a frontier system by Q3 2026

Decidability Q3 2026 (public scoreboard event). The threshold that moves computer use from supervised to deployable for narrow back-office. If crossed, the unattended-agent reliability binding constraint relaxes a quarter earlier than this volume assumes; Bet #3 should add an MCP-servers-for-the-systems-back-office-computer-use-agents-need sub-line. Watch the OSWorld leaderboard.

Crux 4 · EU AI Act Article 14 enforcement teeth (late 2026)

Decidability late 2026 (first enforcement actions). Determines whether Bet #1 advisory becomes a $10-billion-plus category or stays niche. Watch Commission guidance and first GPAI fines. The April 2026 Article 14 oversight draft is the most actionable near-term lever.

Crux 5 · Long-term memory — standalone or absorbed

Decidability 12–18 months. Directional answer from this volume: absorbed for consumer / prosumer; standalone-niche for compliance enterprise. Bet #5 folds memory architecture as a service line. Watch Anthropic, OpenAI, Google native memory feature launches; watch Mem0 and Letta funding events.

Part XII · 6 / 12 / 18-Month Action Map for Alex

The seven Bets translate into a sequenced playbook. The sequence is the synthesis of all five framework analyses applied at agent-layer resolution: Bet #1 first to claim a durable position, Bet #2 second to collect equity at a power-holder, Bet #3 third as advisory practice compounding from both. Bets #4 and #5 fold into Bet #1 as modules; Bet #6 is the distribution layer that makes the rest reach; Bet #7 is the long-arc fallback.

Months 0–6 · Plant the procurement flag and run the vertical-agent role search in parallel

- Bet #1 — outline the Procurement Playbook in Week 1 using the Job 6 phase map and the A6 five-questions structure as the table of contents and chapter prompts. Thirty expert interviews in Weeks 2–6 with F1000 InfoSec, Legal, Privacy, AI Governance, Procurement, and business-sponsor counterparties to validate the gaps. Publish in Week 12 with the open downloadable Procurement Rubric attached. Measure the 500-downloads / 50-inbound test at day 60.
- Bet #2 — open the NYC search in parallel from Week 1 with a target list of ten: Sierra (Field-CTO / Head-of-Enterprise-East seat, highest-aligned), Decagon, Glean, Harvey, Hippocratic, Hebbia, Rogo, Clay, Ramp AI org, Runway. Time the offer before the up-round at Hebbia, Rogo, or Augment (the under-funded relative to ARR velocity targets). Open the consulting-alumni pipeline (BCG / McKinsey / Bain AI practice partners in NYC) as a less-competed feed-stock into Sierra / Hebbia / Harvey.
- Bet #6 — publish the operator-translation newsletter Kit v1 by Week 4. Cadence decision by Week 8 (weekly vs. biweekly). Cross-pollinate Bet #1 Playbook downloads with newsletter subscribers; correlation is the right-audience proof.
- Bet #4 — pilot one trajectory cost-and-latency audit as the Bet #1 module test case. Free first audit for one mid-market AI-spending company; case study at month 6.
- Track Crux #1 (Anthropic ARR Q3 2026 resolution) as the single load-bearing Bet #2 timing variable. Sign before resolution if conviction is high; after if wobbly.

Months 6–12 · Commit to whichever survived

- If Bet #1 Playbook clears the falsifiability test (500 downloads / 50 inbound), commit to the advisory phase: 3–5 paid engagements at $25–100K each, with at least two Procurement Audit reports and one Article 14 oversight playbook in deliverables. Decision on SaaS productization at month nine.
- If Bet #2 offer has not landed by month 6 in the target band, expand to the second tier (Decagon NYC, Regal, smaller bands). If no offer by month 9, the bet on Alex's profile is wrong OR the market is more SF-anchored than the data suggests. Reassess.
- If Bet #6 has crossed 5,000 subscribers and 3+ inbound advisory or role conversations, commit harder (weekly cadence, paid tier). If not, kill at month 6 per falsifiability.
- Bet #3 audit of the ten SaaS systems most lacking MCP servers in enterprise GTM (Q3 2026). Pair-with-Cloudflare / Kong / Pomerium as the AI-procurement specialist on gateway deployment. Decide on productization vs. advisory-only at month twelve.
- Track Crux #2 (MCP commons or fork by EOY 2026), Crux #3 (OSWorld 65%), Crux #4 (EU AI Act teeth). Each answer-event reprices one or more bets.

Months 12–18 · Compound around the survivors

- Bet #1 — if SaaS productization decision is GO, capitalize. If NO, sustain the advisory practice at $250–500K annual revenue with three-product bundling (Playbook + Architecture Audit + FinOps for Trajectories).
- Bet #2 — if a vertical-agent role landed, the first twelve months are about delivering on the Confirm + Modify + Conclude gaps for that vertical's primary job. The advisory practice continues at a controlled cadence.
- Bet #3 — productize at month twelve or eighteen if scar tissue justifies; otherwise sustain as advisory.
- Bet #7 — if Bets 1, 2, 3 stall, this becomes the primary path. Build relationships at three NYC funds via RAAIS, Betaworks, FirstMark MAD; active applications Q4 2026.
- Track all five Cruxes monthly. The one that resolves first reprices the next 12 months' sequencing.

Per-quarter check-in template

Status against each Bet: hypothesis still holding? Leading indicators moving the right way? Conviction up, down, or flat? Cruxes resolved in the quarter — which bets re-rank? Risks materialized — which bets need defensive moves? Talent and Capital Flow refresh — what is the NYC hiring market doing? Each quarter should produce a one-page tracker update plus a single-decision focus for the next twelve weeks.

Part XIII · The Agent-Specific Procurement Rubric (Appendix to Bet #1)

This appendix is the agent-specific section of the future Enterprise AI Procurement Operating Standard. Per the Volume III mandate, it must drop into the Playbook as Part XIII or as a standalone rubric document. The structure follows the Job 6 phase map (Define, Locate, Prepare, Confirm, Execute, Monitor, Modify, Conclude) and the five risk-officer questions from the Volume III A6 brief. Each section names what the buyer should ask, what good vendor answers look like, what vapor sounds like, and where to verify.

1. Define — Map the agent's autonomy scope

Ask the vendor: What does this agent do unattended? What requires a human approval gate? What requires sub-agent privilege separation? Where does the agent's decision authority stop? Good answer: a written autonomy scope document with a tool-by-tool authority list ("agent can read invoices, can route to approver, cannot post journal entries without controller sign-off"), with action-confirmation gates documented, sub-agent privilege separated. Vapor: "the agent uses your existing approval flows" without a written map. Verify: ask for the system prompt, the tool schema, and a trajectory replay of a representative task.

2. Locate — Inventory the data and tool reach

Ask: What systems does the agent read? What does it write? What is the OAuth blast radius for a typical task? Where does memory live physically? Who can subpoena it? Good answer: a data-flow diagram with named systems, scopes per tool call, MCP server inventory, memory residency commitment (US, EU, both), Schrems-II posture, GDPR Article 17 forget API with auditable deletion confirmation, HIPAA BAA available if applicable. Vapor: "we follow industry best practices." Verify: ask for a DPA, a memory residency commitment in writing, and a sample deletion-confirmation log.

3. Prepare — Tool-boundary policy and runtime safety stack

Ask: Show me your tool-boundary policy. Which tools can the agent invoke unsupervised? Which require approval? Which are blocked when context is tainted by external content? What is your defense stack for indirect prompt injection via tool returns? Show me your adaptive-adversary test results (not public-corpus). Good answer: a five-layer defense stack — treat tool returns as untrusted (tag bytes, degrade high-privilege when planning context is tainted); action-confirmation gates for write/pay/send/escalate or run in reversible sandbox; output validation and schema enforcement; sub-agent privilege separation; continuous red-team (Promptfoo, Garak, Lakera Red, or equivalent). Adaptive-adversary numbers between 60–80%, honestly disclosed. Vapor: 99% PI detection without adaptive-adversary qualification. Verify: ask for the most recent red-team report and the adaptive-adversary methodology.

4. Confirm — Pre-flight evaluation evidence

Ask: Show me a signed reproducible eval report for this agent against my use case, with model pin, dataset hash, harness version. Show me your trajectory regression set. How big is it? How was it curated? How often is it refreshed? Good answer: a written signed report tied to a specific model pin (e.g., claude-opus-4-5-20251101), dataset hash, harness version, and a regression set of 200–1,000 trajectories curated from production traffic with monthly refresh cadence. Inspect AI compatibility is a plus; OpenTelemetry GenAI conventions for trace ingest are table stakes. Vapor: "our internal benchmark." Verify: ask to reproduce one score on the buyer-side; treat unreproducible scores as untrusted.

5. Execute — Action-rollback story

Ask: Show me the action-rollback story for every write tool. If the agent writes the wrong value into our system of record, what is the recovery path? Good answer: write tools wrapped in a reversible-action layer (database row versioning, idempotency keys, dead-letter queues for irreversible side effects with human approval before execution), audit log accessible to the buyer in real time, alerting on anomalous write rates. Vapor: "we have logging." Verify: ask for a sample audit log and the recovery procedure for a synthetic incident.

6. Monitor — Trajectory observability and silent-failure detection

Ask: How do I detect a trajectory that completes-as-judged but writes the wrong value to a system of record? What is your silent-failure rate? How do I instrument my own observability against your agent? Good answer: trajectory observability exposed to the buyer via OpenTelemetry GenAI conventions, sample-based human review of agent decisions, written silent-failure rate from production with the methodology, and integration hooks for the buyer's SIEM. Vapor: "you can use our dashboard." Verify: ask for an exported sample trace and confirm it parses in the buyer's observability stack.

7. Modify — Non-engineer intervention capability

Ask: Can a non-engineer line manager redirect the agent mid-task or correct a policy without an engineering ticket? How long does it take to ship a policy correction? Good answer: a cockpit UI built for the line manager (CX lead, FP&A director) with policy editing, prompt editing, and approval-flow editing without code; ship-time measured in hours not days. Vapor: "you can file a support ticket." Verify: ask for a live demo from a non-engineer.

8. Conclude — Counterparty evidence and audit pack

Ask: Show me the EU AI Act / NIST AI RMF GenAI Profile / ISO 42001 conformity-assessment tie-out. Show me how this evidence stays fresh through annual review. Good answer: a written tie-out document mapping the vendor's controls to each named framework with named control IDs and reproducible evidence per control. Counterparty-specific evidence packs available for each of the buyer's six approval lanes (InfoSec, Legal, Privacy, AI Governance, Procurement, business sponsor). Annual refresh process documented. Vapor: "we have SOC 2." Verify: ask for the tie-out and walk it line-by-line.

Scoring guidance

Rate each of the eight sections on a five-point scale (5 = excellent / verified / signed evidence; 1 = vapor / unverifiable / unwilling to disclose). A vendor scoring below 3 average should not pass an F1000 AI Governance review on agent-specific procurement as of May 2026. The buyer should expect that approximately twenty percent of agent vendors today clear an average of 3.5 or higher on this rubric; the remaining eighty percent need either a roadmap commitment in writing or a contractual SLA backed by exit terms. The Procurement Playbook will maintain a vendor-readiness leaderboard updated quarterly as part of the public artifact.

Closing note

This rubric is the buyer-side complement to the vendor-side eval and observability roadmap (Stratum VII). The procurement-readiness gap in the agent layer is not that buyers do not know what to ask; it is that no canonical source compiles the questions in agent-specific form and validates the answers. The Volume III observation is that the canonical source is what Bet #1 ships. The rubric above is the demo. The Playbook is the full document. The SaaS productization is the option, not the first move.

VISUAL ATLAS

Substrate · Volume II

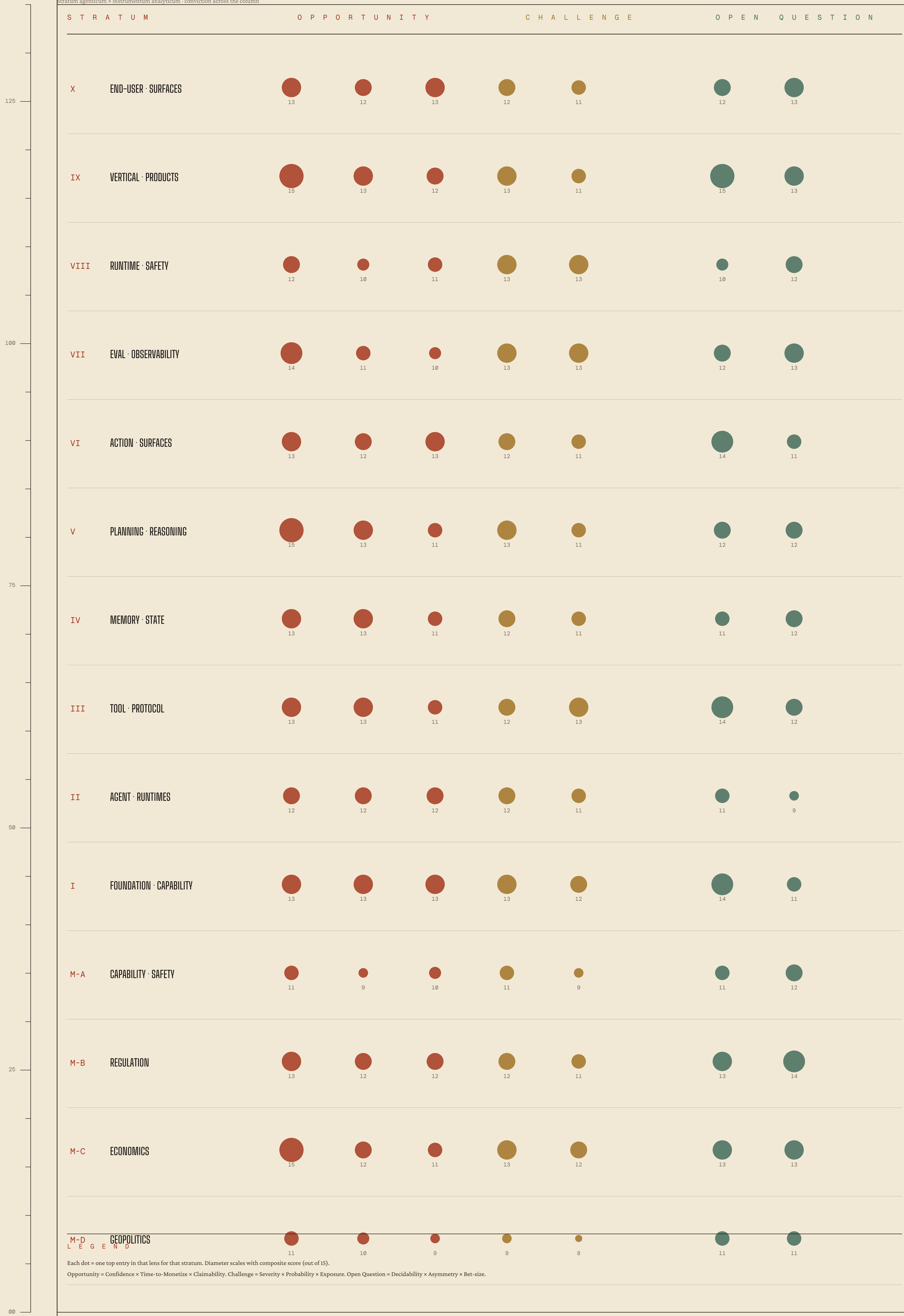
Plates VII–XI · OCQ heat · Wardley · 7 Powers · JTBD · Action Portfolio

Section V of VII

OCQ HEAT MAP · AGENT

Fourteen agent strata × three lenses · scored opportunity, challenge, open question.

Stratum agenticum × instrumentum analyticum · conviction across the column



L E G E N D

Each dot = one top entry in that lens for that stratum. Diameter scales with composite score (out of 15).

Opportunity = Confidence × Time-to-Monetize × Claimability. Challenge = Severity × Probability × Exposure. Open Question = Decidability × Asymmetry × Bet-size.

VIII

WARDLEY MAP · AGENT STACK

Four anchored agent user-needs · dependencies cascading · evolution stage along the X-axis.

Cartographia agenticum · stage of evolution in 2025–2026

1 · MCP gateways · Custom to Product · H2 '26
2 · Computer use · Genesis to Custom · mid-late '26
3 · Voice substrate · Product to Commodity · mid-late '26
4 · Memory · forced binary · H2 '26
5 · Procurement-grade controls · Genesis to Custom · Q4 '26
6 · Eval / obs consolidation · Q4 '26 – H1 '27
7 · Foundation labs walking up-stack · continuous

7 POWERS · AGENT

Per-stratum mapping of the seven Helmer powers · who holds them · direction of travel.
Septem potestates Helmeri applicatae stratis agenticis

STRATUM		SCALE	NETWORK	COUNTER-POS.	SWITCHING	BRAND	CORNERED	PROCESS	HOLDERS
X	END-USER SURF.		·	○	●	○	●	○	MSFT Copilot · Apple AI · WhatsApp
IX	VERTICAL PROD.	·	○	·	●	○	·	●	Sierra · Harvey · Glean · Hippocratic
VIII	RUNTIME SAFETY		·	·	○	○	·	·	Lakera niche; mostly absorbed
VII	EVAL / OBS			·	○	○		○	LangSmith · Braintrust · Inspect AI
VI	ACTION SURF.	○	·	·	○	·	○	·	LiveKit (CR) · Browserbase (SC)
V	PLANNING / RES.						○	●	Inside labs only
IV	MEMORY / STATE			·	○	·		·	Zep compliance niche
III	TOOL / PROTOCOL		●	●	○			·	MCP gateways · Anthropic CP win
II	AGENT RUNTIMES								Thinnest layer · no powers
I	FOUND. CAPABILITY		·	·	○	○	●	●	Anthropic / OAI / Google · talent CR
M-A	CAP. SAFETY			·	○	●	●	●	Anthropic ASL most defensible
M-B	REGULATION			●	○		●	●	Mistral CP · Palantir · Bet #1
M-C	ECONOMICS	●			·			·	Hyperscaler scale only
M-D	GEOPOLITICS			●	·		●	·	Mistral · Manus (CN data CR)

LEGEND · POWERS

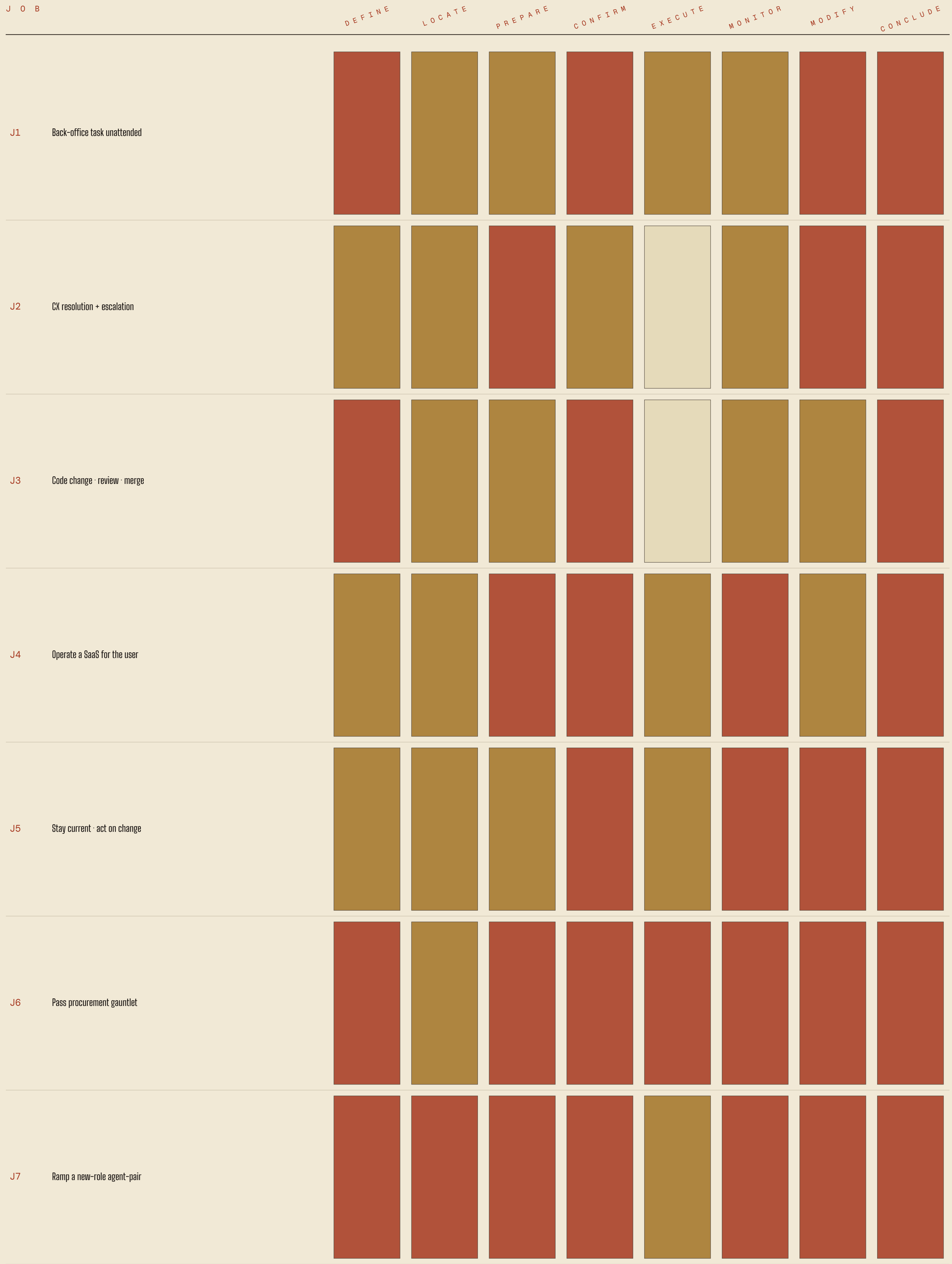
● strong & durable ○ present, partial · weak absent

Discipline: mindshare is not a power. ARR is not a power. A power requires benefit + barrier.

JTBD CANVAS · AGENT

Seven ecosystem-level jobs the agent stack is hired for · Ulwick eight-phase map · Conclude is universally unserved.

Septem opera quaesita · phases octo · Concludere universaliter desertum



UNDER-SERVED PATTERNS (CROSS-JOB)

1 · Conclude is universally unserved — signed evidence, warm handoff, provenance, change-log, decision-routing, agent taper

2 · Modify by a non-engineer — replay UIs for engineers exist; cockpit UIs for line managers / CX leads / FP&A do not

3 · Confirm (pre-flight) — PRMs internal at labs but not exposed to buyers; the difference between 25Kand250K ACV

HEAT MAP

Over-served · Execute on code (14+ funded); Locate / knowledge-search (consolidating to 2–3); Execute on inbound chat.

well-served partial under-served

ACTION PORTFOLIO · 6 / 12 / 18

Seven refreshed Bets sequenced across three windows. Bet #1 first (claim power) · Bet #2 second (collect equity) · Bet #3 third (compound).

Septem electa per tres fenestras temporis · ordo agendi

BET	MONTHS 0–6 · PLANT	MONTHS 6–12 · COMMIT	MONTHS 12–18 · COMPOUND
1 Procurement Operating Standard 5/5	Publish open Playbook (wk 12) 30 expert interviews AI Gov sign-off audit	3–5 paid engagements Procurement Audit reports Article 14 playbook	SaaS productization decision $250–500K annual rev Three-product bundle
2 Vertical Agent GTM Role 5/5	NYC search · 10 targets Time offer before up-round Consulting alumni pipeline	Sign at chosen vertical Close Confirm + Conclude gap Anthropic ARR resolution	Deliver job · vertical Compound from inside Re-rank by Crux #1 outcome
3 MCP-Native Practice 4/5	Audit 10 SaaS · gateway pair Pair w/ Cloudflare · Kong Reframe to advisory	Productize or sustain advisory Gateway-adjacent positioning Watch fork Crux #2	Compound · advisory + Playbook Decide on platform vs. practice
4 FinOps for Trajectories 4/5	Free first audit · 5 cos Case study Per-trajectory rubric	Fold into Bet #1 module Decide scale or absorb	Decay window honest Bedrock auto-routing watch
5 RAG + Memory Architecture 4/5	Bundle as audit Memory architecture line Vendor-neutral guidance	3 architecture audits Three-product practice	Sustain · selective Track CRUX #5
6 Operator Newsletter 3/5	Kit v1 · wk 4 Cadence decision · wk 8 JTBD Job 5 angle	5K subs / 3 inbound · or kill Conclude-phase translation	Sustain · paid tier Distribution layer for #1–#3
7 VC Operating Partner 3/5	Background networking RAAIS · Betaworks · MAD No applications	3 NYC funds · relationships Watch principal-to-operator	Primary path if 1–3 stall Active applications Q4 '26

SEQUENCING · THE VOL III DELTA

The five-framework convergence reorders the prior addendum: Bet #1 first (claim Process Power at Meta-B), Bet #2 second (collect equity at a power-holder), Bet #3 third (advisory practice compounding from both). Bets 4–5 fold into Bet #1 as modules; Bet #6 is the distribution layer; Bet #7 the long-arc fallback.

CRUXES THAT RE-RANK EVERYTHING

C1 · Anthropic ARR 24Bvs30B · Q3 '26

C2 · MCP commons vs fork · EOY '26

C3 · OSWorld 65% on frontier · Q3 '26

C4 · EU AI Act Article 14 teeth · late '26

C5 · Standalone memory absorbed vs niche · H2 '26

THE WORKING SURFACE

The Living Tracker

Volume III · seven Bets · talent + capital flow · cruxes · risks

Section VI of VII

AI Agents — Living Tracker (Volume III)

> *Sub-tracker for the agent layer specifically. Companion to and feeder of `output/ai-stack/OCQ_TRACKER.md`. The parent tracker tracks the full AI stack; this one tracks the agent-layer slice at higher resolution.*

Last updated: 2026-05-12 **Owner:** Alex **Companion artifacts:** - `AI_AGENTS_SUBSTRATE.pdf` (Plates I–VI · the agent column) - `AI_AGENTS_SUBSTRATE_VOL2.pdf` (Plates VII–XI · framework plates) - `AI_AGENTS_MASTER_PLATE.pdf` (Plate M · the agent seen whole) - `AI_AGENTS_REPORT.docx` (Foundation Reference · Vol III) - `AI_AGENTS_ADDENDUM.docx` (Decisions Playbook · Vol III) - Parent: `output/ai-stack/OCQ_TRACKER.md`

How to use this tracker

Same operating cadence as the parent OCQ_TRACKER. Three sections, three cadences:

Section	Cadence	Owner action
A. Bets — agent-layer deltas to the parent 7	Monthly	Update conviction, refresh leading indicators, mark dead modules
B. Agent-specific talent + capital flow	Bi-weekly	Add high-signal moves into / out of agent-specific companies
C. Agent-specific cruxes and risks	When triggered	Each crux has a measurable answer-event; log date and re-rank

The point is **disciplined attention on agent-layer leading indicators** that move the seven Bets. Where the agent-layer view changes a Bet, the change is annotated here AND mirrored in the parent tracker.

A. THE SEVEN BETS — AGENT-LAYER REFRESH

The seven Bets carry over from the parent tracker. Volume III refreshes each at agent-layer resolution. Significant deltas annotated below.

Bet #1 — Enterprise AI Procurement Operating Standard (agent-specific)

Delta from parent: Adds the seven agent-specific overlays the parent tracker does not name — tool-boundary policy, indirect-prompt-injection adaptive red-team, action-rollback documentation, sub-agent privilege separation, signed reproducible eval reports, EU AI Act Article 14 human-oversight tie-out, sectoral overlays. Anchors the EU AI Act Article 14 April 2026 draft as the most-actionable near-term lever. **Conviction: ★★★★★ — refreshed UP from parent** on the strength of five-framework convergence. The 7 Powers screen names procurement Process Power at Meta-B as the only durable position Alex can claim rather than rent. Wardley places procurement-grade controls at Genesis = the unclaimed flag. JTBD Job 6 phase map IS the Playbook table of contents. **Sequencing change:** Now first, not third. **The single biggest delta in Volume III. Next action:** Outline the Playbook using the Job 6 phase map and the A6 five-questions structure. Thirty expert interviews Weeks 2–6. Publish Week 12 with the open downloadable Agent Procurement Rubric (see AI_AGENTS_ADDENDUM.docx Part XIII) attached.

Bet #2 — Vertical Agent GTM Leadership Role

Delta from parent: NYC target list expands from {Sierra, Decagon, Glean, Harvey, Hippocratic, Augment} to **add Hebbia, Rogo, Clay, Runway, Ramp AI org**. Consulting alumni (BCG / McKinsey / Bain AI practice) surfaced as a third, less-competed feed-stock alongside Stripe / Ramp / Datadog / Snowflake. **Anthropic ARR resolution Q3 2026 is the single load-bearing valuation variable. Conviction:** ★★★★★ — held. **Next action:** Time the offer before the up-round at Hebbia, Rogo, or Augment (the under-funded relative to ARR velocity). Sign **before** Anthropic ARR resolution if conviction high; **after** if wobbly.

Bet #3 — MCP-Native Enterprise Integration Practice (reframed)

Delta from parent: Reframed from "build MCP servers for enterprise GTM SaaS" to "advisory + gateway-adjacent practice." The durable power is at the gateway control plane (Cloudflare, Kong, Pomerium) — Switching Costs via auth / audit / rate-limit / secret-injection — not in productized servers, which commoditize the moment Salesforce or HubSpot ships first-party. Alex's claimable position is Process Power flavor (operational scar tissue) + Branding flavor (canonical voice via the Playbook), not software moat. **Conviction:** ★★★★ — held with reframe. **Next action:** Q2 2026 audit of ten SaaS systems most lacking MCP servers, filtered by which Job 4 / Job 1 outcomes each unlocks. Pair-with-Cloudflare / Kong / Pomerium positioning.

Bet #4 — Inference Cost Optimization (split into per-token + per-trajectory)

Delta from parent: Splits into two layers — per-token FinOps (the parent framing) and **per-trajectory FinOps** (the agent-specific reframe). Per-trajectory window is sharper: planner-executor split is the durable architectural idea, but hyperscaler auto-routing default-on by H2 2027 narrows the advisory window to ~12 months. **Conviction:** ★★★★ — held. **Next action:** Free first audit for one mid-market AI-spending company; build case study; **fold into Bet #1 as a module** rather than scale as standalone bet by Q3 2026 unless adoption signals say otherwise.

Bet #5 — Enterprise RAG + Memory Architecture

Delta from parent: Folds memory architecture as a service line per Crux #5 directional answer (consumer/prosumer absorbed; compliance-enterprise niche-standalone). No separate "memory practice" bet. **Conviction:** ★★★★ — held. **Next action:** Bundle as "Enterprise AI Architecture Audit" alongside Bet #1's Playbook + Bet #4's FinOps. Three-product practice, one buyer.

Bet #6 — Operator's Translation Newsletter

Delta from parent: Confirmed as the **distribution layer for Bets #1–#3** per JTBD analysis (Job 5: stay current + act). Angle sharpens around Conclude-phase translation ("what should you do next") rather than executive-brief summarization. **Conviction:** ★★★ — held. **Next action:** Kit v1 Week 4. Cross-pollinate Bet #1 Playbook downloads with newsletter subscribers (correlation = right-audience proof).

Bet #7 — VC Operating Partner / Platform Path

Delta from parent: The principal-to-operator pattern (Carey Lai Insight → Sierra Apr 2026; 3–5 similar moves) confirms platform and operator paths converging at the senior level. **Watch as fallback only. Conviction:** ★★★ — held; ★★★★ if becomes primary.

B. AGENT-SPECIFIC TALENT + CAPITAL FLOW

B1. Senior moves into agent-specific companies (rolling 12 months)

See `framework_analysis/B8_talent_capital_flow.md` for the full table. Updates here bi-weekly.

May 2026 read-through: Stripe / Ramp / Datadog / Snowflake → Sierra / Decagon / Glean / Hippocratic migration **accelerating**; March 2026 was the heaviest single month yet (8+ named LinkedIn moves into top three). NYC concentration at historic peak Q2 2026. Over-funded underperformers (11x, Mem, Magic.dev, Adept residual) losing senior talent to segment winners — supply rising while top equity rooms tighten.

Secondary pattern surfaced by Vol III (data forces, frameworks miss): - **MBB consulting → vertical agent enterprise GTM** is a third, less-competed feed-stock. Hebbia sources MBB heavily (Sivulka public). **Alex isn't currently positioned for it; 30-day push to MBB AI-practice alumni in NYC could open Hebbia / Sivulka pipeline directly.**

B2. Capital events ($50M+ agent-platform rounds, M&A, infra commits)

Key recent (Q1–Q2 2026): - Mar 2026 — Sierra additional $350M @ $10B rumored (The Information); highest agent-platform mark - Mar 2026 — Mistral C €600M @ €11B; first credible non-US enterprise-agent platform - Mar 2026 — Cognition acquires Codeium ~$3B (after OpenAI–Windsurf collapsed Jul 2025) - Feb 2026 — Harvey E $300M @ $5B; Hippocratic C $150M @ $2B; Factory B $100M @ $750M - Jan 2026 — Lovable A $200M @ $1.8B; Clay growth $100M @ $1.5B; Rogo B $50M @ $400M; Perplexity E $500M @ $18B - Apr 2026 — ElevenLabs D rumor $5B+; Galileo C rumor ~$60M - Q2 2026 — Ramp growth rumored $20B+; Hebbia C rumored

B3. Agent-layer ARR watchlist (May 2026)

Company	T12 → RR May'26	Disputed?	Note
Anthropic (parent)	$1B Dec'24 → **$24–30B** Apr'26	YES	Claude Code = inflection driver; **Bet #2 valuation crux**
Sierra	$0 → $175M+ Q1'26	low	Taylor public 4× YoY
Decagon	$1M 2024 → $80M+	low	Sierra peer
Glean	$50M → $300M+	low	Search-agent leader
Harvey	$25M → $100M+	low	Legal NYC tripling Q3'26
Hippocratic	$5M → $50M+	low	"1,000 agents deployed"
Abridge	$50M → $120M+	low	Fastest healthcare-AI
Cursor	<$1M → $500M+ RR	low	Anthropic-dependent
Lovable	$0 → $80M Q1'26	confirmed	Fastest EU SaaS ever
Hebbia	$30M → $50M+	inf	NYC HQ; raise overdue
Rogo	<$5M → $30M+	inf	NYC; should clear $1B Q3'26

Company	T12 → RR May'26	Disputed?	Note
Clay	\$20M → \$80M+	low	Only durable RevOps winner
Augment	\$5M → \$40M	low	Under-funded ratio
Mistral Le Chat Ent.	€30M → €100M+	inf	EU sovereign
11x	\$20M → \$20M flat	YES	Synthetic SDR ceiling

B4. NYC peak Q2 2026 — companies hiring Alex's exact profile

Sierra · Decagon · Glean · Hebbia · Harvey · Rogo · Clay · Ramp AI org · Hippocratic · Runway. Comp band \$250–400K base + 0.05–0.40% equity (inf). Window approximately 6–9 months before Anthropic ARR resolution + first vertical-agent IPO posture reprice equity bands.

C. AGENT-LAYER CRUXES & DECISION TRIGGERS

The five unresolved questions specific to the agent layer. When answered, re-rank bets and update parent tracker.

#	Crux	Decidability horizon	Answer-event to watch	Re-rank consequence
C1	**Anthropic ARR — \$24B or \$30B**	Q3 2026	Audited disclosure / leak	Lower bound = vertical-agent vals compress 20–30%; upper bound = Bet #2 timing accelerates
C2	**MCP — commons or silent fork**	EOY 2026	Major-vendor proprietary tool-use schema announcements; A2A adoption outside Google	Fork = Bet #3 productized form dies; commons = advisory practice accelerates
C3	**OSWorld 65% on a frontier system**	Q3 2026	Public scoreboard event	Crossed = computer use moves from supervised to deployable for narrow back-office a quarter earlier

#	Crux	Decidability horizon	Answer-event to watch	Re-rank consequence
				than this volume assumes
C4	**EU AI Act Article 14 enforcement teeth**	Late 2026	First GPAI fine; Commission Art. 14 enforcement actions	Teeth = Bet #1 becomes $10B+ category; paper = stays niche but defensible
C5	**Standalone memory — absorbed or niche**	H2 2026	Mem0 or Letta acquisition vs. independent Series B at higher val	Directional: consumer/ prosumer absorbed; compliance enterprise niche-standalone (already folded into Bet #5)

D. AGENT-LAYER STRUCTURAL RISKS

#	Risk	What it threatens	Watch
R1	**Foundation labs walking up-stack into verticals**	Bet #2 — ChatGPT Business connectors, Claude for Work, Gemini Workspace agents continuously compress vertical-agent vendor moats	Connectors / Skills / Apps store launches; named-customer deflection from Glean / Cursor
R2	**MCP silent fork via Responses-API proprietary path**	Bet #3 productized form — production OpenAI agents bypassing MCP for latency	OpenAI Responses-API extensions outpacing MCP-spec evolution; A2A adoption outside Google
R3	**Anthropic ARR resolution downside**	Bet #2 valuation + timing — $24B lower bound compresses Sierra $10B → $6–8B	Q3 2026 disclosure or leak
R4		Bet #4 window + standalone eval / obs	AWS Bedrock auto-optimization feature

#	Risk	What it threatens	Watch
	Hyperscaler bundling of runtime safety / eval / sandbox / routing	category — Bedrock auto-optim, Vertex routing, Azure routing default-on plausible H2 2027	launches; Vertex / Azure routing default-on announcements
R5	**EU AI Act paper-tiger outcome**	Bet #1 advisory TAM — small / procedural enforcement late 2026	Commission guidance; first GPAI fine size

E. UPDATE LOG

Date	Change
2026-05-12	Initial Vol III creation. Seven Bets refreshed at agent-layer resolution. Bet #1 sequenced **first** (delta from parent). Bet #3 reframed to advisory + gateway-adjacent. NYC target list expanded for Bet #2 (+ Hebbia, Rogo, Clay, Runway, Ramp AI). MBB consulting feed-stock surfaced as Bet #2 secondary path. Five Cruxes and five Risks refreshed for agent layer.
next update	*Bi-weekly: refresh B1 talent + B2 capital. Monthly: bet status. Trigger-based: cruxes + risks.*

F. NOTES FOR THE NEXT SESSION

The parent `OCQ_TRACKER.md` should be updated with the following delta annotations:

1. **Section A Bet #1:** Note "Volume III refresh: Bet #1 now sequenced first based on five-framework agent-layer convergence. See AI_AGENTS_TRACKER.md §A and AI_AGENTS_ADDENDUM.docx Part XI."
2. **Section A Bet #2:** Update NYC target list to add Hebbia, Rogo, Clay, Runway, Ramp AI org.
3. **Section A Bet #3:** Note reframe from productized servers to advisory + gateway-adjacent.
4. **Section A Bet #4:** Note split into per-token and per-trajectory FinOps; per-trajectory window narrowed.
5. **Section A Bet #5:** Note memory architecture folded in as service line per Crux #5 directional answer.

6. **Section C Cruxes:** Add agent-specific C3 (OSWorld 65% on a frontier system) as a new agent-layer crux.
7. **Section B5 NYC snapshot:** Update to “NYC vertical-agent hiring at historic peak Q2 2026; 10+ targets simultaneously.”

Working surface for the next 18 months at the agent-layer resolution. Update the parent tracker on the same cadence. Both documents read together; neither is the whole.